

Renal Replacement Therapy (Dialysis and Transplantation) in Pediatric End-Stage Kidney Disease

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KEY POINTS

- ESKD in children has broad clinical consequences, including a major impact on growth and development.
- Most common causes of ESKD in children are CAKUT, acquired glomerular disease, and genetic diseases.
- Although kidney transplantation is the preferred modality of RRT, the majority of children require at least temporary dialysis.
- Choice of dialysis modality depends on medical issues, but also on age and geographical and social considerations.
- Some complications are unique to or more severe in pediatric dialysis patients, such as catheter malfunction, vascular damage, and stenosis and infections.
- Advances in kidney transplantation over the past decades have transformed the field, enabling children with ESKD to live close to normal lives, with constantly improving outcomes.
- Posttransplantation infections may be more common in children, particularly viral infections.
- Clinical data on many medications and interventions are not available in the pediatric ESKD population and require further study.

Until the middle of the twentieth century, end-stage kidney disease (ESKD) was a lethal condition, with a life expectancy of at most weeks to months. The developments over the subsequent 2 decades in both dialysis and transplantation completely transformed the lives of patients with chronic kidney disease (CKD), dramatically reducing mortality and morbidity. As technologies and surgical techniques developed, these treatments became available for children and later infants as well.

Treating pediatric ESKD patients poses unique challenges to the medical team and the health care system, which must address not only the disease itself, but also the many manifestations that affect patients' lives and families. Reynolds et al. reported 20 years ago, that compared with the general population, adult survivors of pediatric ESKD were less socially mature, had lower educational qualifications, fewer had intimate relationships outside the family, and more were unemployed. Two decades later, changing this unfavorable state remains the major goal for providers treating children with ESKD.¹

EPIDEMIOLOGY OF END-STAGE KIDNEY DISEASE

There are many barriers to accurate assessment of the global epidemiology of end-stage renal disease (ESRD) in pediatric patients. Approximately 80% of pediatric renal replacement therapy (RRT) is provided to patients who live in high-income countries, who account for only 12% of the global population. Registries, such as the North American Pediatric Renal Trials and Collaborative Studies (NAPRTCS), the European Society of Paediatric Nephrology, the European Renal Association and European Dialysis and Transplant Association (ESPN/ERA-EDTA), which can provide robust information, exist only in those developed countries. The information from developing countries is derived from surveys conducted among health-care providers or from reports of patients admitted to tertiary centers. Both these methods result in underestimation of ESKD because much of the population has limited access to health services. Furthermore, in countries in which registries exist, there are differences in the age span reported as pediatric ESKD. Information from high-income countries show an incidence of ESKD of 15 per million age-related population (pmarp) in the United States, 9.5 pmarp in most West-European countries and Australia, and only 4.0 pmarp in Japan.^{2–7} This is less than 10% of the incidence of ESKD in adults in the same countries. On the far end of the spectrum, Nepal, Nigeria, and some Eastern European countries report an incidence of less than 1 pmarp.^{6,8} This very low incidence probably reflects underdiagnosis rather than true rarity of ESKD. The incidence, in all registries, is higher in adolescents, due to new cases of acquired kidney disease typical to this age group, as well as the accelerated progression of CKD during adolescence. Rapid physical growth and possibly also hormonal factors are responsible for the decline in kidney function observed in adolescents with CKD. Higher incidence in males, due to the higher prevalence of congenital anomalies of kidneys and urinary tract (CAKUT) in boys compared with girls, is also a finding common to all registries and reports.^{9–13} The ancestry composition of each one of the registries may also affect the incidence of ESKD

reported, suggesting yet undefined genetic susceptibility factors. In the United States, African-American children have a twofold increased incidence of ESKD compared with Caucasian children.³ High-risk variants in *APOL1* gene may partially account for this observation, as they appear in about 16% of African-American children and not in other ethnic groups. CKD patients with high-risk *APOL1* variants present with later-onset focal segmental glomerulosclerosis (FSGS; 89% of CKD patients with high-risk variants, compared with 25% with low-risk genotype), with lower glomerular filtration rate (GFR) at diagnosis, rapid decline in GFR, and higher risk of uncontrolled hypertension (HTN).^{14,15} Racial differences are also seen in other countries. In the United Kingdom the incidence is higher in patients who originate from Southeast Asia, and in Australia and New Zealand, the incidence was higher in indigenous populations.^{16,17} In Kuwait, the incidence of CKD is high (17 pmarp), probably due to the increased frequency of autosomal recessively inherited diseases among populations with a high rate (40%–50%) of consanguineous marriages.¹⁸ A study from Jordan showed similar findings—among children with ESKD in Jordan, 23.2% were attributed to hereditary disease, and 8% were because of autosomal recessive primary hyperoxaluria type 1 (PH1) alone (compared with the US incidence in the period of 2011–2015 of 12.5% hereditary diseases and 0.2% with PH1).^{3,19} Over the years there is an increase in both incidence and prevalence of pediatric ESKD in Western countries (United States Renal Data System [USRDS], 2017).²⁰ This may not indicate a real biological change in disease incidence, but instead may result from the inclusion of individuals (such as infants), who in the past would not have been thought to be eligible for RRT. In addition, improved childhood RRT survival rates may increase the ESKD prevalence.

ETIOLOGY

A major obstacle to accurately comparing the various etiologies leading to ESKD worldwide is the definition of a given entity, which may vary by reporting center (Table 74.1). This may be due to the introduction of a more accurate diagnosis based on genetic or other diagnostic study, which may be expensive and thus not widely available, or to adherence to previous dogmas that are now regarded as inaccurate. These include FSGS, which is a histological diagnosis encompassing different disease processes, or reflux nephropathy, which may be more accurately described as hypodysplastic kidneys with vesicoureteral reflux (VUR). In addition, a specific patient's diagnosis may occasionally fit more than one etiologic category; for example, autosomal dominant FSGS may be recorded under FSGS or as a hereditary disorder. A strictly defined and universally accepted CKD etiologic classification is often lacking. In spite of these caveats and lack of concise etiologic definitions, CAKUT remains the leading primary diagnosis in pediatric patients with ESRD worldwide, responsible for 24% to 39.5% of patients with ESKD (Table 74.1).^{3,5,7,9,20,21,24} Of note, this is proportionately lower compared with the incidence of CAKUT at earlier CKD stages, reflecting the relatively slow progression rate of this subgroup of diseases, at least during childhood and early adolescence. Hypoplastic–dysplastic kidneys with or without VUR and obstructive uropathies are the main categories of

Table 74.1 Main Causes of End-Stage Kidney Disease in Various Registries and Reports Worldwide

Source	USRDS	ANZDATA	Italkid	ERA-EDTA	Iran	Japan	UK
Years of data collection	2011–2015	2010–2015	1990–2000	2005–2012	1993–2006	1998	2015
Number of patients	7340	327	263	1973	120	582	769
Age, years	<21	<18	<20	<15	<14	<20	<16
Registry type	Incidence	Incidence	Prevalence	Incidence	Incidence	Prevalence	Prevalence
CAKUT (%)	1617 (22%)	105 (32%)	104 (39.5%)	680 (34.5%)	32 (26.7%)	208 (35.7%)	267 (34.7%)
Hypodysplasia ± VUR	905 (12.3%)	79 (24%)	27% ^a		21 (17.5%)	198 (34.0%)	145 (11.4%)
obstructive uropathy	712 (9.7%)	26 (7%)	12% ^a		9 (7.5%)	10 (2%)	
Glomerulonephritis (%)	1603 (21.8%)	84 (24%)	28 (10.6%)	320 (16.2%)	8.3%	177 (30.4%)	88 (11.4%)
FSGS	849 (11.6%)	24 (7%)	16 (6.1%)		5.0%	112 (19.2%)	
Hereditary (%) autosomal recessive	921 (12.5%)	28 (9%)–6%	66 (25.1%)	410 (20.7%)	57 (47.5%)	101 (17.4%)	139 (18.1%)
	470 (6.1%)		57 (21.7%)		47.5%	73 (12.5%)	

^aEstimated.

ANZDATA, Australian and New Zealand Dialysis and Transplant Registry; CAKUT, congenital anomalies of kidneys and urinary tract; ERA-EDTA, European Renal Association and European Dialysis and Transplant Association; FSGS, focal segmental glomerulosclerosis; USRDS, United States Renal Data System; VUR, vesicoureteral reflux.

Data adapted from ANZDATA Registry. Chapter 11: Paediatric report. In *40th Report: Australia and New Zealand Dialysis and Transplant Registry, Adelaide, Australia*. 2018. Available from http://www.anzdata.org.au/anzdata/AnzdataReport/35thReport/2012c11_paediatric_v1.9.pdf [Last accessed June 14, 2019]; ESPN ERA-EDTA. *ESPN/ERA-EDTA Registry*. Available from <https://www.espn-reg.org/index.jsp?p=hom> [Last accessed June 14, 2019]; Hattori M, Sako M, Kaneko T, et al. End-stage renal disease in Japanese children: a nationwide survey during 2006–2011. *Clin Exp Nephrol*. 2015;19:933–938; Ardissino G, Daccò V, Testa S, et al. Epidemiology of chronic renal failure in children: data from the Italkid project. *Pediatrics*. 2003;111:e382–7; Hooman N, Esfahani ST, Mohkam M, et al. The outcome of Iranian children on continuous ambulatory peritoneal dialysis: the first report of Iranian national registry. *Arch Iran Med*. 2009;12:24–8; USRDS Annual Data Report. Chapter 7: ESRD among Children, Adolescents, and Young Adults. *Am J Kidney Dis*. 2018;71:S383–S416; UK Renal Registry. 18th annual report. *Nephron* 2016;132:1–366.

CAKUT, and the differences in listed diagnoses between various reports and registries are often semantic rather than biological in nature. In most countries, the second most frequent category is glomerulonephritis (8.3%–30.4%), of which the most common entity is FSGS, especially in Japan and the United States, where it is the most common cause of ESRD among adolescent African-Americans. Inherited diseases comprise the next category, although there are variations in which diseases are classified under this heading in different reports. The frequency varies from 19% (Australia and New Zealand) to 47.5% (Iran).^{2,21} In countries where there is a high rate of consanguineous marriages, autosomal recessive inherited diseases comprise a much higher proportion of ESKD etiology. Specific diagnoses include cystinosis, familial steroid-resistant nephrotic syndrome (SRNS; mostly due to mutations in *NPHS1*, *NPHS2*, and *PLCE1*), autosomal recessive polycystic kidney disease, PH1, nephronophthisis, and others.

CLINICAL CONSEQUENCES OF PEDIATRIC END-STAGE KIDNEY DISEASE

In children with ESKD, the treatment goal is not only to prolong survival and to enable a life as normal as possible, but also to promote normal long-term growth and development.

As life expectancy is much longer in a young child with ESKD, compared with a patient in his sixth or seventh decade of life, the child may go through the cycle of CKD–dialysis–transplantation more than once. This also has implications on the choice of treatment modalities (e.g., type of dialysis, vascular access) and the priority given in allocation of organs for transplantation.

Before describing the various modalities of RRT in children, the main medical issues of pediatric ESKD and their management will be discussed.

NUTRITION AND WEIGHT

Nutrition has a critical role in childhood. It significantly affects both linear growth and neurocognitive development. Malnutrition, which is the result of poor appetite, vomiting, and decreased intestinal absorption, is common in children with ESKD. Close monitoring is crucial when the patient is younger than 1 year (every 2–4 weeks), decreasing in frequency as the patient grows. Dietary intake should be assessed periodically. In patients on a formula-based diet, this includes formula type, caloric value, protein content, volume, and duration of each meal. Height and weight are plotted on appropriate growth curves and Z-scores are calculated. Head circumference is measured regularly in children younger than 3 years. Normalized protein catabolic rate can be calculated for patients on dialysis. The individual nutritional plan is based on Kidney Disease Outcomes Quality Initiative (KDOQI) guidelines with adjustments to achieve weight gain and linear growth targets, while addressing clinical and laboratory issues. The majority of younger children fail to meet the energy and protein requirements by oral feeding alone and need supplementation by nasogastric tube or gastrostomy.

Initial recommended caloric intake is the age-related dietary reference intake (DRI) but is adjusted according to weight gain. Protein intake recommendation is 100% to 120% of the age-related DRI, with somewhat higher intake in peritoneal dialysis (PD) patients to compensate for peritoneal losses.²⁵ In contrast to adults, a Cochrane Review found that protein restriction has no impact on delaying progression of renal

failure.^{26,27} Another study raised concern about detrimental influence of protein restriction on linear growth.²⁸

Dietary restrictions may include fluid, sodium, phosphorous, and potassium, and special infant formulas are adapted to the needs of patients with CKD. Human milk also has the adequate electrolyte composition for the infant with CKD, as it is relatively low in potassium and phosphorus. However, because infants' diet is fluid based, it may ultimately cause volume overload. This requires in some cases concentrating the formula to achieve a higher caloric intake in a smaller volume (80–120 Kcal/100 mL vs. the usual 67 Kcal/100 mL). It may also influence the dialytic modality selected or the frequency of the dialysis sessions. Conversely, in several renal diseases (e.g., dysplastic kidneys, nephronophthisis, and cystinosis) patients may be polyuric due to a renal concentrating defect even when they reach ESKD. This makes water handling and nutrition easier.

LINEAR GROWTH

Growth is severely impaired in children with ESKD. Almost half of children with ESKD reach a final adult height below the third percentile if growth hormone (GH) is not prescribed.²⁹ A longer period with ESKD and short stature at disease onset are associated with decreased final height. Short stature affects body image and may have an impact on self-esteem. Patients, including young adults, are often treated as younger than their chronological age. In the context of chronic illness this leads to overprotection, low expectations, and delayed independence, and may subsequently affect future socioeconomic status and formation of intimate relationships.

When assessing patient height, it should be remembered that there are differences between races and in growth curves, with normal values available only for some (developed) countries. In countries with nonhomogenous population, constructing growth charts for the whole population creates inaccuracies that may lead to inappropriate clinical decisions.³⁰

Many factors contribute to impaired linear growth:

- Low energy and protein intake: this is especially significant in early childhood.
- Impaired GH-IGF-1 (insulin-like growth factor 1) axis.
- Additional factors include water and electrolyte imbalance, anemia, mineral bone disease, and metabolic acidosis.³¹

Correcting these abnormalities may improve the likelihood of achieving normal growth.

Beyond correction of the metabolic abnormalities and providing the best possible nutritional support, treatment includes recombinant human GH (rhGH) therapy.

The use of GH in kidney failure is based on experimental and clinical evidence demonstrating that GH insensitivity can be overcome by supraphysiologic doses of exogenous GH.³² This treatment was proved to be effective in a study based on NAPRTCS data: of 5122 dialysis patients, 33% received rhGH treatment. Catch-up growth was achieved in 11% of the treated dialysis patients (compared with 27% of patients in earlier stages of CKD). The mean increase was 0.5 standard deviation score (0.8 standard deviation score in earlier CKD stages).³³ In addition, Cochrane database meta-analysis found increased height velocity (+3.9 cm/year after the first year), without advance in bone age.³⁴ In the

NAPRTCS 2011 report children younger than 6 years of age had improvement in Z score of +0.63 compared with −0.1 in the untreated patients. Older children had milder improvement of +0.26 Z scores.³⁵

ELECTROLYTE IMBALANCE

Hyperkalemia is commonly found in children with ESRD, due to decreased urinary excretion. Treatment includes dietary potassium restriction and the use of sodium polystyrene sulfonate. Formula adapted for infants with CKD, low in potassium, phosphorous, and calcium, is commercially available. Further reduction in potassium content of formula-based diet can be achieved by pretreatment of the food by sodium polystyrene sulfonate.³⁶ This method is associated with increased sodium intake, which may lead to hypervolemia and HTN.

Patiomer, a potassium-binding polymer, has been studied in adults with CKD.³⁷ Pediatric data on the drug are not presently available, although studies are ongoing in children.

METABOLIC ACIDOSIS

Chronic metabolic acidosis is treated in children because of its impact on bones' linear growth and structure. The effect of acidosis on bone health and growth is multifactorial, including increased osteoclastic and decreased osteoblastic activity; blunted action of the GH-IGF-1 axis on bones; decreased 1,25-(OH)₂ vitamin D production, and hence less intestinal calcium absorption; and alteration of homeostatic relationships between vitamin D, parathyroid hormone (PTH), and ionized calcium. Metabolic acidosis is treated with bicarbonate salts with target bicarbonate serum concentrations of 20 to 22 mEq/L.

ANEMIA

Hemoglobin starts to decline in children when estimated GFR (eGFR) is lower than approximately 50 mL/min/1.73 m².³⁸ Diagnosis of iron deficiency may be challenging, because many patients with ESRD have low transferrin levels, due to malnutrition or chronic inflammation, causing the calculated transferrin saturation to be seemingly normal. Ferritin concentrations, by contrast, may be high as ferritin is an acute-phase reactant and increases in inflammatory states. Additional data including mean corpuscular volume, red blood cell distribution width, reticulocyte count, and a blood smear may assist in diagnosis. Other assays such as hepcidin and soluble transferrin receptor are not in routine clinical use. The physiologic hemoglobin concentrations vary with age and gender as do the recommended target levels: above the 5th percentile of the specific age. Treatment includes iron supplementation and erythropoiesis-stimulating agents (ESAs). Target iron levels are higher than in the healthy population: transferrin saturation greater than 20% and ferritin greater than 100 ng/mL (to optimize the effect of ESA). The relative ESA dose (units/kg/week) is higher in children and may even reach 1250 erythropoietin units/kg/week in infants.³⁹ In very young children undergoing chronic hemodialysis (HD) there is an obligatory blood loss in the dialysis tubing, which is comparable (in absolute volume units) with adults, but when calculated per kilogram body weight, it is much higher and may reach a total volume almost equivalent to three units of

blood (27 mL/kg) per month.³⁹ Other causes for failure of ESAs include osteitis fibrosa cystica, chronic inflammation, malnutrition, hemolysis, and rarely carnitine deficiency or aluminum toxicity. Studies in adults on dialysis found that excessive correction of anemia (even within the physiologic hemoglobin level) is associated with increased mortality; therefore it is recommended not to exceed hemoglobin levels of 13 g/dL. The pathophysiology of this phenomenon is unclear and has not been validated in children.

CHRONIC KIDNEY DISEASE-MINERAL BONE DISEASE

Although the pathophysiology of CKD-mineral and bone disorder (MBD) and the pathological subtypes in children are similar to those in adults, both the management and the consequences differ, as linear growth is an important factor in children. In addition, not all medications used in treating CKD-MBD are approved for children. Monitoring of various CKD-MBD-associated parameters includes measurement of serum calcium, phosphorous, and PTH levels every 1 to 3 months, vitamin D every 3 to 6 months, and bone radiography annually, although no single laboratory value is proven as a reliable marker of bone disease type and severity. As normal blood calcium and phosphorous concentrations and DRI are age dependent, the target levels and nutritional recommendations vary also with age. Age-adjusted hypophosphatemia should be avoided as it can result in hypophosphatemic rickets, often seen in preterm infants with insufficient phosphorous intake or in children with phosphaturia due to proximal tubulopathy. In CKD stage 5 the KDOQI 2017 guidelines recommend phosphorous restriction depending on blood levels.⁴⁰ However, there is no evidence that the restriction is effective outside the research setting. It may also be that a modest increase in PTH concentration is an appropriate adaptive response that contributes to phosphaturia in CKD and may therefore be beneficial for maintenance of normal serum phosphate levels as GFR declines. Phosphorous restriction may not only be ineffective, but also it is a goal very difficult to achieve: there is great variability in phosphate absorption in the general population, phosphate content is not stated on processed food labels, and low-phosphate food (raw, organic, unprocessed) is more expensive. Balanced phosphorous is expected to result in lower PTH and increased 1,25-dihydroxyvitamin D (calcitriol) levels, as well as better bone morphology and reduced risk for CKD-associated vasculopathy without linear growth restriction.^{40–43} The choice of phosphate binders is limited in children. The first line remains calcium-based phosphate binders. Sevelamer HCL was associated with worsening of metabolic acidosis, which may impair linear growth.⁴⁴ The newer sevelamer carbonate addresses the issue of acidosis but is US Food and Drug Administration (FDA) approved for use in children only above 6 years of age (not tested at lower age).^{45,46} The use of lanthanum carbonate is not recommended in children because lanthanum deposits in bones, including growth plates, and the consequences on the developing bone are yet unknown.

Vitamin D monitoring and supplementation are important in patients at all CKD stages. The use of activated vitamin D analogs, such as calcitriol, in adults is limited to CKD stage 5D, but in children their use may be considered to maintain serum calcium levels in the normal range in CKD stages 2

to 5D.⁴⁰ The ESPN CKD-MBD work group published somewhat different guidelines, in which the treatment with activated vitamin D analogs should be started if hyperparathyroidism is persistent despite therapy with vitamin D, aimed at normalizing serum calcium concentration and decreasing PTH (although the target PTH values at each CKD stage are not widely agreed upon^{47,48}). There is no advantage to any analog over the other.⁴⁹ Finally, calcimimetics, which increase the sensitivity of the parathyroid calcium receptors, are not yet approved for use in children.

Deformities of the growing bone and impaired linear growth are complications that are typical in the pediatric age group, in addition to vascular calcifications (VCs) that are also well described in the adult patient population. As abnormal linear growth is due to a number of factors whose relative contribution remains to be determined, the consensus is to treat whatever is treatable. Of note, it has been shown that correcting active vitamin D metabolite deficiency alone can increase growth rate.³¹ This was possibly achieved due to maintaining PTH levels at near-normal values.⁵⁰ Skeletal deformities in CKD, similar to the phenotype of vitamin D-deficient rickets, may be very debilitating but are amenable to medical therapy with marked improvement and even complete resolution (Fig. 74.1).

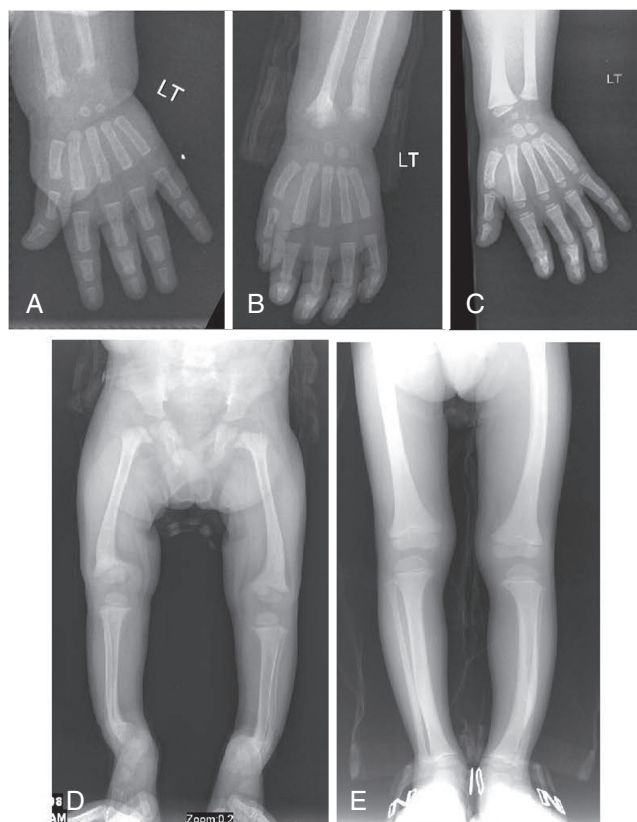


Fig. 74.1 Severe renal osteodystrophy in a boy with end-stage renal disease with resolution obtained with medical treatment: (A) Age 1 year: severe rickets, absorption of the metaphyseal edges of the radius and ulna. Delayed bone age (6 months). (B) Age 2 years: deformation in the radius and ulna bones, widening of the metaphyseal edges, healing rickets. (C) Age 4 years: normal bone structure. Bone age is 3½ years. (D) Age 2 years: deformations with angulations of the femur, tibia, and fibula bones – healing phase. (E) Age 4 years: marked improvement, although some angulation can still be seen.

CARDIOVASCULAR

Cardiovascular system (CVS) disease is common in children with ESRD. Because clinical entities such as ischemic heart disease and valvular heart disease are rare in children, there is an opportunity to better define the development and progression of CKD-related CVS disease. There are some distinct although interrelated clinical features.

HYPERTENSION

HTN, defined as blood pressure (BP; systolic or diastolic) above the 95th percentile for age, gender, and height percentile or treated HTN, is found in 51% to 79% of ESRD patients.^{51–54} In one of these studies 21% of the hypertensive children were not treated at all, and 74% of the treated patients had uncontrolled HTN. It was more frequent in HD patients compared with PD or transplanted children (63.8%, 54.6%, and 26.6%, respectively).⁵³ Other risk factors for HTN were age under 3 years, short time on RRT, acquired (as compared with congenital) kidney disease, and very low body mass index (BMI). Younger hypertensive patients are more likely to be untreated. This may be due to underestimation of the measured BP as HTN: for example, in a 2-year-old, average-height girl the 95th percentile of BP is as low as 104/59 mm Hg. Other explanations are that this group of patients tends to be hypervolemic due to a predominantly liquid diet and, in addition, correct assessment of dry weight is difficult and constantly changing with the patients' growth. Short time on RRT as a risk factor for HTN may be due to a patient not yet accustomed to fluid intake limitation or too short follow-up for adequate BP control. Patients with congenital kidney diseases (mostly dysplastic kidneys) tend to be polyuric or salt losing, and thus normotensive. The 90th percentile of BP measurements should be used as the definition of HTN as well as the treatment goal in high-risk patients, including children with CKD.⁵⁵ Using this lower cutoff, more children with CKD are defined as hypertensive and may therefore be undertreated. In conclusion, HTN, with its extremely detrimental effects (as detailed later), is frequently overlooked, undertreated, or even untreated in pediatric patients with ESRD.

VASCULAR CALCIFICATION

Milliner et al. first noted the high prevalence of VC as part of systemic calcinosis in autopsies of pediatric patients with uremia, on dialysis, or after renal transplantation: 43 of 120 patients (36%) had systemic calcinosis; 30 of them (83%) had VC. A positive correlation was found with the use of vitamin D and peak calcium–phosphorous product (but also age and male gender), which are now well-known risk factors.⁵⁶

In vivo, VC can be measured anatomically by duplex studies measuring carotid intima media thickness (cIMT) or by electron beam computed tomography enabling quantification of VC. Functional studies are also available and measure arterial stiffness by various modes.⁵⁷

As opposed to adults, where two different vascular calcifying processes coexist (intimal-atherosclerotic plaque and medial), in children with CKD medial calcification is the main pathology. VC is associated with hyperphosphatemia, high calcium $\times$ phosphorous product, incremental dose of vitamin D and calcium (as phosphate binder), as well as reduced calcification inhibitors such as fetuin A, all leading to calcification and

concomitant osteoblastic differentiation of vascular smooth muscle cells. There is increased vascular stiffness rather than narrowing of the arterial lumen despite the mineral deposition. These may in turn cause HTN, increased pulse pressure, and later left ventricular hypertrophy (LVH) and relative cardiac ischemia. cIMT has also been found to be thickened in children with ESRD, though not in all studies.

The primary process, endothelial dysfunction, is found in early stages of CKD in children and worsens as disease progresses. Subsequently, medial VC appears and worsens, both quantitatively and functionally, with the increase in CKD stage.^{41,42,58,59,59a} In young adults with ESRD since childhood, increased arterial stiffness is found.⁶⁰

Possible strategies to prevent VC include lowering calcium $\times$ phosphorous product, reduction of calcium-containing phosphate binder dose, lower calcitriol intake, reducing PTH levels, optimal BP control, and shortening of time on dialysis. This may be achieved by reducing dietary phosphate intake, using noncalcium-containing phosphate binders, rarely parathyroidectomy, intense antihypertensive therapy, intensified dialysis, and early transplantation.⁶¹ The following two studies, which explored the effects of various interventions, are of special interest: Shroff et al., in an observational rather than an interventional study, showed that patients with time-integrated PTH levels of less than twice the upper normal limit had normal vessels, measured by cIMT and stiffness, compared with those with higher PTH levels.⁵⁸ Hoppe et al. demonstrated that intensified dialysis improved all variables associated with VC, in addition to improved BP control and, surprisingly, better quality of life and school attendance.⁶² Anatomic and functional VC measures were not tested.

Classic atherosclerosis has an accelerated course in adults with ESRD and this disease process starts even among the healthy population in childhood. The only prevalent modifiable risk factor for atherosclerosis is HTN. The dyslipidemia observed in children with ESRD is not the typical atherosclerosis risk profile, and is characterized by high triglycerides, low high-density lipoprotein (HDL) but usually normal total and low-density lipoprotein (LDL) cholesterol in CKD patients, as opposed to low HDL and high total and LDL cholesterol.⁶³ Dyslipidemia of CKD also does not correlate with current CVS disease; therefore it is not clear whether lipid-lowering therapy is appropriate.^{60,64} Diabetes, smoking, and obesity are all rare before transplantation in children.

CARDIAC PATHOLOGY

Left Ventricular Hypertrophy

LVH is found in 30.4% to 63% of pediatric ESRD patients.^{54,63,65} It is correlated (in different studies) with systolic BP, anemia, gender, and PTH levels. The pathophysiology is complicated and includes HTN, volume overload, anemia, and the uremic milieu per se. Important components of the uremic milieu are the endogenous cardiotoxic steroids, digitalis-like substances whose secretion is augmented in CKD patients as an adaptation to the uremic electrolyte–water imbalance.⁶⁶ Another pathogenic factor that has been shown to induce LVH is elevated serum fibroblast growth factor-23 (FGF-23) levels.²¹⁶ LVH in itself may lead to relative ischemia and diastolic dysfunction. In adults, LVH is considered a significant risk factor for mortality, although this has not been directly proven in children.

Diastolic Dysfunction

Diastolic dysfunction prevalence may be as high as 22.5% to 43.4% in ESRD patients.^{54,59} It is associated with left ventricular mass index (left ventricular mass corrected for height to power of 2.7), calcium–phosphorus product, anemia, HTN, and PTH levels.^{63,65,67} The pathophysiology may be poor compliance of the hypertrophied left ventricle, myocardial fibrosis, and myocardial ischemia.

Arrhythmia

The common arrhythmias seen in children with CKD are ventricular and supraventricular extrasystoles, first-degree arteriovenous (AV) block, or sinus tachycardia and transiently prolonged QTc.^{68–70} However, the incidence of these arrhythmias is not higher than in the general pediatric population.⁶⁹ Life-threatening arrhythmia is rare, and is usually associated with severe electrolyte abnormalities.

Valvular Disease

Valvular disease has a reported incidence of 6% to 15%, depending on age and race.⁶⁹ The specific pathology may be not structural but secondary to overload (mitral regurgitation) or to pulmonary hypertension (PHT; tricuspid regurgitation), which are frequent findings in HD patients.⁷¹ Aortic valve calcification, without hemodynamic significance, was found in 12% to 25% of patients in a single report.⁶³

Overall, the burden of cardiovascular disease in children with ESKD may have been overestimated. According to the USRDS 2017 annual report, the 1-year cardiovascular hospitalization rate declined by approximately 30% between the periods of 2005 to 2009 and 2010 to 2014.³ This is unlikely to be due only to improvement in treatment, but rather changes in semantic definitions (e.g., is HTN a cardiovascular problem?).

NEURODEVELOPMENT

There are a number of reasons for neurodevelopmental compromise in children with ESRD. Some pediatric kidney diseases are a part of a syndrome or systemic disease which may have an effect on the central nervous system and on psychomotor development (e.g., syndromes such as Bardet-Biedl, Joubert, trisomy 21, Galloway Mowat, Wilms tumor, aniridia, genitourinary anomalies, and intellectual disability (WAGR), and more). Chromosomal microarrays to detect genomic imbalances were performed in children enrolled in the Chronic Kidney Disease in Children study. Genomic imbalances were detected in 7.4% of children with CKD. In the vast majority of these cases, the genomic lesion was unsuspected based on clinical assessment.⁷² Other pediatric kidney diseases can be associated with prematurity (Finnish-type nephrotic syndrome) or with oligohydramnios and hypoplastic lungs, and hence possible perinatal hypoxia (autosomal recessive polycystic kidney disease). Uremic toxins, HTN (or BP fluctuations during HD), and chronic anemia can cause further damage to the developing brain. Motor skills development can also be delayed because the sick infant is not free to move and play while connected to the dialysis or feeding machine. Various catheters including feeding tubes and gastrostomy may interfere with rolling over and crawling and there is less sensory stimulation during hospitalization or dialysis. Muscle tone is diminished and renal osteodystrophy

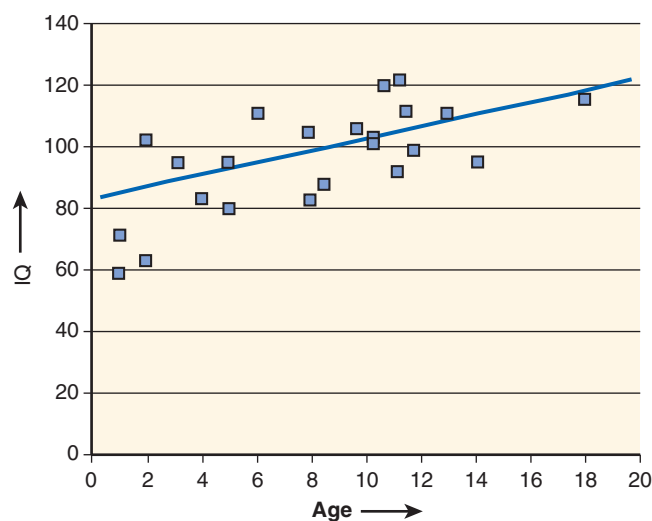


Fig. 74.2 Distribution of intelligence quotient (IQ) scores by age at onset of renal failure (creatinine > 2.5 mg/dL). The earlier the onset of chronic renal failure the lower the IQ score: $R = 0.49$; $P = 0.029$. (Modified from Lawry KW, Brouhard BH, Cunningham RJ. Cognitive functioning and school performance in children with renal failure. *Pediatr Nephrol.* 1994;8:326–329.)

may be painful and prevent the needed physical activity for normal development. Some medications frequently given to CKD patients, including aminoglycoside antibiotics (limited use, only in anuric patients) and furosemide, are potentially ototoxic. Several studies over the last 30 years showed that patients with CKD have compromised cognitive functions. This can be seen even in the early stages of CKD.⁷³ Specifically, studies found lower intelligence quotient (IQ) levels (~10 points), verbal or written language difficulties, impaired memory, decreased executive functions, and inefficiency in key neurocognitive domains.^{74–78} Particularly affected are children who reached ESKD at a younger age (Fig. 74.2),^{75,79} children with longer dialysis vintage,^{76,80,81} more severe HTN or hypertensive crisis events,^{80,82} as well as children with comorbidities.⁸³ Sensorineural hearing loss was found in 14% to 18% and ischemic lesions of variable severity on magnetic resonance imaging (MRI) were seen in 18% to 33% of the patients.^{80,83} Despite this, 61% to 79% of the children with ESKD attend regular school and have academic achievement comparable with their siblings.^{76,77,80} Long-term follow-up in adults treated since childhood for ESKD found impaired schooling and lower IQ (9.2–10.4 points less than controls).⁸⁴ Current ESKD treatment modality (dialysis or transplantation) was not associated with decreased IQ score. However, dialysis for more than 4 years was associated with 3.4 higher risk for decreased IQ. The authors conclude that it is not clear whether the lower IQ and achievements are the result of lower potential or lack of school attendance.

QUALITY OF LIFE

Quality of life of children on RRT was assessed in only a few studies.⁸⁵ Quality of life scores are lower than the general population norms in all domains. Generally, patients perceive their quality of life as better than their parents do.⁸⁶ This may be due to the fact that patients are unaware of an

alternative reality, developing a defense mechanism, or reflecting parental overprotection. The disadvantage is that parents' lower rating may decrease the patients' or their parents' motivation, and thus may adversely affect their outcomes.

MORTALITY

Mortality rate in pediatric patients with ESRD is 30 to 74 times higher than healthy age-adjusted population,^{84,87} with overall survival (in Australia) of 79% at 10 years and 66% at 20 years.⁸⁸ Factors associated with higher death rates are earlier era, younger age at initiation of RRT, and mode of treatment (dialysis mortality rate higher compared with transplantation). These factors were repeatedly confirmed in other studies.^{3,4,24} Most deaths occurred within the first year of life. Similar results were presented in an all-European study and in the USRDS database: overall 5-year survival was 90% for patients beginning RRT during the years 2006 to 2010, with the worst results for patients aged 0 to 4 years (85%) and the best for the 10- to 13-year-old group (96.5%).⁸⁹ Analysis of the USRDS data by type of RRT found 5-year survival of 97% for transplanted patients, 81% for HD patients, and 83% for PD patients (all somewhat better than results achieved in the period 2001–2005). Finally, there are also racial disparities, with Hispanic children having the lowest mortality rates, followed by non-Hispanic white children, and non-Hispanic black children with the highest mortality rates.⁹⁰

The main causes of death were documented as cardiovascular disease and infection, responsible for 45% and 21%, respectively, according to the Australian and New Zealand Dialysis and Transplant Registry (ANZDATA).⁸⁸ Similar findings in the United States showed death from cardiovascular disease of 27.1% to 39.3% and from infection of 9.7% to 22.5%.⁹¹ A decrease over time of death rates from both these causes was demonstrated: CVS mortality decreased in patients younger than 5 years from 36.3 per 1000 person years in 1990 to 1994 to 22.6 in 2006 to 2010 (adjusted hazard ratio [HR] 0.54, confidence interval [CI] 0.47–0.63), and in older patients declined from 16.2 to 9.3 per 1000 person years (HR 0.66, CI 0.61–0.70). According to the USRDS the 1-year adjusted cardiovascular mortality rates were lower and declining over time: 7.5 per 1000 patients in 2010 to 2014, compared with 14 per 1000 patient years in 2005 to 2010 (47% reduction). This was noted in all age groups and for all treatment modalities. A similar trend was found for infection-related mortality: a decline from seven per 1000 patient years in 2005 to 2009 to four in 2010–2014. The decrease in the cardiovascular mortality is (at least in part) due to better definition of the cause of death. A different point of view was presented by Kramer et al., studying young adults who started RRT as children. The 5-year survival from the 18th birthday was 95.1% (CI 93.9–96.0), with an average life expectancy of 63 years for young adults with a functioning graft and 38 years for those remaining on dialysis.⁹²

CARDIOVASCULAR MORTALITY

Cardiovascular mortality is the most frequent reported cause of death: 14% to 41% of all pediatric deaths, depending on

age and race.^{52,86,88,92,93} Although CVS disease has a major impact, it is conceivable that the percentages are an overestimation and in fact, death may not be caused directly or exclusively by a disease of either the heart or blood vessels. All the studies are, by nature, retrospective and rely on physicians' report. The most frequent cause of death is "cardiac arrest" (25%–52%), a very nonspecific diagnosis, which may well be the final common pathway of various primary causes of death.^{86,92} In a single study where the authors directly reviewed all patients' files, the report includes under the heading of CVS death, cerebrovascular accident (58% of cardiac deaths), congestive heart failure (15%), and other diagnoses which may be the consequence of HTN, heparinization, and volume overload.⁸¹ In addition, most of the data available are on children who reached ESKD 2 or 3 decades ago. There is marked decrease not only in the absolute death rate but also in the percentage of deaths attributed to cardiac causes over the years: from 44.4% in 1972 to 1981 to 33.3% in 1992 to 1999. This was also shown in the period of 1990 to 2010: in patients younger than 5 years at initiation of dialysis CVS mortality decreased from 35.3% in 1990 to 1994 to 22.6% in 2005 to 2010⁸⁸ or, according to the USRDS, from 29% in 2005 to 2009 to 15% in 2010 to 2014 (same trend, lower percentages, in other age groups). A study from the Netherlands points to a shift from cardiovascular disease to infections as the main cause of death at long-term follow-up in patients with CKD—all these maybe reflecting a more accurate diagnosis.⁹⁴ However, more specific causes of death must be defined. This is not merely semantic, as accurate determination of causes of death can help focus efforts aimed at decreasing death rate.

Nevertheless, to underscore the high risk for morbidity and mortality, the American Heart Association classified pediatric CKD patients in the same high-risk group as children with homozygous familial hypercholesterolemia, type 1 diabetes mellitus, and postorthotopic heart transplantation.⁵⁵

MEDICATIONS

Any medication theoretically needs to be specifically approved for its use in children, after testing its pharmacokinetics, pharmacodynamics, bioavailability, efficacy, dosing, effects on growth, and development at various ages. Many routinely used drugs have not been through this process, never received such formal approval for use in children, and yet are widely used based on clinical experience accumulated and lack of alternatives. Newer medicines approved for adults are often not yet adequately tested in children, for example, some long-acting ESAs, whereas others have raised concerns over safety, such as calcimimetics. Some drugs used for treatment of patients with CKD are contraindicated in children due to possible side effects specific to children (e.g., lanthanum carbonate, which has been shown to accumulate in bone growth plates).

RENAL REPLACEMENT THERAPY

All available RRT modalities can be used in children. Transplantation is the preferred long-term modality, but it is not always initially possible.

ADVANTAGES OF RENAL TRANSPLANTATION OVER CHRONIC DIALYSIS IN CHILDREN

SURVIVAL

Survival is longer following transplantation in all age groups and in both genders (from 6 months posttransplantation and on): The adjusted relative risk of death in a transplanted child compared with a child who remained on dialysis at 30 months is 0.26 (95% CI 0.11–0.46; $P < .001$; Fig. 74.3).⁹⁵ Similar results are presented in the USRDS 2017 report: in the 2010 to 2014 period the all-cause mortality rate per 1000 patient years is 24, 43, and 9 for children on HD, PD, and postrenal transplantation, respectively (a decline from the 2005–2009 period: respective rates were 51, 43, and 13 per 1000 patient years). The 5-year survival in the same report is 96% for transplanted patients, compared with 83% for PD and 81% for HD patients. In a report from the ANZDATA registry⁸⁸ HR was calculated using HD as the reference value, PD was equivalent to HD, but transplantation had reduced HR of 0.15 to 0.30 (dependent on era tested).

Several risk-associated variables are improved in transplanted children compared with children treated with dialysis: BP values above the 95th percentile were significantly more prevalent in patients on HD compared with transplant recipients (as well as those on PD) with odds ratios of 2.48 and 1.59, respectively.⁵³ In addition, there is slower coronary artery calcification after transplantation as compared with HD patients.⁹⁶ cIMT does not progress and sometimes regresses after kidney transplantation.⁹⁷

QUALITY OF LIFE

Parents of patients who underwent renal transplantation (but less so the children themselves) noted a positive impact of transplantation on almost all the domains of health-related quality of life questionnaire.⁹⁸ Another study found that

patients after renal transplantation and their parents reported better health-related quality of life compared with patients on chronic HD.⁹⁹

NEURODEVELOPMENT

Neurodevelopment is improved following renal transplantation. This was shown as early as 1984 and later in a longitudinal study, showing improved mental processing speed and better sustained attention.^{82,100} Comparative studies demonstrated better language performance and school grades in transplanted patients compared with children on dialysis.⁷⁵ Furthermore, the number of dialysis years correlated with worse neurocognitive outcome.^{76,80,101}

EXHAUSTION OF ACCESS SITES

Because life expectancy is longer than the graft survival in children, the patient may pass more than once through the cycle of dialysis–transplantation. Being exclusively on dialysis as mode of RRT will exhaust access sites (vascular and peritoneal) sooner.

COST

Kidney transplantation is more cost effective than any other RRT modality. In a meta-analysis of articles published between 1968 and 1998, after adjusting to changing price levels, in-center HD was found to be the most expensive modality (steady over the years between \$55,000 and \$80,000 per life year saved [LYS]) compared with kidney transplantation (\$10,000 per LYS, becoming more cost-effective over time).¹⁰² Another study¹⁰³ calculated cost-effectiveness of RRT in Austria: the cost of the first year of treatment was €43,600 for HD, €25,900 for PD, and €51,000 for renal transplantation. In the third year of treatment, medical costs declined to €40,600 for HD, €20,500 for PD, and €12,900 for the transplanted patient. In some developing countries where most people do not have health insurance, dialysis is not a valid option

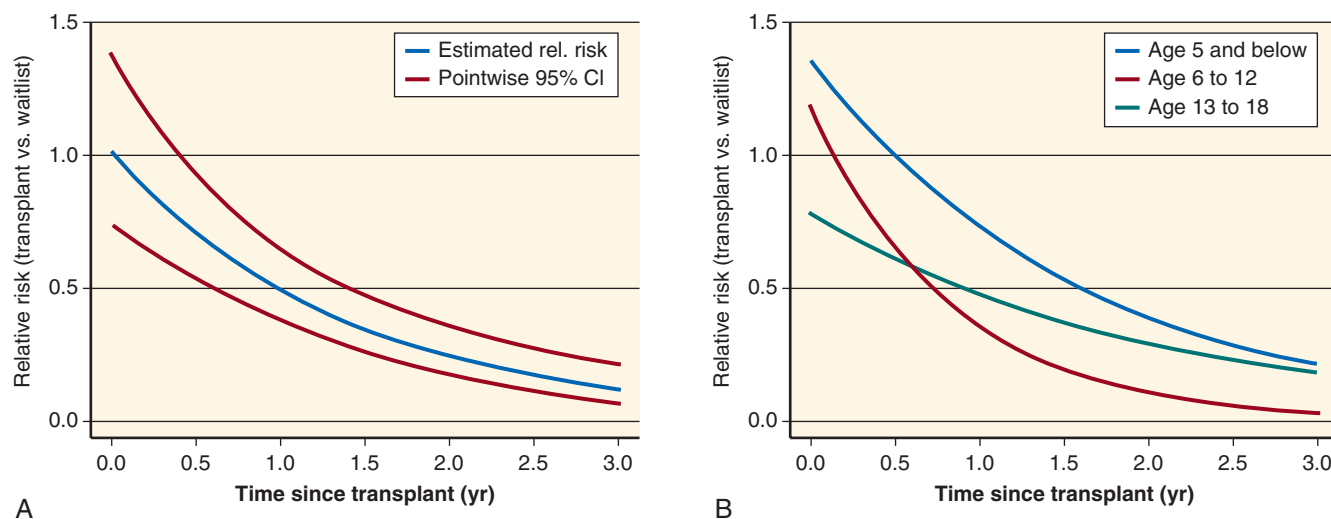


Fig. 74.3 Estimated relative risk of mortality by time since transplantation. (A) The figure is estimated from a model including all ages. (B) Model-based estimates stratified by age. CI, Confidence interval. (Modified from Gillen DL, Stehman-Breen CO, Smith JM, et al. Survival advantage of pediatric recipients of a first kidney transplant among children awaiting kidney transplantation. *Am J Transplant.* 2008;8:2600–2606.)

due to its high price as well as the limited professional services. Transplantation remains the only option, although at times with catastrophic financial ramifications for the families.¹⁰⁴

Usually, a minimal weight of 8.5 to 10 kg is required to achieve good surgical results, necessitating postponement of transplantation in smaller infants. Although the pediatric patient often has potential willing and suitable donors, primarily the parents, this is not always the case. Donor issues such as ABO incompatibility, positive crossmatch, medical conditions or unwillingness to donate a kidney, and sometimes patient size or medical reasons preclude or delay transplantation. For all these reasons in the great majority of children, dialysis is regarded as a necessary but temporary interim solution until transplantation is possible.

COMPARING HEMODIALYSIS WITH PERITONEAL DIALYSIS

If preemptive transplantation is not an option, mode of dialysis (HD or PD) has to be selected. There is a notable absence of comparative studies proving superiority of one method of dialysis over another. Of special interest are infants for whom the widespread recommendation is to treat with PD preferentially and to initiate HD only if PD is not feasible. Nevertheless, in a study based on the ESPN/ERA registry in which 1063 infants were included, 13.7% were initiated on HD rather PD. The median age was 4.5 months (interquartile range [IQR] 0.7–7.9) and the median weight was 5.7 kg (IQR 3.7–7.5). The HD and PD groups had similar characteristics at initiation, and prospective mortality and time to transplantation were the same.¹⁰⁵ Another small study showed an advantage for daily nocturnal HD over PD.¹⁰⁶ The treatment burden on the patients' families is extreme, whichever dialysis mode is chosen. Parents of patients on RRT, in addition to their normal parental roles,

become high level health care providers, and are required to solve problems and seek information, and provide financial resources and practical skills. Caregivers are in continuous need of emotional, psychological, and financial support.⁸⁵ These stresses cause fatigue, as well as disruption of work and social life.¹⁰⁷ Thus once a patient approaches ESKD, the family's abilities and socioeconomic and psychological background must be carefully evaluated by a multidisciplinary team, including nephrologist, dialysis nurse, social worker, and psychologist. PD enables more liberal fluid intake, which is especially important in infants with fluid-based nutrition. In-center HD mandates traveling at least thrice weekly to the dialysis center, which is time-consuming, costly, and not available in some places. PD enables better school attendance than HD: according to the 2011 NAPRTCS report 74% to 85% of PD patients receive full schooling in the first 3 years (the rate is steady from initiation of dialysis to 36 months later) but only 47% to 59% of HD patients attend school regularly (and the rate steadily declines the longer the patient is on HD). Sometimes, PD may be the only choice in locations where there are no available HD centers at all, as occurs often in developing countries, or if the patient lives in remote area where pediatric HD service is lacking, even in developed countries.^{108,109} Rarely, in very-low-birth-weight infants PD again is the only possible option.¹¹⁰ However, with PD the treatment burden is on the caregivers to be compliant and capable of performing dialysis at home, with the additional risk of parental stress and burnout. Taken together, the final decision of HD versus PD is based upon center experience, patient and family choice, socioeconomic issues, compliance, and treatment availability.

The first treatment modality used for children with ESKD in developed countries according to the registries is presented in Table 74.2. Preemptive transplantation is performed in 14% to 24.4% of patients. However, there is variability between centers; in the Norwegian RRT program, for example, 51%

Table 74.2 First Treatment Modality for End-Stage Kidney Disease Pediatric Patients in the Main Registries Worldwide

	ESPN/ERA-EDTA	ANZDATA	UK Renal Registry	NAPRTCS
Year(s) of data collection	2015 (incidence)	2010–2015 (incidence)	2015 (prevalence)	1992–2010 (prevalence)
N	512	327	769	9198 ^a
Age limit, years	<15	<18	<16	<18
Preemptive transplant	96 (18.8%)	64 (19.6%)	23% ^{b,c}	26% ^a
Hemodialysis	213 (41.6%)	108 (33.0%)	33% ^b	46% ^a
Peritoneal dialysis	202 (39.5%)	155 (47.4%)	44% ^b	26% ^a

^aApproximate percentages and total number (N): N calculated from absolute preemptive transplants number plus number of first course of dialysis.

^bData for treatment at 3 months from initiation of renal replacement therapy (RRT).

^cPreemptive transplantation includes both living (15%) and deceased donors (9%). Includes all living transplantations until 3 months from initiation of RRT.

ANZDATA, Australian and New Zealand Dialysis and Transplant Registry; ERA-EDTA, European Renal Association and European Dialysis and Transplant Association; ESKD, end-stage kidney disease; ESPN, European Society of Paediatric Nephrology; NAPRTCS, North American Pediatric Renal Trials and Collaborative Studies.

Modified from ANZDATA Registry. Chapter 11: Paediatric report. In *40th Report: Australia and New Zealand Dialysis and Transplant Registry, Adelaide, Australia*. 2018. Available from http://www.anzdata.org.au/anzdata/AnzdataReport/35thReport/2012c11_paediatric_v1.9.pdf [Last accessed June 14, 2019]; ESPN ERA-EDTA. ESPN/ERA-EDTA Registry. Available from <https://www.espn-reg.org/index.jsp?p=hom> [Last accessed June 14, 2019]; UK Renal Registry. 18th annual report. *Nephron* 2016;132:1–366; North American Pediatric Renal Trials and Collaborative Studies (NAPRTCS). *Annual Transplant Report*. Boston, MA: NAPRTCS; 2014.

of children undergo preemptive transplantations (from living donors) and the median duration of dialysis treatment before first transplantation is 3 months.¹¹² The cause of the disparities between the reports reflects differences in attitudes to kidney donation, traditions and religious beliefs, as well as health care system and financial issues.

The choice of selection of HD versus PD is greatly influenced by patient age: according to the NAPRTCS 2011 report, of 927 patients initiating dialysis at the age of 0 to 1 years 857 (92%) were on PD, compared with 552 of 727 (76%) in the 2- to 5-year age group, 1373 of 2125 patients (65%) in the 6- to 12-year age group, and 1648 of 3250 (51%) in the 13- to 18-year age group who were on this dialysis mode. In Australia and New Zealand similar proportions are seen: 89.8% of infants were initially treated with PD, with declining percentage to 34.5% of adolescents.²⁰ Similar trends were noted in the United Kingdom: at age 0 to 2 years, 80% were on PD and at age 16 to 18 years only 10.4%.

The proportion of patients under the age of 1 year at initiation of dialysis has steadily increased and doubled from 10% to around 20% of all pediatric dialysis patients from 1990 to 2010.¹¹³ This is explained by the fact that more nephrologists were encouraged to treat younger patients as their survival improved. The same trend was found in the UK renal registry: from the 1996 to 2000 period to the 2006 to 2010 period the percentage of patients at age 0 to 2 increased from 13.7% to 17.6% (the absolute figures are from 70–104 per year) while the age group of 12 to 16 decreased by 2.3%.^{4,16}

In the United States there is a steady trend of increased use of HD instead of PD, from 34% of patients treated with HD in 1991 to 54% in 2010. This trend was not found in the UK where a relatively constant percentage, around 30%, of the pediatric dialysis patient population has been on HD since 1996. Similarly, in Australia and New Zealand, 30% to 40% of all dialysis patients were on HD during the period 2006 to 2015.

HEMODIALYSIS

Given the complexity of the treatment with HD in children, a hospital providing this service needs to have a multidisciplinary team with expertise in the various facets of this modality. This includes pediatric nephrologists, dialysis nurses, dietitians, invasive radiologists, vascular surgeons, pediatric surgeons, vascular laboratory, all the subspecialties in pediatrics, as well as social workers, educators, psychologists, and other therapists.

There are several modifications of the HD procedure distinct for children and several specific medical issues that must be addressed in addition to tailoring the dialysis prescription for every child according to size.

VASCULAR ACCESS

The main forms of vascular access are central venous catheter (CVC), AV fistula, or rarely AV graft. AV fistula is the recommended choice because of the lower rate of complications including infection and malfunction, and longer access longevity; median survival time is 3.14 years for AV fistula (95% CI 1.22–5.06) compared with 0.6 year for a CVC (95% CI 0.2–1.00).¹¹⁴ Additional advantages attributed to AV fistulas

including better clearance and higher albumin and hemoglobin levels are less convincing.¹¹⁵ CVCs cause more damage to blood vessels and the resulting central venous stenosis may preclude future CVC insertion or ipsilateral AV fistula creation. This is important in children, who are expected to require RRT for a longer time, often also for a second dialysis period after failure of kidney transplant, and therefore blood vessel preservation is crucial.¹¹⁶ Therefore AV fistula is the preferred vascular access, with the exception of children weighing less than 20 kg, because of technical difficulties of vascular surgery. Although some exceptions exist, AV fistula is rarely created in patients weighing less than 15 kg.¹¹⁷ When HD is planned for a short time, bridging for PD, or awaiting a potential living donor to complete their evaluation, CVC should be used.

In practice, reviewing the pediatric registries demonstrates a different picture. A survey in Europe found that the use of CVCs was more frequent than recommended in prevalent pediatric HD patients: 40% had AV fistula or graft versus 57% who had CVCs.¹¹⁸ Only in children over 15 years of age, there were more patients with AV fistula as an access. In the ANZDATA 2009 registry all children under 9 years had CVC as their prevalent access. Although in adolescents prevalence of AV fistula was higher, there is a decreased trend over time in Australia from 78% in 2005 to only 46% in 2008 and 25% in 2014.¹¹⁹ According to the NAPRTCS 2011 report, the majority of patients on HD use CVCs: 78.7% of all incident pediatric HD patients had a CVC, and a clear increase in this trend was observed, from 73% in 1992 to more than 90% in 2010. However, in many cases this is probably due to the choice of HD using CVC access by patients who are expected to require dialysis for a short time. Time on dialysis is shorter in children with CVCs: at 3 months $20.7\% \pm 1.0\%$ of those with CVCs have discontinued dialysis compared with $7.2\% \pm 1.6\%$ for AV fistula and $7.0\% \pm 2.0\%$ for AV grafts. At 24 months the comparable figures are $71.9\% \pm 1.2\%$, $50.0\% \pm 3.2\%$, and $57.5\% \pm 4.3\%$. In addition, dialysis is terminated sooner in HD compared with PD patients: within 6 months $30.0\% \pm 1.0\%$ of HD patients have discontinued treatment compared with only $18.9\% \pm 0.6\%$ of PD patients.³ Other reasons for increased use of CVC in pediatric HD include lack of venipuncture experience in small HD units, the ease of achieving an immediately functioning CVC with the advancement of invasive radiology, and the preference for painless connection to the dialysis machine. Not all centers have vascular surgeons experienced in construction and maintenance of AV fistulae in children, a factor which is crucial to fistula patency and function, and therefore CVCs may be used more often in such units.¹²⁰ AV fistula needs time to mature before it can be used, usually longer in children than adults.¹²¹ In the majority of cases appropriate planning can overcome this issue.

CVCs should be adjusted to patient size. Catheter diameter of 8 Fr is the smallest tunneled catheter size available, and can be used even in patients as small as 3 kg of weight. This is also the smallest size that can provide long-term adequate flow to ensure sufficient solute and fluid removal, as the flow is proportional to the fourth power of the CVC diameter. The distance between the arterial and venous ends should ideally be as far as possible to reduce recirculation (and urea clearance) but close enough to ensure that both are within the right atrium, preventing sealing of the catheter

ends by the venous wall. However, 8-Fr size CVC is almost the same size as the venous lumen of the small child, increasing the risk of vascular damage and stenosis. Patients weighing more than 18 to 20 kg can accommodate 10-Fr catheters. Ultrasound and fluoroscopic guidance are used to prevent catheter malposition, reducing the primary failure rate to almost zero. Although there are no published data in children, it is recommended to follow the experience in adults using the internal jugular (IJ) veins as the preferred access, as the risk of central venous stenosis is reduced in this site. Anecdotally, in newborns who need short-term HD, as in the case of inborn errors of metabolism, the umbilical vein can be used as a temporary access.

The main CVC complications are infections and malfunction. Infections are more common in younger patients, possibly due to close proximity of the exit site to the infection source: diapers and gastrostomy tubes.^{122,123} Infection rate is usually 1.5 to 4.8/1000 CVC days, although an exceptionally favorable report documented a rate of 0.5/1000 CVC days.^{124–128} Malfunction is the result of thrombosis, fibrin sheath formation, vascular stenosis, or mechanical damage to the CVC. If local fibrinolysis fails, the CVC must be replaced. A major complication with long-term sequelae is central venous stenosis. It occurs more often in younger patients, with small-diameter veins, with prolonged use of CVCs, with many reinsertions and possibly also depending on CVC locations. The worst outcome is for subclavian veins, followed by left IJ vein, whereas the right IJ vein has the smallest chance of becoming obstructed. Clinically, stenosis is often asymptomatic until an ipsilateral AV fistula is constructed, but sometimes superior vena cava syndrome may develop.¹²⁹

AV fistulas, and rarely AV grafts, have a much lower complication rate, resulting in fewer hospitalizations, less revisions, and have higher access longevity.¹³⁰ In a small survey, although with significant age difference between groups, the hospitalization rate for access-related issues was 0.44% versus 3.1% per year in the AV fistula versus the CVC groups.¹³¹ Because fistula thrombosis/malfunction is the main complication, and this is usually the result of vessel stenosis, routine surveillance of the fistula by Doppler ultrasound performed by an experienced professional should be performed. Vascular surgeons and invasive radiologists should be available to detect and treat stenosis or thrombosis.

The average AV fistula blood flow is 800–1200 mL/min. In adults, this is significant compared with the normal cardiac output (CO): normal adult CO is 5.2–8.6 L/min and the flow through an AV fistula will increase CO by 9% to 23%. A similar flow in a child, required for adequate dialysis, may increase the CO of a child with a 0.7-m² body surface area (BSA), whose normal CO is 2.1 to 3.5 L/min, by 23% to 57%. Theoretically, this may lead to high-output heart failure, especially in face of the high cardiac morbidity in dialysis patients; in practice this rarely becomes clinically significant.

PHT is another complication of adults on HD, and the presence of an AV fistula accounts, at least in part, for this process.⁷¹ High pulmonary blood flow, a direct result of the AV fistula, may cause PHT.^{132,133} Other contributing factors include endohormonal imbalances such as high endothelin-1 and low nitric oxide levels. This has not been tested in the pediatric HD population. PHT is partially reversible, after kidney transplantation, but may have significant sequelae including increased mortality.

DIALYSIS APPARATUS

TUBING

Tubing should be adapted to the patient's size. The extracorporeal volume, consisting of the arterial and venous segments and the dialyzer, should not exceed 10% of the patient's blood volume. For patients weighing less than 6 kg this goal cannot be achieved even with the smallest available tubing, and priming of the extracorporeal system is needed, either with whole blood or with packed red blood cells diluted with normal saline to an estimated hematocrit of 33% to 36%. Although blood units can be divided so a single unit can be used for four consecutive HD sessions, there still is a high exposure to blood products. Iron overload may occur despite the fact that at the end of each session the blood in the extracorporeal system is not routinely washed back to the patient, unless the patient requires a transfusion.

Exposing a patient to many units of blood increases the risk of viral infection and of human leukocyte antigen (HLA) sensitization. However, the incidence of sensitization in infants treated with chronic HD necessitating multiple transfusions has not been systematically studied.

In patients weighing more than 6 kg the tubing is primed with either 5% albumin solution (in smaller patients) or physiological saline solution in those weighing more than 10 kg. In those patients, the blood in the tubing is washed back to the patient at the end of each session. This procedure deserves special consideration as the blood volume filling the tubing is equivalent to 0.5 to 1 weight-adjusted blood unit. Even with the slowest possible pump speed the whole volume will be washed back within 2 to 3 minutes, which is exceptionally fast. This may cause an abrupt increase in the right atrial pressure and if the patient has a patent foramen ovale (PFO) it would increase the risk for possible micro-thrombi paradoxical embolism. Despite this concern, as well as the known relative cognitive deficiency in patients on dialysis and the frequent (18%–33%) finding of ischemic lesions on brain MRI studies in pediatric ESRD patients, this point has not been explored.^{80,83} In adult patients there was no evidence of faster cognitive deterioration with PFO on HD compared with those without PFO, although atrial pressures during blood return at the end of dialysis may not be as high in adults.¹³⁴

DIALYZER

The dialyzer should also be adjusted to patient size to ensure adequate clearance with minimal extracorporeal blood volume, and therefore hollow fiber dialyzers are used. The dialyzer size is determined by the patient's BSA.

HEMODIALYSIS MACHINE

The HD machine, to provide dialysis for children, must be compatible with the use of small tubing, be adjustable to low pump speed, and have a volumetric function. The latter is crucial as the ultrafiltrate has to be measured directly, because small inaccuracies in volumes removed that are negligible in adults can have a major hemodynamic effect on the small child.

BLOOD FLOW

Blood flow speed is a major determinant of solute clearance, but excessively high flow can cause cardiovascular instability,

which might manifest as pallor, irritability, vomiting, or altered mental status. Blood flow is determined according to body size, and is usually not higher than 10% of blood volume in milliliter/minute.

DOSE OF DIALYSIS

According to National Kidney Foundation KDOQI 2006 guidelines, dialysis should be initiated once the eGFR falls below 15 mL/min/1.73 m², or earlier, if other metabolic derangements, volume overload, or medical signs and symptoms are refractory to conventional treatment. Practically, there are variations in eGFR at dialysis initiation among nephrologists.¹³⁵

Once dialysis is started, the recommendation states that solute clearance should be greater than that recommended for adults ($Kt/V > 1.2$, urea reduction ratio $> 65\%$), given that it should support higher protein intake, needed for adequate growth and development.¹³⁶ However, there is little observational information and no randomized studies regarding the correct dose of dialysis for children. Hence HD is arbitrarily delivered thrice weekly, for 3 to 4 hours each session. However, the true adequacy test is based on clinical parameters, which include in the short term, optimal balance of electrolytes, BP, and volume, and in the long term, adequate weight gain, linear growth, neurodevelopment, and quality of life. Several small studies suggest that delivering more dialysis can improve BP control, hyperphosphatemia, obviate dietary restriction, and improve appetite and general well-being, linear growth, and even school attendance.^{82,137–142} Intensive dialysis may also attenuate vascular disease, which is a major concern.⁶¹ This goal can be achieved by either nocturnal home HD, performing 5–7 sessions per week, hemodiafiltration, or a combination of both. The theoretical basis for these observations was established in a study by Daugirdas et al.¹³⁶: In the standard calculation, the dose of dialysis is scaled to urea distribution volume (V), which is similar to total body water. If, however, dialysis is scaled to BSA, similar to calculation of GFR, because BSA is relatively higher in children, this would require increased dialysis time for smaller children. When these calculations were applied to two of the studies of intensive dialysis, the patients were indeed found to have a much higher weekly dialysis dose.^{140,142} Despite this, intensive dialysis is still not widely used, and it is premature to recommend a general increase in the dose or frequency of dialysis in all pediatric patients. Randomized studies with clinical endpoints are necessary. Widespread changes in clinical practice would have substantial financial ramifications. Daily or nocturnal home HD is not available in most countries for children.

FLUID REMOVAL

Total body water percentage changes by age: from 80% in a newborn to 65% in a 12-year-old child and to 55% to 60% in adults (female and male, respectively), and moreover the extracellular volume falls from 52% in infancy to only 18% to 20% in adults.¹⁴³ For this reason the allowance of interdialytic fluid gain is larger and also the tolerance of dialysis fluid removal is better in infants (up to 8%–10% of body weight) compared with adolescents (up to 5%). An additional challenge is that the child's weight

constantly changes, so the dry weight has to be reassessed frequently, especially in the very young ages. There is no single reliable marker of correct dry weight, and thus it is estimated as the lowest weight not causing symptomatic hypovolemia.

Infants receiving fluid-based nutrition often need to have HD performed 4 to 5 days weekly in order to enable adequate caloric intake without volume overload. This also provides more clearance, as discussed earlier.

COMPLICATIONS

Complications associated with vascular access for HD are detailed earlier, and include catheter-related infections, malfunction, and in the long term, central venous stenosis. AV fistula complications include fistula malfunction and thrombosis, or rarely high-output heart failure or PHT. In smaller children, there is obligatory exposure to multiple blood products, which may increase the risk of infection or HLA sensitization.

HYPOTENSION

Hypotension is a common complication during dialysis. It is associated with excessive or rapid fluid removal, which often exceeds 5% of body weight, both in younger children on liquid diets and in adolescent patients who are not adherent to their fluid restriction requirements. Hypotension may appear without warning, particularly in infants, and manifest as pallor, irritability, vomiting, or altered mental status. Accurate measurement of ultrafiltration (UF) volume is crucial in small children, as small volumes constitute a relatively large percentage of their blood volume. In patients with repeated hypotensive episodes, sodium profiling, osmotic agents such as mannitol, lowering dialysate temperature, or (off-label) use of the alpha 1 receptor agonist, midodrine, can sometimes be helpful. More frequent dialysis sessions may be necessary for large interdialytic weight gain, to reach dry weight and avoid hypotension during dialysis.¹⁴⁴

DISEQUILIBRIUM SYNDROME

Disequilibrium syndrome is now a rare, yet potentially dangerous complication of HD. It is usually precipitated by overly rapid urea removal causing a discrepancy between the osmolality of the plasma and that of the brain cells, causing fluid shift into the brain.¹⁴⁵ Seizures are more common in children with disequilibrium syndrome than in adults. Prevention is aimed at a gradual reduction of urea, especially in new HD patients and when urea is very high. This is achieved by selection of an appropriately small dialyzer and limiting blood flow and session length for the first few treatments. Mannitol infusion, or in cases of HTN and hypervolemia, high dialysate glucose or slightly increased dialysate sodium concentration may also be helpful, by preventing the rapid decrease in plasma osmolality. Blood flow rate should be slowed if mild symptoms such as nausea, vomiting, or headache appear, and dialysis stopped if more significant neurological manifestations appear.¹⁴⁶

HYPOTHERMIA

Hypothermia may occur if the dialysate is not warmed, particularly in small children, as the dialysate flow is generally constant regardless of patient size.

HYPOPHOSPHATEMIA

Hypophosphatemia may be caused by HD, particularly in younger children, when kidney function is normal (e.g., when HD is done as emergency treatment for inborn errors of metabolism) and when intensive dialysis is performed (for fluid removal purposes or daily dialysis for ESRD due to primary hyperoxaluria). Hypophosphatemia is attributed to excessive phosphate clearance due to high flow of dialysate relative to patient weight.¹⁴⁷ This can be treated either by adding phosphate (e.g., sodium phosphate in enema preparations) to the dialysate concentrate, or by slowing the dialysate flow.

HYPOGLYCEMIA

Hypoglycemia may occur during HD as described in adults.¹⁴⁸ It is associated with the use of glucose-free dialysate and high dialysate flow, leading to excessive glucose clearance. Infants, especially if catabolic, may be at a higher risk for hypoglycemia and plasma glucose should be monitored during dialysis if suspected.

PERITONEAL DIALYSIS

PD is the most common modality of dialysis treatment in children worldwide, especially at young ages.^{3,8,35} The PD prescription can be individualized according to the patient's age, body size, residual renal function (RRF), nutritional intake, and growth-related metabolic needs. As stated earlier, studies comparing outcomes of children treated with HD or PD are scarce. The choice of PD is based on the assumption that PD is better adapted to the lifestyle and diet of the infant or child. Sometimes PD is the default choice due to the lack of experienced HD personnel or equipment. However, family preference and center expertise may direct even newborns to HD. In contrast to HD, PD efficacy is dependent on biologic properties of patient's peritoneal membrane rather than on a synthetic dialyzer. This means that the initial prescription must be modified according to membrane characteristics, which are determined only after initiation of PD. The peritoneum is periodically assessed and prescription modified according to findings, and it may also fail, precluding further PD. Assessing the membrane function is done by peritoneal equilibration test (PET), which is modified according to BSA of the individual as exchange volume may impact test results.

Some contraindications to performing PD are specific to the pediatric patient.¹⁴⁹ These include various abdominal congenital anomalies such as gastroschisis, omphalocele, diaphragmatic hernia or bladder exstrophy, and usually polycystic kidney disease where cysts may literally fill the entire abdominal cavity. In addition, medical issues including abdominal surgery, ventriculoperitoneal shunt, abdominal adhesions, and membrane failure may all preclude the use of PD. Finally, because PD depends on the availability of a compliant and competent caregiver, the lack of appropriate parental support precludes using this technique.

Infants in need of dialysis pose many challenges to the medical team and family. For this reason many nephrologists refrained from treating these patients.^{150,151} However, current USRDS data show that although overall death rate among infants on PD is 20.3%, there is decline in the 2000 to 2014

era compared with the 1990 to 1999 era, with current 5-year patient survival of 74.6% for neonates and 79.3% for older infants.¹⁵² Similar results were found in another study, showing also 92.1% 3-year graft survival for those patients.¹⁵³ As often happens when new treatment modalities or options are introduced, further options are tried and past limits are pushed: PD has been performed in very-low-birth-weight neonates (weight 825 ± 215 g) with acute kidney injury (AKI), with 20% survival.¹¹⁰

ACCESS

A reliable access is critical for successful PD. A double-cuffed Tenckhoff catheter with downward-pointing exit site is associated with a lower infection rate. Specifically in infants or children on PD, the catheter has to be placed as far as possible from sources of infection such as diapers or gastrostomy tubes.¹⁴⁹

PERITONEAL DIALYSIS PRESCRIPTION

TYPES OF PERITONEAL DIALYSIS

All available PD modalities can be used in children. Continuous ambulatory PD (CAPD), which is manual, is usually reserved for special circumstances. However, in many parts of the world, if dialysis is not at all available, this is the only modality used.²¹ Because automated PD (APD) is easier and flexible, this is almost the exclusive modality used in developed countries. Nightly intermittent PD (NIPD), which is APD at night, with no dialysis in the daytime, can be used when there is significant RRF, because fluid removal and solute clearance are limited. The advantages are daytime convenience, lower intraperitoneal pressure (IPP), lower risk of hernia formation, less glucose absorption, and less membrane exposure to glucose. In continuous cycling PD a daytime exchange is added, thus the NIPD advantages are partially abolished, although better fluid and clearance are achieved, making it more suitable for patients with low RRF or with continuous need for efficient fluid removal (e.g., infants with fluid-based diet). Tidal PD is an option recommended when complete drainage either causes pain or is not possible. Only part of the fluid is removed in each cycle. This makes the dialysis cycle less efficient but is partially compensated for by shorter drainage time and thus more cycles.

PERITONEAL DIALYSIS SOLUTIONS

Standard PD solutions contain buffer to control acidosis, electrolytes, an osmotic agent (usually dextrose at varying concentrations) to enable net fluid removal, and water as a carrier. However, long-term membrane exposure to lactate-acidic solution and high dextrose concentration are detrimental, leading to peritoneal fibrosis and reduced function.¹⁵⁴ This is more common in frequent and short cycles as used in APD. For this reason newer solutions have been developed, which have also been tested in children. In addition, there are special considerations in children at varying ages. Recommendations by the European Pediatric Dialysis Working Group include the use of the lowest glucose concentration possible, and fluids with reduced glucose degradation products (GDPs) content whenever possible.¹⁵⁴ In face of a paucity of studies prospectively comparing low GDP solutions or various buffers, evidence-based recommendations cannot yet be given, but

it seems that correction of metabolic acidosis with pH-neutral bicarbonate-based fluids is superior to single-chamber, acidic, lactate-based solutions. Icodextrin (a high-molecular-weight glucose polymer, with high osmotic power) may be applied once daily in a long dwell, in particular in children with insufficient UF. In addition, sodium balance needs to be closely monitored as infants on PD are at risk for UF-associated sodium depletion, while anuric adolescents may have salt and water overload. Dialysate calcium should be adapted according to individual needs as the child grows to maintain positive calcium balance.¹⁴⁹ The use of amino acid-based PD fluids (designed to compensate for excessive protein loss during PD) in children is well tolerated but has little anabolic effect.

EXCHANGE VOLUME

The volume of dialysate infused is adjusted to patient size, 1000 to 1200 mL/m² for patients 2 years of age or older, about two thirds of that volume for younger patients.¹⁴⁹ As in adults, volumes that are too small can lead to rapid solute equilibration and thus inadequate UF, and too high a volume may result in increased IPP, with increased lymphatic absorption (causing reduced UF), physical discomfort, dyspnea, hernia formation, and gastroesophageal reflux.

DWELL TIME

The length of each exchange depends on PD modality; it is short in APD compared with CAPD. Membrane properties, as detected by PET, guide the subsequent adjustment of the initial prescription, shortening dwell time in patients with higher solute transport, to prevent loss of osmotic gradient due to solute equilibration and reduced UF.

ADEQUACY OF DIALYSIS

No large-scale prospective studies have been performed in children that assess the correlation between solute removal and clinical outcomes. This is the result of the rarity of ESKD in children and the relatively shorter time on dialysis in children compared with adult patients. Furthermore, longer life expectancy in pediatric RRT precludes the assessment of the effect of dialysis prescription changes on survival. In areas where there are no pediatric data, adult clinical practice guidelines serve as a minimal standard. Clinical and laboratory assessment should be done at least once a month (more often if clinically necessary). Kt/V should be measured 1 month after initiating PD and subsequently at least every 6 months, and RRF is assessed by 24-hour urine collection every 3 months. Adequacy is evaluated clinically, seeking signs and symptoms of uremia: HTN, pulmonary congestion, pericarditis, hyperkalemia, hyperphosphatemia, and worsening school performance and general well-being. Additional laboratory data may indicate inadequacy of dialysis (serum solute concentrations, and hemoglobin and albumin levels), and finally, calculating Kt/V. Target Kt/V is 1.8 or greater per week, while V, or total body water, should be calculated using age- and gender-specific nomograms.¹⁴⁹ Prescription should be modified if needed, taking into account also preservation of RRF, patient convenience (IPP, fitting with the child's daily routine), and lowest possible dialysate dextrose concentration to achieve sufficient fluid removal.¹⁵⁴

COMPLICATIONS

INFECTION

Peritonitis is a major complication in patients on PD, which may result in long-term membrane failure. The reported rate by NAPRTCS in 2011 is one episode per 18.8 PD months, higher in infants (1/15.3 months) compared with adolescents (1/21.2 per months), probably due to close proximity to contamination sources such as diapers or gastrostomy tubes in the young ages. Similar results were found in Australia (1/16.9 months), but with no correlation with age.¹⁵⁵ A report of the Italian Registry of Paediatric Chronic Dialysis found somewhat better results; 1/20.7 months in infants and 1/28.3 in older children.¹⁵⁶ In a survey including 47 centers in 14 countries 44% of the peritonitis events were due to gram-positive bacteria, 25% due to gram-negative bacteria, and 31% were culture negative.¹⁵⁷ Less than half of the patients had fever higher than 38°C, less than half had abdominal pain, and 70% had marked effluent cloudiness. Of 482 children, 420 (89%) made a full recovery, and 39 (8.1%) had to discontinue PD permanently, due to UF problems, adhesions, uncontrolled infection, and secondary fungal infection. As such, infections are the most common reason for modality change for patients on PD. In another study, including 501 peritonitis events from 44 pediatric dialysis centers in 14 countries, significant regional variability was found for both bacteria type and sensitivity.¹⁵⁸ This makes recommendation for global treatment guidelines more complex. Permanent discontinuation of PD varied considerably, from 0/18 episodes (Argentina) to 9/46 episodes (20%) in Eastern Europe. Of note, a recent Cochrane study assessing evidence in PD-associated peritonitis (including adults) concluded that the data available are poor. In general, review conclusions were based on a small number of studies with few events, in which risk of bias was generally high; interventions were heterogeneous, and outcome definitions were often inconsistent. There were no randomized controlled trials evaluating optimal timing of catheter removal and data for APD were absent.¹⁵⁹

Other infections include exit site and tunnel infections. Preventive measures include aseptic handling of the catheter, daily cleansing, immobilization, and applying topical antibiotics and intranasal mupirocin application. Interestingly, an alternative measure of preventing exit site infection with antibacterial honey found noninferior results compared with intranasal mupirocin.¹⁶⁰

TECHNIQUE FAILURE

Catheter Malfunction

Catheter malfunction due to its migration or occlusion may preclude effective dialysis. Technical solutions such as the use of specific types of catheters and omentectomy at the time of catheter insertion may reduce the incidence.

Ultrafiltration Failure

UF failure is a main cause of technique failure and leads to volume overload. High solute transport, as detected by PET, causes rapid dissipation of the osmotic gradient needed for UF. Peritonitis events and long-term exposure to GDPs in the standard PD solutions are the main causes. Filling volume that is too high can, in addition, lead to UF failure due to

lymphatic absorption, whereas low exchange volumes may falsely seem as peritoneal failure, because with low volumes there is rapid solute equilibration.

Fluid Leaks

Fluid leaks secondary to high IPP are more common in infants, often manifesting as hernias. Recurrent hydrothorax is due to pleura-peritoneal connections and the negative pressure produced on inhalation. Pericatheter exit site leak is usually an early event after insertion, and is usually managed by postponing start of PD or using smaller volumes initially. Leak into the abdominal wall causes subcutaneous edema.

ENCAPSULATING PERITONEAL SCLEROSIS

Encapsulating peritoneal sclerosis (EPS) is a rare complication with high fatality rate, characterized by extensive intraperitoneal fibrosis that causes UF failure occasionally with symptoms of bowel obstruction. It was found in 14 of 712 (1.9%) pediatric PD patients registered in the Italian registry of chronic dialysis.¹⁶¹ Eleven of the patients were treated with PD for longer than 5 years. Because the disease is insidious and progresses slowly, five (36%) were diagnosed when no longer on dialysis. In 71% of the patients, high dextrose solutions were used over the 6 months prior to diagnosis. Peritonitis rate was not higher than in patients without this complication. A second study found the same rate of EPS, but a higher infection rate relative to those without this complication, and no correlation with dialysate type.¹⁶²

ADHERENCE TO TREATMENT

Patient compliance was evaluated in 51 patients in a single-center study, using automatic PD device memory cards.¹⁶³ Variables tested were number of sessions per month, duration of each session, number of cycles per session, and volume of PD solution instilled. Only 55% were adherent in all four variables; males and African-American patients were more likely nonadherent. Nonadherent patients tended to skip whole sessions or instill reduced volumes more than shortening sessions or reducing the number of cycles. Apart from being aware of the gap between prescribed and provided PD treatment, this can shed light on the challenge of being compliant with complex treatment, the aspiration for some “treatment holidays,” and the discomfort of large-volume instillation.

WORLD DISPARITIES

Gross national income (GNI) has major impact not only on the availability but also on practices of PD. In a study of the International Pediatric Peritoneal Dialysis Network Registry close associations were found between GNI and percentage of infants on dialysis, lack of diagnosis of the primary disease, prevalence of APD, and more advanced techniques such as biocompatible solutions, enteral feeding, calcium-free phosphate binders, ESAs, and vitamin D analogs.¹⁶⁴ This led to higher death rate from infectious complication (although peritonitis incidence was not higher), lower hemoglobin and calcium levels, higher PTH, lower height percentile, and overall higher mortality.

CHANGE OF DIALYSIS MODALITY

The main reasons for termination of PD are transplantation, transition to HD, death, and rarely withdrawal of treatment or recovery of renal function.

The NAPRTCS 2011 report provides information about termination of 4687 PD courses in the registry (excluding deaths): 19.4% still had a working access, 53.7% were transplanted, and 16.0% changed modality. The main reason for modality change was excessive infections (43%), 8.8% did so because of family choice, 8.2% due to access failure, and 26.7% due to other medical issues.³⁵ It is not clear what the percentage of UF failure was. It is interesting to note that in HD patients, reasons for modality change were mainly family choice (43.0%) and only rarely excessive infections (7.0%). As PD is much more demanding for caregivers, one would expect the opposite ratio, reflecting caregiver burnout.³⁵ The answer to this may be the following: half of HD patients who change modality do so within 3 months and two thirds within 6 months, over a total 36 months' observation period. PD patients, by contrast, change modality at a steady pace over the period. This implies that most modality changes from HD to PD are actually planned in advance, and only insufficient time for Tenckhoff catheter insertion and PD learning time mandated starting HD temporarily. In addition, some patients live very far from pediatric HD centers, and therefore have no option of modality change to HD.

There are several other reports from around the world demonstrating transition of pediatric PD patients to HD: In a report from Taiwan¹⁶⁵ eight of 29 (27.5%) changed modality—two due to refractory peritonitis, three inadequate dialysis, and three UF failure. In Turkey, of 476 children treated with CAPD, 142 converted later to APD. Ninety-five percent remained on PD at 1 year and 69% at 5 years,¹⁶⁶ whereas 13.7% changed to HD. In a report from Iran, outcomes improved by era but are still poor²¹: between 1998 and 2001 the annual death rate was 60%, 13% switched to HD, 11% were transplanted, and only 4% stayed on PD. In the next era (2002–2006) the death rate decreased to 23%, 15% switched to HD, 10% were transplanted, and 50% were still on PD.

RENAL REPLACEMENT THERAPY FOR CHILDREN WITH INBORN ERRORS OF METABOLISM

Some inborn errors of metabolism manifest in early infancy and may lead to death or permanent, severe neurological damage due to accumulation of neurotoxic abnormal metabolites. Long-term outcome depends upon quick diagnosis and removal of the toxins by diet (reducing their synthesis) or medications (redirecting the toxins to alternative metabolic pathways). These modalities are frequently insufficient in the short term and necessitate more aggressive treatment to reduce metabolites to a nonhazardous range, especially in neonates. All RRT modalities have been used for metabolite clearance in various diseases, such as urea cycle defects, maple syrup urine disease, and methylmalonic acidemia. In the past, acute PD was the only technological solution for clearance of toxic metabolites, but HD, continuous venovenous HD, and continuous venovenous hemofiltration

(CVVH) are now possible even in neonates and have been found to be superior in clearance rate.^{167–170} Comparison of recent outcomes with historical cohorts shows clearly better outcomes. However, due to the rarity of these diseases and their detrimental outcome, there are no comparative studies of long-term outcomes of the various treatment modalities.

CONTINUOUS RENAL REPLACEMENT THERAPY

Over the past 2 decades, continuous RRT (CRRT) has become the preferred modality for the management of children with AKI and volume overload, who may be hemodynamically unstable.^{171–173} CRRT allows provision of diffusive and convective clearance separately or in combination. Diffusion refers to the movement of molecules down a concentration gradient across a semipermeable membrane and convection describes the movement of dissolved solutes with water in response to transmembrane pressure. Both mechanisms provide similar clearance of small molecules, although larger ones would be better cleared by convection. The nomenclature of CRRT stems from the access type and the mode of clearance. Although CRRT was originally developed based on combined arterial and venous access, namely, continuous arteriovenous hemofiltration, current practice uses pump-driven venovenous access (CVVH). CVVH, in and of itself, provides convective clearance through high UF rates. In order to prevent volume depletion, most of the ultrafiltrate is replaced by electrolyte-containing fluid. Continuous venovenous hemodiafiltration combines both diffusive and convective clearance by counter-current infusion of dialysate and net UF to maintain euvolemia. The choice of a given modality is often center dependent.

The shift in the epidemiology of AKI in developed countries impacts on the use of CRRT as it is delivered to critically ill patients with AKI. In the 1980s the leading causes of AKI were primary renal diseases, including hemolytic uremic syndrome (HUS), sepsis, and burns.¹⁷⁴ The Prospective Pediatric Continuous Renal Replacement Therapy (ppCRRT) registry enrolled 344 patients from 13 centers in the United States from 2001 to 2005, ranging from 1 day to 25 years. The most represented diagnoses include sepsis, stem cell transplantation, cardiac disease, liver disease, and malignancies.¹⁷²

INDICATIONS

In the ppCRRT, 46% received CRRT to treat fluid overload (FO) and electrolyte abnormalities, 29% for isolated volume overload, and additional 3% to obviate the need for fluid restriction and allow better intake of nutrition or the administration of blood products. An additional 6% were treated for inborn errors of metabolism, primarily hyperammonemia, or intoxications/overdose.¹⁷⁵ The rationale for attention to FO as a trigger for initiation of CRRT has been the understanding of the potential deleterious effect of excess fluid on pulmonary function and the need to provide critically ill children with adequate caloric intake. Several studies have recently shown an association between the degree of fluid accumulation from intensive care unit (ICU) admission to

CRRT initiation and mortality.¹⁷⁶ Percent FO is calculated according to the following formula¹⁷⁷:

$$\text{Fluid overload (\%FO)} = \frac{\text{Fluid intake (L)} - \text{Fluid output (L)}}{\text{Intensive care unit (ICU) admit weight (kg)}} \times 100$$

COMPARISON WITH HEMODIALYSIS AND PERITONEAL DIALYSIS

Although CRRT shares similar principles with HD, both the blood and dialysate flows are significantly slower, resulting in a lower hourly clearance rate. This is compensated for by extending the clearance time: over 24 hours, CRRT provides solute clearance comparable to a 4-hour HD session. The main advantage of CRRT in the critically ill child is the ability to maintain hemodynamic stability. CRRT and PD are continuous in nature, but the former provides much greater daily clearance rates.

TECHNICAL CONSIDERATIONS

Double-lumen catheters are usually used and the diameter size is selected based on the patient's BSA. One must reconcile the need for adequate blood pump flow rates with the desire to limit vessel trauma. Although femoral catheters are four times more often used than IJ ones, the latter result in a far better circuit survival.¹⁷⁸ Furthermore, femoral catheters, unless located in the inferior vena cava (IVC), are significantly affected by patient movement and may require the patient to be sedated or even paralyzed for proper use. Taken together, the preferred site for vascular access is in the right IJ vein with the distal tip of the catheter placed in the right atrium of the heart. The recommended blood flow is 3 to 10 mL/kg/h with a relatively higher flow in the very young child if an adult-size device is used. Although no prospective study has randomized pediatric patients to different effluent dose targets, the common practice is to use a dose of approximately 2000 mL/1.73 m²/h to achieve good metabolic control within 24 hours of CRRT initiation.¹⁷³ One should be cautious, as a recent study has demonstrated that hypotension after the connection to CRRT occurred in 49.7% of the connections a median of 5 minutes after the beginning of therapy.¹⁷⁹ In previous years, lactate-based fluids were used, resulting in lactic acidosis, cardiac dysfunction, and hypotension. Bicarbonate-based dialysate and replacement fluids are currently considered standard of care.¹⁸⁰ Large volumes of replacement fluid, containing varying amounts of electrolytes, are administered. This requires close monitoring of the patient laboratory profile to assure stable acid-base and electrolyte balance.

Activation of the clotting cascade, impacting circuit longevity, results from the contact of blood with the artificial filter as well as from slow blood flow and small catheters used. This mandates the administration of anticoagulants: Although a single pediatric study demonstrated that heparin and citrate are equally efficacious with more bleeding episodes occurring with the former,¹⁸¹ most adult studies favor regional citrate, together with systemic calcium infusion, for better longevity of the circuit and fewer bleeding episodes. Citrate is metabolized to bicarbonate in the liver and should be cautiously

used in patients with liver failure. In a multicenter pediatric survey, it was shown that citrate was used in 56% of children and heparin in 37%.¹⁷³ In a recent retrospective study it was demonstrated that hemofilter survival was higher in the citrate group than in the heparin group. Bivariate analysis showed that the hemofilter coagulation risk was significantly increased when heparin was used, regardless of hemofilter size and pump flow.¹⁸²

DOSING OF CONTINUOUS RENAL REPLACEMENT THERAPY

There are no robust studies in children exploring optimal CRRT dosing, but the recently reached consensus in adults is that there is no benefit to delivering high-volume CRRT reaching greater than 20 to 25 mL/kg/h clearance.¹⁸³ Attention should be paid, though, to the potential discrepancy between the prescribed and the delivered doses of CRRT, particularly those that are due to technical difficulties which are prevalent among the very young patients. Higher intensity of CRRT may lead to hypophosphatemia, increased amino acid loss, and augmented drug clearance rate.

CONTINUOUS RENAL REPLACEMENT THERAPY OUTCOMES

Patient outcomes are largely dependent on the underlying disease state and comorbid conditions. According to the ppCRRT registry, overall mortality was 42%, with higher rates in children with liver failure or liver transplant, pulmonary disease, or lung transplant and in stem cell transplant recipients, ranging from 55% to 69%.¹⁸⁴ FO at the initiation of CRRT is an independent risk factor for mortality: patients with greater than 20% FO were 8.5-fold more likely to expire.¹⁷⁷ Of note, this incremental increased risk of mortality was greater than that conferred by multiorgan failure (defined as receiving one vasoactive medication and invasive mechanical ventilation), sepsis, or malignancy, none of which were modifiable risk factors. Patients with over 20% FO demonstrated similar illness severity to those with 10% to 20%, inferring that the most severely fluid overloaded group was not sicker and there was probably no reason to administer the excessive fluid they received. A prospective multicenter study in children, addressing the effect of early initiation of CRRT on outcome measures, including mortality, is currently required.

PEDIATRIC KIDNEY TRANSPLANTATION

Dialysis and kidney transplantation developed in parallel, and gradually became available for pediatric patients, offering long-term survival to children with ESKD. The first case series of kidney transplantation in children was described in the mid-1960s by Starzl et al.¹⁸⁵ Over the subsequent decades, tremendous advances in pretransplant and posttransplant patient care, operative techniques, immunosuppression, and infection prophylaxis and treatment have transformed the field, enabling children with ESKD to undergo transplantation and live a normal life with constantly improving outcomes.¹⁸⁶

Although chronic dialysis provides clearance of uremic toxins, and together with medications and strict dietary

limitations, enables long-term survival of patients with ESRD, renal transplantation offers significant advantages, particularly in children. A well-functioning kidney graft can provide all the functions of a normal kidney, in addition to efficient clearance of metabolic waste products and fine-tuning of water and electrolyte balance, as well as secretion of hormones involved in systemic and renal hemodynamics, erythropoietin and calcitriol. There is a significant survival advantage for children with a functioning graft, compared with children remaining on chronic dialysis, with expected life span close to the general age-matched population (Fig. 74.4).^{22,187} Until the quest for tolerance in transplantation is successful, long-term immunosuppression must still be universal and close monitoring of patients is critical in order to achieve optimal outcomes, including improved growth, development, and rehabilitation.

Several national and international databases collect information on pediatric patients undergoing kidney transplantation. The NAPRTCS database has been collecting data on pediatric kidney transplantation recipients since 1987, and was later expanded to include children with CKD.¹¹¹ The Organ Procurement and Transplant Network (OPTN) has been collecting data on all organ transplantations in the United States since 1986, including pediatric renal transplantation.¹⁸⁸ The USRDS is the US database on ESKD, including kidney transplantation.²² These registries provide current information about the status of pediatric renal transplantation in the United States, and extensive analysis has been performed on the data collected. Other registries providing information on pediatric renal transplantation include ESPN/ERA-EDTA in Europe, with the latest annual report providing data from 35 countries, the United Kingdom renal registry, the ANZDATA, and others.^{23,187,189}

INCIDENCE, PREVALENCE, AND ALLOCATION

In 2015, 928 children under the age of 18 years were added to the United States waiting list for a kidney, with a prevalence of 1509 children listed at the end of that year, out of a total of 97,680 patients listed (1.5%).¹⁸⁸ Table 74.3 shows data on age, race, gender, and ESKD etiology at time of transplant in the United States. Of the 18,597 kidney transplants performed in 2015, 718 were in children (3.8%), a significantly higher proportion than those waitlisted. Of these, 33.7% of pediatric kidney transplants were from a living donor, down from 50.1% in 2005. The most frequent living donor was a parent (78% of living donors in 2014),¹¹¹ the rest were siblings, other family members, and living unrelated donors. Whereas there was an increase in living kidney donation to pediatric patients between 1987 and 2001, with a peak of 64% of kidney transplants, there has been a gradual decrease in this trend in the years between 2001 and 2015 in the United States. Changes in allocation policy of deceased donor kidneys, which favor pediatric patients and shorten waiting times, may have resulted in a decrease in living donation. Specifically in the United States, the implementation of the “Share 35” policy in 2005 gave priority to patients under 18 years to be offered a kidney from young deceased donors. However, further analysis has shown that the decline in living donor rates, particularly parental donation, predated this policy, and may also be affected by changes in patient demographics.¹⁹⁰ In the United States, a new national deceased donor

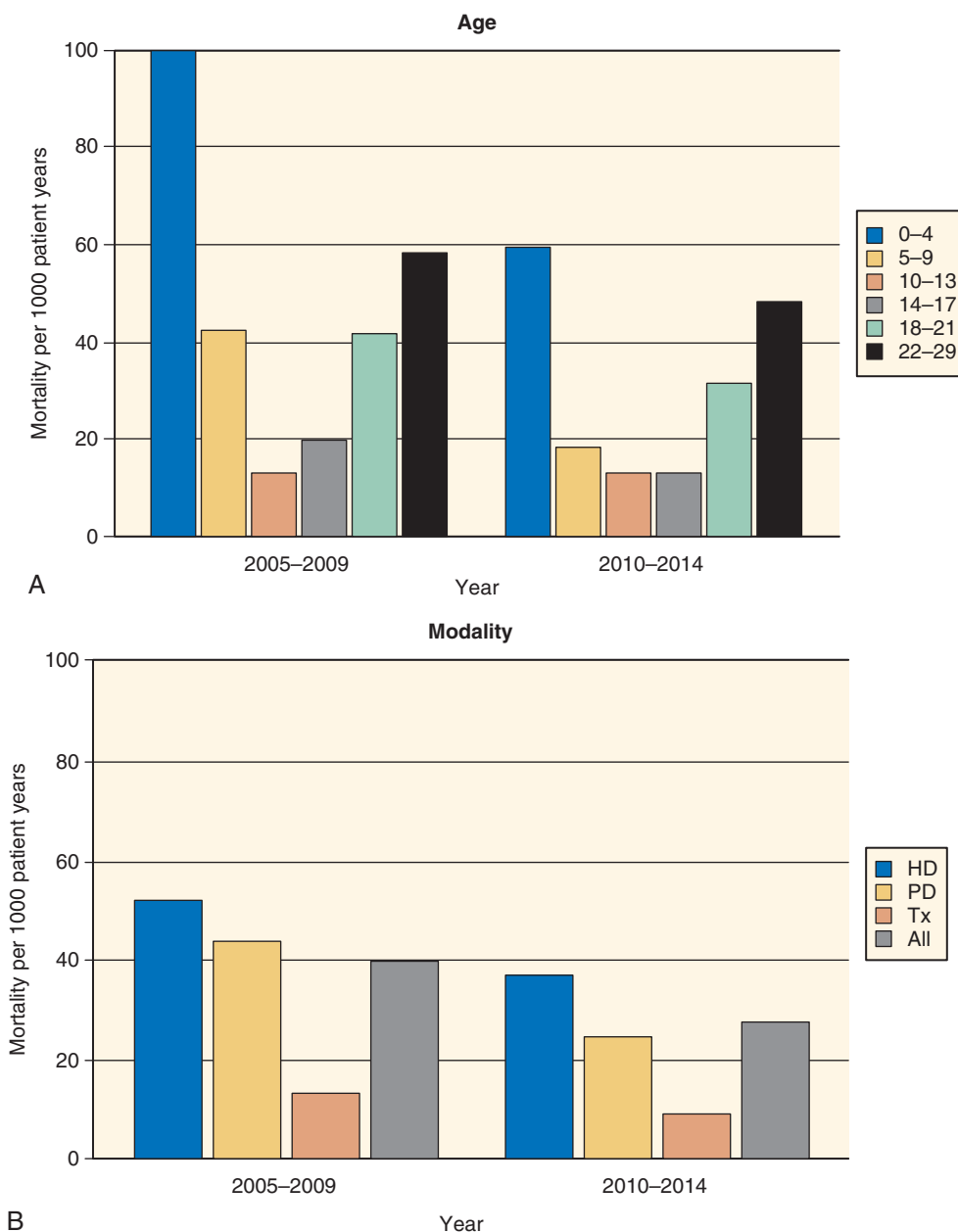


Fig. 74.4 One-year adjusted all-cause mortality rates in incident pediatric patients with end-stage kidney disease by (A) age with comparison to young adults (aged 0–29 years), and (B) modality (aged 0–21 years only), 2005 to 2009, and 2010 to 2014. *HD*, Hemodialysis; *PD*, peritoneal dialysis; *Tx*, kidney transplantation. (From Report., U. A. 2017 USRDS Annual Data Report. Chapter 7: ESRD among children, adolescents and young adults. *Am J Kidney Dis.* 2018;71(S1):S383–S416.)

kidney allocation system was implemented in December 2014. Its major goals were to decrease disparities in transplant access and to align expected survival of the allograft with the expected survival of the recipient. This system allocates the top 20% of kidneys in the kidney donor profile index to candidates in the top 20% of expected posttransplant survival. Priority is given to pediatric candidates, as well as patients with prior living donors, zero-HLA mismatch kidneys, and calculated panel-reactive antibodies (PRAs) 98% and up.¹⁸⁸ The change in policy may, however, result in a reduction in transplant rate for children, but further observation of these trends is necessary before firm conclusions can be drawn.¹⁹¹

In Europe, data collected from 32 countries demonstrated that 43% of pediatric kidney transplants were from living donors. Most countries implement an allocation scheme which prioritizes pediatric patients on the waiting list, in particular with regard to younger donors.¹⁹² The rate of pediatric kidney transplantation, waiting time, and donor source varies considerably between the different European countries. A positive correlation between the national gross domestic product per capita and the rate of pediatric kidney transplantation was noted.¹⁹³ In countries with a high pediatric priority policy the percentage of children with ESRD with a functioning graft was significantly higher.¹⁹³ Access to transplantation differed according to ethnic groups, with black and Asian children

Table 74.3 Age, Race, Gender, and End-Stage Kidney Disease Etiology at Time of Transplant in the United States

Recipient and Transplant Characteristics		N	%
Total		11,186	100
Sex	Male	6606	59.1
	Female	4580	40.9
Race	White	6605	59
	Black	1911	17.1
	Hispanic	1910	17.1
	Other	760	6.8
		1769	15.8
Underlying diagnosis	Kidney aplasia/hypoplasia/dysplasia		
	Obstructive uropathy	1713	15.3
	Focal segmental glomerulosclerosis	1308	11.7
	Reflux nephropathy	576	5.1
	Chronic glomerulonephritis	344	3.1
	Polycystic disease	339	3
	Medullary cystic disease	305	2.7
	Congenital nephrotic syndrome	289	2.6
	Hemolytic uremic syndrome	288	2.6
	Prune belly	279	2.5
	Familial nephritis	247	2.2
	Cystinosis	225	2

As of January 1, 2014, 12,189 renal transplants had been reported for 11,186 pediatric patients. This represents 586 new transplants and 554 patients with their first registry transplant since the 2010 Annual Report. Other diagnoses, not listed, with less than 2% of total for each include idiopathic crescentic glomerulonephritis, membranoproliferative glomerulonephritis, interstitial nephritis, lupus nephritis, immunoglobulin A nephritis, granulomatosis with polyangiitis, renal infarct, oxalosis, Wilms tumor, Drash syndrome, membranous nephropathy, sickle cell nephropathy, as well as 6.2% falling into the category of unknown etiology. Kidney dysplasia, hypoplasia, reflux nephropathy, and obstructive uropathy can also be grouped together as congenital anomalies of kidneys and urinary tract.

Modified from North American Pediatric Renal Trials and Collaborative Studies (NAPRTCS). *Annual Transplant Report*. Boston, MA: NAPRTCS; 2014.

less likely to receive a kidney transplant than white patients, mostly due to decreased living donation.¹⁹⁴

Preemptive kidney transplantation, meaning no dialysis performed prior to transplantation, is the modality used in 25% of pediatric ESRD patients in the United States and 27% in Europe.^{111,192} Worldwide, preemptive transplantation is mostly from living donors.

Renal transplantation in infants under the age of 1 year is uncommon, reported at 0.8% of pediatric transplants in the NAPRTCS database.¹¹¹ As noted earlier, deceased donor kidneys from younger donors are often allocated preferentially to children. However, utilization of deceased donor kidneys from infants and very small children holds unique challenges, as they are more prone to vasospasm and graft thrombosis, urinary complications, and possibly glomerular hyperfiltration.

Recent data have shown good results in children receiving kidneys from small pediatric donors, either a single kidney or both donor kidneys en bloc.^{195–197} One study from Europe reported a significant increase in allograft size after transplantation and superior graft function in the recipients of pediatric kidneys.¹⁹⁶

Although allocation systems usually try to match pediatric recipients with the best donors, a recent study showed excellent 3-year graft survival in children receiving a kidney after circulatory death of the donor.¹⁹⁸

PREPARATION OF THE RECIPIENT FOR TRANSPLANTATION

CHOICE OF DONOR

Living donor kidney transplantation offers better graft survival than deceased donor transplantation, though other factors, such as donor age, HLA matching, and sensitization play a significant role.¹⁹⁹ Kidney grafts from middle-aged living donors afford improved long-term allograft survival compared with younger deceased donor kidneys.²⁰⁰ However, data from the Collaborative Transplant Study showed that well-matched deceased donor kidneys compare favorably with living donor grafts with many HLA mismatches.²⁰¹ The question of whether it is advantageous to perform living or deceased donor transplantation first has been studied, as most pediatric kidney transplant patients will eventually require a second transplant. For most candidates, living donor transplantation as the first treatment option provides the best long-term benefit, except possibly for children who are highly sensitized, for whom deceased donor transplantation first, while they have priority as children on the waiting list, may be an advantage.²⁰²

TIMING

When the patient approaches KDOQI stage 4 CKD, preparation of the patient for transplant, as well as evaluation of potential living donors should be pursued. However, graft survival is not indefinite, and no benefit has been demonstrated for transplantation prior to reaching ESKD.²⁰³ Preemptive transplantation is associated with better patient and graft survival when compared with children who receive dialysis for longer than 1 year.²⁰³ Other considerations, particularly morbidity on dialysis, neurocognitive development, and decreased linear growth also favor transplantation as the first modality of RRT. Certain situations may preclude preemptive transplantation, such as active nephrotic disease, which is also a hypercoagulable state, presenting a high risk of thrombosis during transplantation. To be a candidate for transplantation without dialysis, the patient must have some minimal residual kidney function and urine volume.

Two age groups require special consideration. Very small children may have inferior results due mainly to technical difficulties and graft thrombosis. Some specialized centers, with greater experience in this age group, have demonstrated excellent results.¹⁹⁹ The alternative, remaining on dialysis for a prolonged period, also carries a high morbidity and mortality, particularly in this young age group.¹¹¹ Most units refer young patients for transplantation when they reach a weight close to 10 kg, but the risks and benefits must be

carefully considered in each case. The other potentially complex age group is adolescents. There is a higher rate of acute rejection episodes, shorter graft survival, and lower creatinine clearance in this group.¹¹¹ This is mostly due to nonadherence, which is more frequent in teenagers, and may manifest as missed doses of immunosuppressive medications, more extensive “drug holidays,” missed clinic visits, and other high-risk behaviors.^{204–206}

CONTRAINDICATIONS

Some clinical situations may present a temporary or permanent contraindication to transplantation. Malignancy should postpone plans for transplantation until complete remission is achieved, chemotherapeutic medications are discontinued, and no relapse is evident. High-dose immunosuppressive therapy after transplantation may increase the risk of cancer recurrence due to enhanced development of micrometastases. The period between completion of therapy for cancer and listing for transplant will depend also on the type of malignancy and its characteristics.²⁰⁷ Immunosuppressive medications also impair the body's ability to fight infection; therefore active infection is a temporary contraindication to transplantation and latent infections should be sought out and addressed before proceeding to transplant. Severe comorbidities or profound neurodevelopmental delay may render a child unsuitable for transplant. However, intellectual disability does not impact patient or graft survival, and should not be a barrier to transplantation in itself.^{208,209} Each child should be evaluated individually to assess the impact of transplantation on life expectancy, quality of life, and rehabilitation, taking into account the wishes of the family. Recalcitrant nonadherence to medication or dialysis schedule, diet, and clinic appointments may also be a temporary contraindication to transplantation.

RECIPIENT EVALUATION

A comprehensive evaluation of the pediatric kidney transplant candidate is crucial to achieve the best results. The goals of this workup are to provide an optimal plan for each individual patient, reduce complications, and increase long-term patient and graft survival. The main elements of the pretransplant evaluation are summarized in Table 74.4. Some of the important issues are discussed in the following sections.

INFECTION

A history, physical signs, or laboratory findings suggesting active infection need to be addressed before subjecting the patient to major surgery and high-dose immunosuppression. The respiratory tract, teeth, skin, dialysis access exit site, and other sites of possible chronic infection should be carefully examined, as even a minor infection may be exacerbated by immunosuppressive therapy. Testing of serology for cytomegalovirus (CMV) and Epstein–Barr virus (EBV) to assess the risk of posttransplant viral disease are important in planning monitoring and antiviral prophylaxis posttransplant. Many pediatric patients are seronegative for these viruses, and should be monitored frequently while waitlisted, to avoid inadvertently transplanting during a subclinical primary infection. Vaccinations should be reviewed and completed before transplantation, as patients will be at increased risk

Table 74.4 Evaluation of Pediatric Candidates for Renal Transplantation

Medical history	End-stage renal disease etiology, family history of renal or other disease, biopsy results, previous transplant history, dialysis access and prescription, medication list, compliance with medical regimen, urological interventions, urine output, other comorbidities, allergies, previous procedures, transfusions, growth charts
Physical examination	Weight, height, body mass index and percentiles, blood pressure and pulse, general well-being, complete physical examination
Specialist assessments	Pediatric nephrologist, transplant surgeon, urologist, anesthesiologist, transplant coordinator, nurse, social worker, dietitian, pediatric dentist. As indicated: psychologist, cardiologist, hematologist, pulmonologist
Laboratory tests	Blood type, complete blood count, electrolytes, blood urea nitrogen, creatinine, calcium, phosphorus, liver enzymes, protein, albumin, lipids, iron, parathyroid hormone, thyroid, fasting glucose and HbA1C, prothrombin time/partial thromboplastin time, urinalysis, 24-hour urine collection for creatinine clearance and proteinuria
Serology	Epstein–Barr virus (immunoglobulin [Ig]G Epstein–Barr nuclear antigen and IgM), cytomegalovirus (IgG and IgM), hepatitis B virus (surface Ag and Ab), hepatitis C virus, human immunodeficiency virus, varicella, purified protein derivative test
Histocompatibility testing	Human leukocyte antigen typing – class I (A,B), class II (DR, DQ), panel-reactive antibodies, donor-specific antibodies, crossmatch with donor (T- and B-cell crossmatch, by complement-dependent cytotoxicity with antihuman globulin or flow cytometry)
Imaging	Chest X-ray, abdominal ultrasound including the kidneys, bladder, and postvoiding volume Cardiac echo and electrocardiogram. Doppler study of abdominal and pelvic arteries and veins Voiding cystourethrogram, urodynamic testing, or other if indicated
Vaccinations	Diphtheria, tetanus, pertussis (DTaP), hemophilus influenza type B (HiB), inactivated polio vaccine (IPV), hepatitis A virus (HAV), hepatitis B virus (HBV), varicella, measles, mumps, and rubella (MMR), pneumococcus conjugate and polysaccharide, influenza Selected patients: meningococcus, human papillomavirus
Social	Evaluation of the family, support systems, financial issues, school

of infection after transplantation. In addition, response to vaccines may be attenuated by immunosuppressive medications, and live vaccines are not recommended posttransplant.^{209a} However, even with pretransplantation vaccination, immunity may wane in some recipients when they receive immunosuppression.²¹⁰ Higher pretransplant titers of hepatitis B antibodies may protect against titer loss after transplantation.²¹¹ The regular pediatric schedule should be given, including pneumococcus, influenza, and varicella, as well as other vaccines depending on age and exposure. Close contacts and household members of these children should complete age-appropriate vaccination schedules to increase the child's indirect protection.²¹² Despite these recommendations, the vaccination coverage in pediatric kidney transplant candidates is incomplete, and efforts need to be made to improve vaccination rates before kidney transplantation.²¹¹

MALIGNANCY

Although less common in children, a basic workup to rule out malignancy includes history, physical examination, chest X-ray, and abdominal ultrasound, as well as routine laboratory tests. Children with a history of cancer should be evaluated by a pediatric oncologist to assess remission and risk of recurrence and to help determine the timing of transplantation.²⁰⁷

UROLOGICAL ISSUES

The most frequent cause of ESRD in younger children is CAKUT.¹¹¹ Structural and functional evaluation of the urinary tract is important before transplant, as the kidney allograft may be compromised by an abnormal urinary tract, with obstructed drainage, massive reflux, or a small defunctionalized bladder. Investigation should begin with an ultrasound of the kidneys and urinary tract, including imaging of the full and postvoid bladder. In patients with an abnormal urinary tract, further tests like a voiding cystourethrogram and urodynamic evaluation provide further information. When the child has prolonged anuria, it is sometimes difficult to assess bladder capacity and function. Surgical intervention to alleviate obstruction, correct massive reflux if needed, or occasionally bladder augmentation can be performed prior to or during transplantation. However, the capacity of a small defunctionalized bladder may increase after renal transplantation and enable normal voiding without compromising graft function, thus avoiding the need for bladder augmentation.²¹³ Nephrectomy of native kidneys is indicated in cases of persistent nephrotic syndrome of presumed immunological etiology (to enable prompt and accurate diagnosis of recurring nephrotic syndrome), chronically infected kidneys, very large kidneys in polycystic kidney disease (to enable more physical space for the graft and alleviate discomfort), and sometimes for uncontrolled HTN.²¹⁴ In cases of congenital nephrotic syndrome, particularly the Finnish type, mortality due to infection and thrombosis is high; therefore bilateral nephrectomy is usually performed when the child has reached a size compatible with transplantation, even if renal function is good or even normal.²¹⁵

CARDIOVASCULAR ISSUES

Congenital heart disease may coexist with CKD in some patients, which requires consultation with a pediatric cardiologist and consideration of surgical correction prior to transplantation. HTN is very common in ESKD, in many cases

causing LVH, with diastolic and occasionally systolic dysfunction. Treatment with antihypertensive medications and avoiding chronic volume overload in dialysis patients are important, though optimal control of HTN is often difficult to achieve. However, FGF-23 which is elevated in CKD, induces LVH independent of BP. Therefore LVH may not be completely correctable as long as CKD persists.^{216,217} Cardiomyopathy may improve significantly in children undergoing renal transplantation and LVH may improve.^{218,219}

Evaluation of abdominal vasculature prior to transplantation is beneficial in planning anastomoses and anticipating significant surgical challenges. Children who have had previous abdominal or pelvic surgery, CVCs in the lower body, including neonatal umbilical arterial or venous catheterization, and patients with hypercoagulable state should have Doppler studies of the large arteries and veins prior to transplantation. Abnormal studies can be followed by formal venography.²²⁰

RISK OF RECURRENCE

The disease process that caused ESKD may affect the graft. Many immune-mediated kidney diseases have a tendency to recur due to continuation of the underlying immunological process or persistence of a humoral factor (FSGS, membranoproliferative glomerulonephritis [MPGN], systemic lupus erythematosus, immunoglobulin A [IgA] nephropathy). Genetic mutations causing formation of an abnormal protein product or enzymatic deficiency are highly likely to lead to rapid disease recurrence in the transplanted kidney (atypical HUS [aHUS], primary hyperoxaluria).

Recurrence is particularly frequent in idiopathic FSGS, with a rate in children of 30% to 80%, depending on the report. Nephrotic syndrome and graft dysfunction may appear immediately after transplantation, and prompt intervention is needed. In these cases pretransplant nephrectomy of native kidneys, if the patient still has proteinuria, will assist in timely diagnosis of recurrence and enable early treatment.²¹⁴ However, FSGS is a histological diagnosis, which may be due to one of several underlying pathophysiological processes. Conversely, SRNS may have histological findings compatible with minimal change nephropathy or other nonspecific findings, and yet have clinical features indistinguishable from FSGS. The PodoNet registry, a large multinational study of childhood-onset SRNS, demonstrated posttransplant disease recurrence in only 4.5% of children with a genetic diagnosis compared with 25.8% of patients without.²²¹ Children who developed secondary steroid resistance were more likely to have posttransplantation recurrence than those who never responded to steroid treatment.²²² Higher risk of recurrence is seen in young children and in those who already had recurrence in a previous graft.^{223,224} To help quantify the risk of recurrence, it is important to clarify, prior to transplantation, whether FSGS is genetic, due to mutations in genes encoding components of the glomerular filtration barrier.

MPGN also carries a high risk of recurrence, which affects graft survival. The risk of graft loss at 5 years is significantly higher in children with FSGS or MPGN compared with children with renal dysplasia.²²⁵ Other glomerular diseases, such as lupus nephritis and IgA nephropathy, may recur but are not usually associated with graft loss.

aHUS can cause ESRD and has a high rate of recurrence in the graft.²²⁶ It is often caused by various disorders of

complement regulation, including mutations in complement factor H or I, membrane cofactor protein, and others. The specific cause of the disease should be investigated before considering transplantation, in order to assess the risk of recurrence and guide treatment. Living related donors should be screened for the causative mutation, if known, prior to approval for donation. Disease recurrence can be prevented or treated with eculizumab (discussed later).²²⁷

PH1 is a disorder of glyoxylate metabolism in which an enzymatic defect in the liver causes overproduction of oxalate and deposition of calcium oxalate in the kidneys and in other organ systems once advanced renal failure develops. Presentation in infancy is associated with nephrocalcinosis and early CKD. Renal transplantation alone usually results in massive oxalate deposition in the new kidney and early graft failure. Combined liver and kidney transplantation (or liver transplantation alone if renal function is not severely affected) is curative, correcting the underlying enzymatic defect.²²⁸

SENSITIZATION

Blood type, HLA typing, PRAs should be tested in each transplant candidate. If a living donor is being considered, donor-specific antibodies (DSAs) should be identified and quantified if PRA is positive. A negative direct crossmatch by complement-dependent cytotoxicity with antihuman globulin is crucial to avoid hyperacute rejection. Crossmatch by the flow cytometry method is more sensitive, but the clinical implications of a positive test are less clear-cut. Low-level DSA may be overcome with desensitization protocols, particularly in a planned living donor transplantation.

TRANSPLANT IMMUNOLOGY

HLA MATCHING AND SENSITIZATION

Transplantation immunobiology is covered elsewhere in detail. The importance of HLA matching between donor and recipient was recognized in the 1960s and 1970s and became the cornerstone of deceased donor allocation policy. A clear association between number of HLA mismatches and graft survival was shown in several large studies.²²⁹ With the introduction of more potent immunosuppression, the significance of HLA matching may have diminished, while other factors such as peak PRAs, cold ischemic time, and donor age retained their relative importance.²³⁰ However, other recent publications have shown that HLA mismatches continue to adversely affect renal allograft outcomes in the present era, as shown by a study from ANZDATA that included children.²³¹ Studies that examined the effect of HLA matching specifically in children have also shown that an increase in number of HLA mismatches is associated with decreased graft survival.²⁰¹ Recently new allocation algorithms have been implemented which prioritize better HLA-matched kidneys in addition to favoring pediatric patients.¹⁸⁸ Both donor age and HLA matching are important in determining deceased donor graft survival and may be offset by each other.²³²

Most pediatric patients with CKD will need to undergo more than one transplant over their lifetime, and HLA matching of the first kidney graft may have an impact on subsequent transplants. More HLA mismatches at first

transplant are associated with HLA sensitization, longer waiting time for a second transplant, and decreased graft survival.^{233,234} However, the benefits of better HLA matching at first transplant on lifetime with graft function, although significant, are relatively small.²³⁵

In addition to HLA matching, PRAs are measured in renal transplant candidates, and if positive, should be characterized by solid-phase assays.²³⁶ The presence of DSA is associated with increased risk of acute antibody-mediated rejection (AMR) as well as chronic AMR and decreased graft survival. When DSAs are further characterized, patients with complement-binding DSA after transplantation are at the highest risk of graft loss and AMR.²³⁷ The deleterious impact of DSA has been demonstrated in children, even when the direct crossmatch between donor and recipient is negative by flow cytometry, though not in all series.^{238,239} Although good HLA matching without DSA is desirable, not all patients will find an appropriate donor. Annual reports show that about 5% of children listed for deceased donor transplantation have PRA greater than 80%.¹⁸⁸ Desensitization protocols, including treatment with combinations of intravenous immunoglobulin (IVIG), plasma pheresis, and rituximab, have shown acceptable intermediate-term outcomes in adults undergoing living or deceased donor transplantation.^{240,241} Pediatric patients are theoretically at higher risk for infectious complications after transplantation; therefore, desensitization in children may cause overimmunosuppression. There are remarkably few case series and clinical trials of desensitization in children. A recent study used IVIG and rituximab desensitization with alemtuzumab induction in highly sensitized children who underwent deceased donor kidney transplantation. Although a higher rate of acute rejection was seen in the sensitized patients, there was no significant decrease in 6-year graft survival and no increase in infectious complication.²⁴²

De novo DSA appears in 35% of children in the first years after transplantation, with a high prevalence of DSA to class II and specifically HLA-DQ antigens.²⁴³ These antibodies are associated with increased risk of AMR and graft dysfunction, regardless of whether they develop in the first year after transplantation or later.²⁴⁴ Serial monitoring of DSA in children after transplant may help to identify patients at risk for adverse outcomes, and guide tailoring of immunosuppression.

Although theoretically kidneys transplanted from maternal donors might be less immunogenic than paternal kidneys, due to fetal-maternal interaction, this does not seem to be the case. A recent study shows that in the current era of immunosuppression, patients receiving a kidney from their mother were slightly more likely to experience acute rejection, consistent with earlier findings from the European Collaborative Transplant Study.^{245,246}

ABO COMPATIBILITY

In the past, ABO incompatibility was considered a contraindication to transplantation, because of the high risk of hyperacute AMR. However, in the past few years there are reports of successful ABO-incompatible renal transplantation in children. ABO-incompatible transplantation in children is rare, and usually involves patients with blood type O or B and low titers of anti-A antibodies, receiving an organ from

blood type A donors. No ABO-incompatible transplants took place in pediatric kidney recipients in 2015 in the United States.¹⁸⁸ ABO-incompatible transplantation is more common in Japan, where most transplants are from living donors. Treatment of these patients with a desensitization protocol has shown long-term outcomes equivalent to the general pediatric transplant population.²⁴⁷ In Europe, data from the collaborative transplant study, which included children, reported that 1420 ABO-incompatible living donor kidney transplants were performed in 2005–2012, with similar long-term patient and graft survival. A slight increase in death due to infection was seen in the first year after ABO-incompatible transplantation.²⁴⁸ Treatment protocols initially utilized pretransplant plasmapheresis, in order to remove anti-A/anti-B antibodies, together with splenectomy or rituximab. Newer regimens include immunoadsorption, using either blood group-specific columns or nonspecific columns, to remove the antibodies, together with rituximab to prevent ongoing antibody production after transplantation. Addition of other antibody induction in these patients did not result in altered graft survival.²⁴⁸ The risk of AMR is associated with higher titers of anti-A/B isoagglutinins, and the intensity of desensitization may be modifiable according to pretreatment titers, thus potentially opening this option in deceased donor transplantation as well.²⁴⁹ Immunoadsorption therapy for ABO-incompatible renal transplantation has been associated with surgical bleeding in some cases.²⁵⁰ Late AMR in the setting of ABO incompatibility is less frequent than AMR in HLA sensitization.^{236,251,252} Despite more intense immunosuppressive regimens, a study in adults did not demonstrate a higher risk of cancer in patients undergoing ABO-incompatible transplantation.²⁵³

IMMUNOSUPPRESSION

The goal of immunosuppressive therapy in pediatric kidney transplantation is to prevent graft rejection while minimizing adverse effects. A wide range of immunosuppressive medications are used in pediatric renal transplantation and are a key factor in preventing rejection and enabling long-term graft function. In general, more information is available from trials in adult transplant patients, and specific data in children do not always exist. Special considerations in children include altered drug metabolism, as well as adverse effects particular to children, including effects on growth and development. The optimal immunosuppression regimen should minimize the risk of rejection, without causing over-immunosuppression or other side effects of medications.

INDUCTION IMMUNOSUPPRESSION

The risk of acute rejection is highest in the immediate post-transplant period, which is why in addition to higher doses of maintenance immunosuppressive drugs, many transplantation protocols include a biological induction agent. These antibody preparations are aimed at depleting T cells or preventing their activation and are administered in the perioperative period.

Antilymphocyte preparations are T-cell depleting agents that can be polyclonal or monoclonal. In addition to their use for induction immunosuppression, they have a role in

treatment of acute cellular rejection. The polyclonal antibody antithymocyte globulin (ATG) contains antibodies to a wide variety of lymphocyte antigens in serum of animals immunized with human lymphocytes. The most frequently used preparation today is the rabbit-derived ATG.

The first monoclonal preparation, introduced in the 1980s, was OKT3, containing antibodies to the CD3 receptor on the surface of T cells. It is no longer in use, due to some severe adverse effects. The anti-CD52 antibody preparation alemtuzumab binds to the CD52 receptor, found on both T and B cells as well as monocytes and natural killer cells, causing cell lysis.

T-cell activation induces interleukin-2 (IL-2) production causing T-cell proliferation; therefore monoclonal IL-2 receptor antibodies such as basiliximab block this pathway, preventing T-cell activation without depletion.

Antibody induction has been shown to improve outcomes in kidney transplantation in adults, compared with conventional immunosuppression alone. However, in children there are less supportive data. Patients who are highly sensitized are more likely to receive antibody-induction therapy; therefore it is difficult to compare efficacy or safety outcomes in retrospective studies. Data from earlier studies of antibody induction, which used maintenance immunosuppression agents that are rarely used today, such as azathioprine and cyclosporine, are also not necessarily pertinent today.

ATG is the most frequently used antibody induction in the United States, particularly in steroid-free protocols and in highly sensitized patients.^{254,255} Alemtuzumab induction has been described in several small series of children, enabling steroid avoidance or tacrolimus monotherapy, with good results.^{256–258} Alemtuzumab has also been used for highly sensitized children in combination with desensitization. Outcomes were similar to nonsensitized patients who received IL-2 receptor blockers. A significant reduction in lymphocyte counts was observed in the alemtuzumab group up to 1 year from transplantation.²⁴² Concerns regarding the risk for posttransplant lymphoproliferative disease (PTLD) have been raised regarding the use of lymphocyte-depleting agents, particularly alemtuzumab. In a large OPTN database study, T-cell depleting agents, alemtuzumab and ATG, were compared with nondepleting induction agents or no induction, and the incidence of PTLD without induction (0.43%) was shown not to be different from that with basiliximab (0.38%) or alemtuzumab (0.37%), but was slightly higher in patients who received ATG (0.67%).²⁵⁹ The population in the study had received higher doses of ATG than employed today, higher doses of maintenance immunosuppression, and less antiviral prophylaxis, suggesting that the ATG induction may not currently be a significant risk factor for PTLD.²⁶⁰

Pediatric studies using the IL-2 receptor blocker basiliximab as induction therapy combined with triple-drug maintenance immunosuppression showed a good safety profile but minimal or no additional benefit compared with no antibody induction.^{261,262} A recent large study using data from the Australia and New Zealand Dialysis and Transplant registry showed that induction with IL-2 receptor antibodies in pediatric patients was associated with at least a 40% reduction in the rate of acute rejection, though no significant decrease in graft loss was found. This was found to be consistent across donor types, initial immunosuppression, and transplant era;

propensity scoring was used to account for confounding factors.²⁶³

Comparing the efficacy of lymphocyte-depleting agents with IL-2 receptor blockers in a prospective trial would require a very large study population. Retrospective data from the OPTN database found that lymphocyte-depleting induction (ATG or alemtuzumab) is more effective at lowering acute rejection rates in African-American pediatric kidney transplant recipients, but no difference was found in other ethnic groups.²⁶⁴

According to the OPTN data report from 2015, most children undergoing kidney transplantation in the United States received antibody induction, approximately 60% were treated with ATG and 33% received basiliximab induction.¹⁸⁸ Children with elevated PRAs (HLA sensitized) were more likely to receive ATG induction.¹⁸⁸ In Australia and in Europe the preferred induction agent is basiliximab, used in 76% to 93% of children according to the 2016 ANZDATA, whereas 4% to 6% were treated with ATG.¹⁸⁹

MAINTENANCE IMMUNOSUPPRESSION

CALCINEURIN INHIBITORS

Cyclosporine

Cyclosporine was introduced as an immunosuppressant for renal transplantation in 1982, revolutionizing the treatment of transplant recipients. Whereas in 1996, 82% of children were treated with cyclosporine 1 month after renal transplantation, consistently less than 3% received cyclosporine since 2009, as it was largely replaced by tacrolimus.¹⁸⁸ Recently, low-dose cyclosporine has been used in pediatric kidney transplant patients together with the proliferative signal inhibitor everolimus, with good 3-year outcomes.²⁶⁵

Cyclosporine doses in children need to be higher per body weight, as it is more rapidly metabolized than in adults. It has been suggested that cyclosporine should therefore be divided every 8 hours in young children and infants; however, this may actually result in lower dose-normalized area under the curve (AUC) throughout the day, in addition to being a drug regimen that is more difficult to follow.^{265a} Drug level monitoring of cyclosporine is important to maintain therapeutic concentrations, while avoiding toxicity. Trough levels are usually kept at 150 to 300 mcg/L initially, and 75 to 125 mcg/L after the first 6 months. Monitoring AUC or 2-hour postdrug level, as an indicator of AUC, may be more effective in avoiding toxic effects. Cyclosporine is available as capsules and oral solution. Absorption of different formulations may vary, and dosing should be guided by serum drug levels.

Cyclosporine is nephrotoxic in a dose-dependent manner. It is associated with HTN, hyperlipidemia, and hyperuricemia, all of which may contribute to cardiovascular disease, as well as their deleterious effect on graft function.²⁶⁷ Cosmetic side effects are a major issue in young patients, including hirsutism, gum hyperplasia, and less frequently coarse facial features, which may contribute to nonadherence, particularly in teenagers. Several drugs interact with cyclosporine, in many cases decreasing its metabolism and potentially causing toxic concentrations. Common examples in children include macrolide antibiotics or azole antifungal medications, which should be given with care, while monitoring calcineurin inhibitor levels. Ingestion of grapefruit also decreases calcineurin

inhibitor metabolism and increases serum levels, mostly by inhibition of enteric CYP3A4, and should be avoided.

Tacrolimus

The use of tacrolimus as part of the immunosuppression regimen has continuously increased since its introduction in the 1990s. In 2015, 96.3% of pediatric kidney transplant recipients received tacrolimus initially.¹⁸⁸ Metabolism varies greatly between individuals, younger children may require higher doses by weight, and all patients should have regular drug level monitoring.²⁶⁸ Other factors also have impact on tacrolimus dose needed to achieve trough blood concentrations. Studies have shown that frequency of various cytochrome P450 CYP3A5 allele variants may account for over 50% of variance of tacrolimus blood concentrations.²⁶⁹ Moreover, the frequency of these alleles is different in African-American patients, which can explain higher doses needed to achieve therapeutic drug concentrations in this population.²⁷⁰ Initial target tacrolimus levels are high (10–15 ng/mL), gradually decreasing to 3 to 6 ng/mL after the first 6 months, depending on the specific protocol.

In the first years after tacrolimus became available, several studies were published comparing tacrolimus and cyclosporine in children. An NAPRTCS study comparing tacrolimus and cyclosporine, both in combination with mycophenolate mofetil and prednisone, showed no difference in rejection or graft survival at 1 and 2 years posttransplant. However, the children in the tacrolimus group were less likely to require antihypertensive treatments, and had a higher GFR.²⁷¹ A similar multicenter study from Europe showed decreased frequency of acute rejection and a higher GFR in the tacrolimus group, with a similar safety profile for both drugs.²⁷² Long-term data showed better graft survival and GFR in the tacrolimus group, as well as lower cholesterol levels.²⁶⁶ A comparison of protocol biopsies in children demonstrated more subclinical acute rejection in those treated with cyclosporine.²⁷³ More recent data have shown a lower prevalence of cardiovascular risk factors in children treated with tacrolimus, compared with cyclosporine or sirolimus.²⁶⁷ As tacrolimus became the standard of care for immunosuppression for pediatric kidney transplantation, large head-to-head studies were no longer performed. A study comparing children treated with tacrolimus and mycophenolate with a historical cohort treated with cyclosporine and prednisone showed superior graft survival in the tacrolimus group.²⁷⁴

Tacrolimus adverse effects are similar to cyclosporine, with a higher rate of posttransplant diabetes mellitus and neurotoxicity for tacrolimus, manifested as tremor or seizures.²⁷⁵ A frequent reason for changing treatment from cyclosporine to tacrolimus in children and adolescents is the lack of cosmetic side effects of tacrolimus. A once-daily modified-release formulation of tacrolimus has been shown to improve patient compliance.²⁷⁶

ANTIMETABOLITES

Azathioprine

Azathioprine was found to be effective in prevention of renal graft rejection in the early 1960s, especially when combined with corticosteroids. Subsequently, studies comparing azathioprine with mycophenolate mofetil showed a significantly lower rate of acute rejection in patients treated with

mycophenolate.²⁷⁷ While in the first years, most children were treated with azathioprine as part of their immunosuppression regimen, in the 2014 NAPRTCS report, only 2.5% to 3% of children received azathioprine since 2002.¹¹¹ Adverse effects of azathioprine include myelosuppression and hepatotoxicity, which might require dose reduction. Therapeutic efficacy, bone marrow toxicity, and liver toxicity of azathioprine and its active metabolite 6-mercaptopurine (6-MP) may be related to their metabolites: 6-thioguanine and 6-methylmercaptopurine. Thiopurine methyltransferase (TPMT) enzyme activity is a major factor determining azathioprine and 6-MP metabolism. Testing for TPMT genotype and enzyme activity is available and can help guide therapy. The myelosuppressive action of azathioprine is potentiated by xanthine oxidase inhibitors such as allopurinol, and if necessary, the dose of azathioprine must be reduced by 50% to 75%. Although azathioprine is still classified by the FDA as category D for use in pregnancy, suggesting positive evidence of risk to the fetus, data accumulated over the past few decades, especially from the National Transplantation Pregnancy Registry, have shown that the incidence of birth defects is similar to the general population.²⁷⁸ Therefore azathioprine might present as an alternative to mycophenolate formulations for female kidney transplant recipients planning pregnancy. In addition, azathioprine is significantly less expensive than mycophenolate.

Mycophenolate

Mycophenolate mofetil, a reversible inhibitor of inosine monophosphate dehydrogenase that downregulates specifically T- and B-cell proliferation, was developed as an alternative antimetabolite to azathioprine, with less bone marrow toxicity. It is rapidly metabolized to mycophenolic acid *in vivo*. It was shown to reduce frequency of acute rejection and improve 5-year graft survival in children compared with azathioprine, in combination with cyclosporine and prednisone.²⁷⁷ Some data suggest that in children with chronic allograft nephropathy (CAN), adding mycophenolate to the immunosuppressive regimen improves graft function.²⁷⁹ In 2015 mycophenolate formulations were used as part of the initial maintenance immunosuppression regimen in 93.2% of pediatric kidney recipients in the United States.¹⁸⁸ Although initially dosing was 600 mg/m² BSA twice daily, some protocols give a lower dose (300 mg/m²) to low-risk patients, with less adverse effects.²⁸⁰

The main adverse effects of mycophenolate mofetil are gastrointestinal, including abdominal discomfort, nausea, and diarrhea. Although there have been reports of mycophenolate-associated inflammatory bowel disease, a recent case series of pediatric liver or kidney transplant recipients with new-onset colitis did not show a link to mycophenolate treatment.²⁸¹ An alternative formulation, enteric-coated mycophenolic acid, has been shown to have fewer gastrointestinal adverse effects, as shown in small studies in children.²⁸² Mycophenolate mofetil is available as capsules, tablets, and oral suspension, as well as intravenous preparation. Enteric-coated mycophenolic acid cannot be made into a suspension, which limits its use in younger children. There is debate over whether mycophenolic acid monitoring is beneficial in prevention of acute rejection. Data suggest that mycophenolate exposure is associated with time from transplantation, graft function, and age, with younger patients requiring higher doses per BSA in order to

reach therapeutic levels.²⁸³ Lower mycophenolate exposure has been shown to be associated with *de novo* DSA formation.²⁸⁴ Mycophenolate has been associated with neutropenia in pediatric kidney transplant recipients, often necessitating dose reduction or temporary interruption.²⁸⁵ Pregnancies in patients exposed to mycophenolic acid are associated with a high rate of birth defects; therefore adolescent girls and women of childbearing potential must be cautioned to use reliable contraception if sexually active while treated with mycophenolate.²⁸⁶

CORTICOSTEROIDS

Corticosteroids have been a cornerstone drug in prevention of rejection of transplanted organs for decades. However, chronic corticosteroid therapy is associated with many adverse effects including HTN, hyperlipidemia, cardiovascular disease, glucose intolerance, osteoporosis and aseptic bone necrosis, cataracts and glaucoma, as well as weight gain, cushingoid appearance, acne, and psychological effects. The cosmetic effects are often a cause of nonadherence in many patients, particularly adolescents. Corticosteroids also inhibit linear growth by several mechanisms of action, including interference with the action of GH and inhibition of bone formation.²⁸⁷ Efforts to reduce the adverse effects of corticosteroids in children undergoing renal transplantation have included using lower doses of prednisone, alternate-day regimens, steroid-withdrawal protocols, and steroid avoidance.

According to NAPRTCS, corticosteroid use was almost universal in children undergoing renal transplantation until the year 2000. Subsequently, their use decreased to 50% to 60%, which has remained constant over the past few years, despite extensive research on steroid-sparing immunosuppression in pediatric kidney transplantation.¹⁸⁸ However, centers that continue to use triple immunosuppression in low-risk patients generally use lower doses than previously used, tapering corticosteroid dose within a few weeks to doses of 0.1 mg/kg or 5 mg/m² or even lower.¹¹¹

Initial single-center studies of steroid-free pediatric kidney transplantation in low-risk patients showed promising results at the turn of the twenty-first century. Larger multicenter randomized controlled studies, in both the United States and Europe, found improved linear growth with no increase in biopsy-proven acute rejection (BPAR) in children on steroid avoidance or early steroid-withdrawal protocols.^{280,288} Improvement in height standard deviation scores was significant only in prepubertal children. Other benefits of steroid avoidance, such as lower rates of HTN and posttransplant diabetes, were inconsistent. Of note, most of these studies used daclizumab induction, which is no longer available, and although small studies have shown similar results with thymoglobulin induction, prospective studies using other IL-2 receptor blockers have not been performed.²⁵⁴ Late withdrawal of steroid treatment in children treated with cyclosporine–mycophenolate immunosuppression also demonstrated improved linear growth with no increase in the rate of graft rejection.²⁸⁹ Two meta-analyses of steroid avoidance or withdrawal studies confirmed the findings of no increase in acute rejection episodes with improved linear growth in prepubertal children.^{290,291} In addition, the pooling of data demonstrated a beneficial effect on HTN and new-onset diabetes in children in the steroid-sparing groups.²⁹⁰ Corticosteroid avoidance

protocols have not been well studied in high-risk pediatric transplant recipients, including non-Caucasians, retransplants, and sensitized patients.

MAMMALIAN TARGET OF RAPAMYCIN INHIBITORS

Mammalian target of rapamycin (mTOR) inhibitors block the proliferative response of T-cells to IL-2 stimulation. This group of immunosuppressive agents also inhibit proliferation of vascular muscle and some cancer cells, and therefore can reduce transplant vasculopathy. This, together with lack of nephrotoxicity, made this group of drugs a promising treatment option for pediatric transplantation. Adverse effects of mTOR inhibitors include impaired wound healing, which can present a problem when used immediately after transplantation, hyperlipidemia, proteinuria, decreased levels of testosterone in some adolescent males, aphthous ulcers, interstitial pneumonia (including *Pneumocystis jirovecii* infections), anemia, and thrombocytopenia.²⁹²

Sirolimus

The first drug in this group was approved for use in adult solid organ transplantation in 2000. Initially it was used in an attempt to avoid calcineurin inhibitors completely, by combination with IL-2 receptor blocker induction, mycophenolate, and steroids. A higher incidence of acute rejection was seen in this protocol as well as increased adverse events.²⁹³ Subsequently sirolimus was used in steroid-sparing protocols, together with low-dose cyclosporine in a study that was terminated due to a higher incidence of PTLT.²⁹⁴ Conversion from calcineurin inhibitor to sirolimus to treat calcineurin inhibitor-induced nephrotoxicity has also been attempted with improved GFR in some children, but with an increase in adverse events.^{295,296} The use of sirolimus in pediatric kidney transplantation peaked in 2002 at 25%, but subsequently decreased to less than 1% in the United States,¹¹¹ with a similar trend seen also in other registries.¹⁸⁹

Everolimus

Everolimus has been studied in several small uncontrolled studies in children, generally instead of mycophenolate treatment and in combination with low-dose calcineurin inhibitors. A study based on data from the Cooperative European Paediatric Renal Transplant Initiative (CERTAIN) registry compared children treated with everolimus with low-dose cyclosporine and steroids with those on standard immunosuppression. There was no increase in biopsy-proven rejection and graft function was similar between the groups, although side-effect profiles differed. The everolimus group had a higher incidence of hyperlipidemia, HTN, and anemia than the control group.²⁹⁷ Further analysis showed a significant decrease in CMV replication and disease in the group treated with everolimus, most likely due to the avoidance of mycophenolate.²⁹⁸ Another study, with a similar treatment protocol with no control arm, also showed promising long-term results. To avoid wound healing problems, most protocols delay everolimus introduction until a few weeks after transplantation.²⁶⁵

OTHER IMMUNOSUPPRESSIVE MEDICATIONS

In most cases, new immunosuppressive agents are studied in adults before trials in children are performed. Belatacept, a costimulation inhibitor, has shown promise in adult renal

transplant patients, demonstrating improved GFR compared with cyclosporine, though with a higher incidence of acute rejection. No studies have been performed in children to date, and a specific concern might be higher frequency of PTLT in EBV-seronegative patients, particularly of the central nervous system. As more children are EBV naïve going into transplantation, the potential risk of PTLT could be even higher in this population.²⁹⁹ A potential advantage is the intermittent intravenous administration, which could help overcome drug nonadherence particularly in adolescent patients.

IMMUNOSUPPRESSION COMBINATIONS

In North America, the most common combination of immunosuppression for pediatric kidney transplantation in the past decade is tacrolimus, mycophenolate, and prednisone (42.3% at 3 years posttransplantation), followed by tacrolimus and mycophenolate without corticosteroids (22.5% at 3 years).¹¹¹ Other combinations include tacrolimus with low-dose prednisone either as dual therapy or with azathioprine. No difference was seen in the frequencies of the immunosuppression combinations between deceased versus living donor recipients.

PERIOPERATIVE MANAGEMENT

Pediatric renal transplantation should always be managed by an experienced team which includes pediatric experts in the fields of transplant surgery, nephrology, anesthesiology, intensive care, and other specialties, to ensure an optimal outcome.

Technical causes are responsible for a higher proportion of early graft loss in young children, primarily from graft thrombosis. Although the graft kidney is usually placed extraperitoneally in the iliac fossa in adults and older children, a midline intraperitoneal approach is often used in youngest patients. Vascular anastomoses of the graft in very small children are usually to the aorta and the IVC, rather than the iliac vessels used in older children and adults, to provide sufficient renal perfusion, particularly when the graft is an adult-sized kidney. The donor renal artery may be the same size as the recipient aorta and the renal vein may be significantly larger than the recipient's IVC. A straight and direct course for each renal vessel to the aorta and vena cava will avoid kinking and decrease the risk of thrombosis. Stenosis or thrombosis of the IVC is a rare finding but may be seen in children who have had CVCs in infancy, abdominal surgery, or a hypercoagulable state, posing a problem for graft anastomosis. This should be evaluated pretransplant by Doppler study or venography, and in case of significant IVC pathology an alternative vein should be selected. There may be an advantage for transplanting a pediatric (deceased) donor kidney in such cases.^{300,301}

The adult-sized kidney graft requires a major increase in CO to maintain good perfusion and avoid acute tubular necrosis or vascular thrombosis. Even with optimal maintenance of recipient intravascular volume, blood flow to the adult-sized donor kidney is significantly reduced at transplantation. Further reduction in blood flow could result in graft ischemia; therefore BP values, which may be relatively high

for the young child, and high-normal central venous pressure need to be maintained during transplant surgery and postoperatively. After anastomosis is completed, release of vascular clamps will cause immediate flow of a large portion of the child's blood volume into the graft, a critical time point when intravascular volume must be controlled. Maintaining relative hypervolemia immediately posttransplant may delay weaning from mechanical ventilation for 1 to 2 days, but is nonetheless of utmost importance to preserve graft perfusion.³⁰² Aggressive hydration to optimize intravascular volume and graft blood flow is important in the following weeks and months, and may require fluid supplements by the nasogastric route.³⁰⁰ Avoiding acute tubular necrosis in the postoperative period also enables early full dosing of calcineurin inhibitors, thus preventing early acute rejection.

IMMEDIATE POSTOPERATIVE PERIOD

In the first few days after transplantation the child is usually treated in the ICU. Close attention is paid to vital signs and fluid management with the goal of optimizing graft perfusion. Brisk urine output, particularly in a patient previously oliguric or anuric, is a useful marker of graft function. Oligoanuria or a significant decrease in urine output, by contrast, needs to be evaluated and treated immediately. Bladder drainage must be ensured by flushing or exchanging the bladder catheter. Ultrasound can assess the kidney, the blood vessels, ureter, and bladder, and provide information on organ perfusion, the presence of a fluid collection suggesting leak or bleeding. Absent blood flow may be due to thrombosis, and necessitate prompt surgical intervention, as do significant bleeding or a urinary leak. Intravenous fluids are given to replace insensible losses and urine output, as well as additional fluids to account for third space losses and to keep central venous pressure on the high side of normal. This approach should maintain good graft perfusion, which in turn will increase urine output. However, if urine output does not increase, intravascular volume overload may result, manifesting as respiratory compromise and HTN.³⁰² Volume overload may respond to diuretics, but dialysis may be necessary in some patients.

Immunosuppression is started at transplantation, including intravenous corticosteroids perioperatively in most cases (except for complete steroid-avoidance protocols). Initiation of calcineurin inhibitors may be delayed for several hours or longer, if there is postoperative oliguria. Prophylactic treatment for infections is started postoperatively. Perioperative antibiotics are usually given to prevent wound infection.³⁰³ *P. jirovecii* prophylaxis, usually with trimethoprim-sulfamethoxazole (TMP-SMX), may also decrease the incidence of urinary tract infection (UTI). Antiviral medication is given to prevent CMV disease, usually oral valganciclovir, with the duration of therapy depending upon pretransplant donor and recipient serologic status and on whether lymphocyte-depleting induction immunosuppression was given.³⁰⁴ Some centers prescribe antifungal prophylaxis, especially in patients with prolonged ICU stay or antibiotic treatment. Low-molecular-weight heparin or aspirin is often given after transplantation to prevent graft thrombosis, particularly in younger patients and in patients with abnormal pretransplant coagulation workup.³⁰⁵

EARLY COMPLICATIONS OF KIDNEY TRANSPLANTATION

DELAYED GRAFT FUNCTION

Delayed graft function (DGF) is usually defined as the need for dialysis during the first week after renal transplantation and is sometimes referred to as acute tubular necrosis.¹¹¹ The most common cause of DGF is postischemic acute injury of the graft, which is associated with deceased donor kidney, prolonged ischemic time, and hemodynamic compromise during and immediately after transplantation. Changes in allocation policies in the United States resulted in deceased donor organs being shipped over longer distances, leading to increased cold ischemic times, with an increase in DGF rates.³⁰⁶ The incidence of DGF is higher in children receiving kidneys from donors after cardiac death (DCD), although pediatric DCD kidney recipients have good renal allograft survival at 3-year follow-up.¹⁹⁸ DGF has been shown to adversely affect long-term graft survival and function, with children who required posttransplantation dialysis having a twofold risk of overall graft loss.³⁰⁷ DGF is also associated with a higher risk of subsequent acute rejection, in both living and deceased donor pediatric kidney transplantation.¹¹¹

Other causes of DGF include graft thrombosis, which can be diagnosed by ultrasound or radionuclide scan, and may require surgical revision. Thrombosis can be prevented by meticulous surgical technique and attention to fluid balance, as well as identifying and treating patients with thrombophilic risk factors prior to transplantation. Use of antibody induction may decrease the incidence of postischemic DGF, although the reasons for this are not clear.³⁰⁸

Urological complications such as urinary leak or obstruction may also cause DGF or acute decrease in urine output, and diagnosis is aided by ultrasound and radionuclide scan. Ureteral stents are often placed at transplantation for 4 to 6 weeks to prevent obstruction, but their use may be associated with increased risk of UTI.³⁰⁹

Hyperacute rejection or accelerated acute rejection superimposed upon postoperative acute tubular necrosis (ATN) may also present as DGF. Hyperacute rejection is usually due to preformed antidonor ABO or HLA antibodies, manifesting as irreversible rapid destruction of the graft within minutes to hours of transplantation. Because blood type and HLA matching have been routinely performed, these reactions are exceedingly rare. Pretreatment of recipients with grafts mismatched for ABO antigens or with DSA with desensitization protocols generally prevents this disastrous outcome.³¹⁰

Recurrence of FSGS can manifest as primary nonfunction, and requires a graft biopsy for diagnosis, in order to guide treatment.²¹³

ACUTE REJECTION

Acute renal allograft rejection is a common complication of pediatric renal transplantation, though its frequency has decreased significantly from over 70% in 1987 to 1990 to 16% since 2007.¹¹¹ In NAPRTCS, a rejection episode is defined by the physician's decision to initiate specific antirejection therapy, rather than BPAR. However, in the latest cohort the frequency of biopsy in reported acute rejections has increased

from 45% in 1987 to 94%. The trend away from clinical diagnosis of acute rejection to BPAR may also impact the overall frequency of reported acute rejection episodes. The probability of acute rejection 1 year after transplantation in children decreased from 52.8% in 1987 to 1991 to 9.3% between 2007 and 2013 for living donor kidneys. In patients receiving a kidney from a deceased donor the probability of acute rejection in the first year dropped from 67.4% to 13.8%.¹¹¹

The lower incidence of acute rejection in recent years is thought to be due to newer, more potent immunosuppressive medications now used. The highest risk of acute rejection is in the first few weeks to months; however, it may occur at any time, even years after transplantation.³¹¹ It is more frequent after deceased donor transplantation compared with living donor, suggesting susceptibility of the graft with longer ischemic time, for example, due to exposure of antigens promoting an immune reaction. Although in earlier patient cohorts risk factors for acute rejection included African-American race and no induction therapy, in the latest NAPRTCS cohort (2007–2013) these factors were no longer significant. This may be a result of differential practices in choice of immunosuppression protocol in African-American children.²⁶⁴ As noted earlier, DGF due to acute tubular necrosis continues to be a significant risk factor for subsequent acute rejection.¹¹¹ Combined liver–kidney transplantation may protect the kidney from acute cellular and antibody-mediated rejection and from decline in kidney graft function.³¹²

Acute rejection may often be clinically asymptomatic. Graft tenderness and fever as manifestations are unusual in the present era of immunosuppression, and acute decrease in GFR may be the only sign. GFR may decrease significantly before an increase in serum creatinine is noted, particularly when baseline creatinine is normal. A prompt workup should include urinalysis and urine culture to rule out infection and glomerular disease, BK virus (BKV) polymerase chain reaction (PCR), and ultrasound to look for obstruction, fluid collection, or vascular compromise. Biopsy should be performed whenever possible to confirm a diagnosis of acute rejection, as other causes of graft dysfunction, such as calcineurin inhibitor toxicity or viral infection including BK nephropathy, may be difficult to distinguish from acute rejection on clinical grounds alone. Graft biopsy is safely performed under ultrasound guidance, and light general anesthesia or conscious sedation is helpful in reducing anxiety and pain. The Banff criteria are utilized to grade acute rejection.³¹³ Acute cellular rejection is characterized by lymphocyte infiltration of tubules (tubulitis) and interstitial inflammation, with involvement of arterial walls in more severe cases. In acute antibody-mediated rejection (ABMR) there is acute tissue injury including findings of microvascular inflammation and often also acute tubular injury, arteritis, or fibrin thrombi in glomerular capillaries. In addition, there is evidence of antibody-mediated complement activation – C4d deposition in the peritubular capillaries of the allograft, although C4d-negative ABMR is recognized. The biopsy findings together with circulating DSAs complete the diagnosis of ABMR. The two types of acute rejection may coexist.³¹³ The question of whether to perform routine surveillance graft biopsies to help tailor immunosuppressive therapy and detect subclinical rejection is unresolved.³¹⁴ Serial monitoring

of DSA may help identify high-risk patients, adjust immunosuppression, and prevent overt antibody-mediated injury.³¹⁵

The cornerstone of therapy for acute cellular rejection is high-dose corticosteroids, most often pulse intravenous methylprednisolone. In steroid-resistant rejection, lymphocyte-depleting agents, particularly thymoglobulin, may be administered. Premedication with methylprednisolone and antihistamine usually prevents a cytokine release syndrome, which may be seen particularly with the first dose of thymoglobulin. Approximately 50% to 60% of children treated for acute rejection will have a complete response, with GFR returning to baseline, and another 30% to 40% will have a partial response. Complete reversal of acute rejection is less likely in repeated episodes.¹¹¹ When acute rejection is diagnosed, the possibility of underimmunosuppression must be addressed. This may be due to underdosing of medications, or more commonly, particularly in teenage patients, nonadherence to immunosuppressive drugs.²⁰⁶ Attention must be given to maintenance immunosuppression, considering adding prednisone to a steroid-free protocol, switching cyclosporine to tacrolimus, or increasing the antimetabolite dose or the target levels of tacrolimus, in order to improve response to therapy and reduce the incidence of recurrent acute rejection.³¹⁶

ABMR is often refractory to conventional antirejection therapy and has a high rate of graft loss. Few randomized controlled studies have been performed regarding treatment of ABMR, in adults or in children, and there is insufficient evidence to guide therapy.^{317,318} The first line of treatment usually includes high-dose corticosteroids, plasma exchange, and IVIG. The goal of plasma exchange and IVIG is removal of DSA from the circulation.³¹⁹ Second-line therapies target antibody production by depleting B cells (rituximab) or plasma cells (bortezomib). Recent studies have shown stabilization of graft function in children with refractory ABMR after treatment with bortezomib.^{320,321} Inhibition of complement with eculizumab to decrease the antibody-mediated injury has been described in a small number of pediatric patients.³²² Less severe cases of ABMR, for example, the presence of DSA with mild findings of ABMR on graft biopsy, but without significant deterioration of graft function, might be treated with corticosteroids and IVIG with or without rituximab. This approach avoids the need for CVC insertion and plasma exchange in lower-risk patients.³²⁰

Acute rejection is the most common cause of graft loss in the first year after transplantation and has a significant negative impact on long-term graft survival.^{199,311,323} Close monitoring is imperative in order to avoid early acute rejection and enable its prompt detection and treatment. Serum creatinine is monitored daily in the first few days after transplantation, followed by at least twice weekly for 2 to 3 months, together with drug concentration monitoring. Later, if there are no complications, the frequency of testing is decreased to once a month after the first 6 to 12 months. DSA should be monitored regularly during the first months after transplantation, and periodically after that. This is especially important in children with pretransplantation sensitization, or at any time if there are signs of graft dysfunction. The appearance of de novo DSA, most frequently DQ antibodies, is a risk factor for subsequent ABMR, which could be prevented by timely intervention.³²⁴ Even stable children and adolescents require regular follow-up at the pediatric

transplant clinic, to help prevent nonadherence and the resulting risk of graft rejection, as well as long-term management of the many other aspects of their health.

INFECTIONS

As more potent immunosuppressive agents were introduced into the treatment of renal transplantation, the frequency of acute rejection decreased, as did hospitalizations for rejection episodes. In the pediatric population in particular, the price for this has been a higher rate of infections and hospitalization for infection.³²⁵ The use of antibody induction and the presence of indwelling vascular or urinary catheters and the presence of posttransplantation neutropenia increase the risk of infection.²⁸⁵ Hospitalization for infection was higher in children after renal transplantation at 47% in the first 3 years, than in adults.³²⁶ In the immediate posttransplant period, the most frequent infections are surgical, UTIs, and pneumonia. Subsequently, in the early months after transplantation viral infections are common, particularly CMV and EBV and BKV nephropathy, which cause significant morbidity and may adversely affect graft function. In pediatric kidney transplant recipients hospitalized for fever, the rate of bacterial illness is high (50%) and warrants a full workup, with consideration for empiric antibiotic treatment.³²⁷

Urinary Tract Infection

UTI is a common complication in children who have undergone renal transplantation, with hospitalization of 16.8% of patients in the first 3 years.³²⁶ The risk of UTI after pediatric kidney transplant has increased over the last 2 decades.³²⁵ Risk factors include a pretransplant diagnosis of obstructive uropathy or VUR and bladder abnormalities.³²⁸ Ureteral stents placed during transplantation may also increase the risk of UTI.³⁰⁹ Although graft dysfunction may occur during allograft pyelonephritis, long-term graft function may not be affected.³²⁹ TMP-SMX prophylaxis for *P. jiroveci* infection may reduce the incidence of UTI, but bacterial susceptibilities vary between medical centers and may render this strategy ineffective. Many clinics routinely culture urine from transplant patients at every visit; however, the indication for treatment of asymptomatic bacteriuria without pyuria remains unclear, as resistant organisms may emerge.³³⁰ Recurrent UTI after kidney transplantation may be associated with VUR, which should be addressed (see the “Urological Complications” section).

Cytomegalovirus

CMV infection in pediatric kidney transplantation has decreased significantly since the widespread use of antiviral prophylactic treatment, yet it remains an important and occasionally severe complication. Up to 20% of pediatric kidney recipients in the current era will develop CMV infection (defined as evidence of viral replication) and 3% to 10% have CMV disease (viral replication with clinical disease).³³¹

The most vulnerable group is children who are seronegative for CMV pretransplant, receiving a kidney from a seropositive donor (D+/R-), who develop a primary infection. Primary infection may also be community acquired in children whose donor was seronegative for CMV.³³¹ Children who receive thymoglobulin induction are also at a higher risk of CMV

infection.³³² According to OPTN data, over half of children undergoing kidney transplantation in 2011 to 2015 had negative CMV serology, whereas only a quarter of adults were CMV negative at transplantation.¹⁸⁸ Secondary infection may also occur, either as reactivation of the host strain or as infection from the donor strain, and may be less severe clinically. The risk of acute graft rejection is higher following CMV infection in kidney transplant recipients and may be associated with increased graft loss.³³³ Decline in kidney graft function was more pronounced in children with CMV replication.²⁹⁸

Valganciclovir has been shown to be at least as effective as oral ganciclovir in children for CMV prevention, and has better bioavailability and easier once-daily dosing.³³⁴ Dosing should be based on BSA and corrected for renal function.³³⁵ There are two CMV prevention strategies: prophylaxis and preemptive antiviral treatment. In adults, the incidence of CMV infection and disease is lower in patients receiving prophylaxis; however, there is no significant difference in graft outcome or death between the strategies.³³⁶ The proportion of CMV-naïve children who undergo transplantation is high, rendering them at higher risk for CMV disease. Most centers use prophylaxis for at least the first few weeks, to reduce viral replication within the incoming CMV-positive graft. Most prophylaxis protocols are 3 to 6 months long, depending on immunosuppression regimen and patient serostatus, but the optimal length of prophylaxis is unknown.³³¹ Viral monitoring is important during prophylaxis and particularly in patients being managed preemptively, or after valganciclovir discontinuation.³³⁷ Antiviral prophylaxis should also be considered when there is intensification of immunosuppression, for example, when treating acute graft rejection. Adverse effects of valganciclovir include hematological toxicity—mainly leukopenia and neutropenia, as well as gastrointestinal discomfort or diarrhea, occasionally necessitating dose reduction or discontinuation of treatment.³³⁸

CMV infection is most common after cessation of prophylactic antiviral therapy. Asymptomatic CMV infection should be treated with intravenous ganciclovir; however, in stable older children and adolescents, oral valganciclovir can be used as initial therapy.³³¹ CMV disease is defined as viremia with clinical symptoms, most commonly fever and malaise, with tissue-invasive disease in some cases, involving the respiratory, hepatic, and gastrointestinal tracts. Allograft dysfunction as well as leukopenia and thrombocytopenia are frequently observed laboratory findings. In invasive disease, treatment is usually started with intravenous ganciclovir, followed by oral valganciclovir, when clinical improvement and reduction in viral load are seen. Although the VICTOR study demonstrated equivalent efficacy of oral valganciclovir to IV ganciclovir in adult transplant recipients, it is not clear if these findings can be extrapolated to the pediatric population. To assist in viral eradication, immunosuppression should be reduced, usually by discontinuation of antimetabolite drug – azathioprine or mycophenolate. Hyperimmune CMV globulin preparation may be given to the most severe cases, though its efficacy is questionable.³³¹ Treatment is continued until after viral eradication is demonstrated, followed by a prophylactic dose for several weeks. Relapse may occur after end of therapy; therefore viral monitoring should be continued for several weeks after treatment cessation, and after reintroduction of the antimetabolite, which is usually at a

lower dose than before CMV infection. Failure to respond to therapy, either clinically or virologically, may be due to viral ganciclovir resistance usually due to viral UL97 mutations. Treatment options include high-dose ganciclovir or switching to foscarnet, which is nephrotoxic, though studies in children are lacking.³³⁹

EPSTEIN-BARR VIRUS INFECTION

Similar to the situation with CMV serology, about half of the pediatric patients undergoing renal transplantation are seronegative for EBV, whereas only 10% of donors are EBV naïve.¹⁸⁸ Seronegative patients receiving grafts from EBV-positive patients are at a higher risk of infection, therefore EBV disease is more common in younger children. EBV infection may be asymptomatic, it can present as an acute viral syndrome, or may be associated with PTLTLD. The risk of developing PTLTLD is higher when primary EBV infection occurs after transplantation. Children who are EBV negative prior to transplantation have a 2.9% incidence of PTLTLD in the first 5 years after transplantation, compared with 0.7% in children who are seropositive.¹⁸⁸ Of the patients with primary EBV infection, adolescents may be more likely to develop PTLTLD and have a poorer prognosis.³⁴⁰

Monitoring viral load in peripheral blood by quantitative PCR, particularly in high-risk patients in the first year after transplantation, is important in order to identify infection and allow early intervention. A prospective study in Germany found that primary infection occurred in 63% of seronegative pediatric patients and reactivation or reinfection in 44% of seropositive patients in the first year, though many of the infections were asymptomatic.³⁴¹ More intensive immunosuppression is associated with a higher rate of symptomatic EBV infection. Prophylaxis with ganciclovir or valganciclovir in high-risk patients was shown to decrease the incidence of EBV, according to some but not all studies.^{342–344} Clinical disease usually presents with protracted fever, lymphadenopathy, tonsillitis, and hepatosplenomegaly. Laboratory findings may include leukopenia, atypical lymphocytes, elevated liver enzymes, and IgM antibodies for EBV, and allograft dysfunction may occur. Diagnosis is confirmed by EBV viral load, which is also used to monitor the course of the disease and response to therapy.

EBV-associated PTLTLD appears to be due to B-cell proliferation induced by infection with EBV. There is continuum between EBV-associated diseases from mononucleosis to full-blown monomorphic lymphoma. Imaging will help in detecting lymphoid masses, which might require biopsy to rule out malignant PTLTLD. Other than chest X-ray and abdominal sonography, computed tomography and positron emission tomography can assist in staging of the disease and can guide biopsy.³⁴⁵ The cornerstone of treatment of EBV infection and prevention of PTLTLD is immunosuppression reduction. Strategies include discontinuation of the antimetabolite and lowering the target levels of tacrolimus, or switching to mTOR inhibitors. Antiviral therapy for persistent EBV viremia with valganciclovir has not been proven to be effective.³⁴⁴ Some children after renal transplantation may have persistent high EBV viral load without ever developing PTLTLD.³⁴²

BK VIRUS INFECTION

BK polyomavirus infection emerged as an important clinical entity in patients undergoing renal transplantation in the

late 1990s, coinciding with the introduction of more potent immunosuppression regimens. BKV infection may result in BKV-associated nephropathy (BKVAN), manifesting as tubulointerstitial nephritis with graft dysfunction and potential graft loss, as well as ureteral stenosis and hemorrhagic cystitis.³⁴⁶ Although previously thought to be rare in children, recent data from NAPRTCS showed a prevalence of 4.6%, similar to that in adults.³⁴⁷ Some recent studies have shown a higher incidence (10%–27%) of pediatric kidney transplant patients with BK viremia detected by screening tests, though overt nephropathy is less common.^{348,349} BK transmission occurs in childhood, when the virus remains latent in the tubular epithelial cells. Immunosuppression after transplantation may cause reactivation and high-rate viral replication in the kidney graft and is therefore considered a consequence of immunosuppression. Young recipients may develop primary BK infection transmitted from the donor kidney. Risk factors for BKVAN include antibody induction, DR mismatch, or treatment for rejection.

The virus appears first in the urine, and if progression occurs, is detected in the blood by quantitative PCR for viral DNA. Histological changes are observed in patients with persistent or increasing plasma viral loads.³⁴⁶ Surveillance is recommended to enable early diagnosis and intervention. Detection of viruria has questionable clinical implications, as it is not correlated with graft damage, but does identify patients at risk for subsequent viremia. Monitoring plasma quantitative PCR periodically, particularly in the early months after transplantation, after acute rejection, and at any time when there is an unexplained elevation of serum creatinine, is the accepted strategy for diagnosis in children.^{350,351} Renal biopsy is performed if there is graft dysfunction. Typical findings include cytopathic changes and tubulointerstitial nephritis, which can be difficult to differentiate from other viral infections or acute cellular rejection, and may even coexist with other conditions in some cases. Immunohistochemical staining for SV40 viral antigen is highly specific for BKV nephropathy.³⁴⁶ Relevant histological findings on kidney biopsy in association with high BK viral load in the blood are diagnostic.

The cornerstone of management of BKV infection is reduction of immunosuppression when significant BK viremia is detected. This is usually done by discontinuation of the antimetabolite or decrease in the target levels of the calcineurin inhibitor, with the goal of prevention of progression to overt nephropathy. The reduction of immunosuppression enables the patient's immune system to clear viremia more effectively by expanding BKV-specific cell immunity and specific immunoglobulins.³⁴⁷ Treatment of established BKVAN is often less successful. Several medications with antiviral activity have been used to treat patients with BKVAN unresponsive to immunosuppression minimization, with varying degrees of success, but no randomized controlled studies have been performed.³⁴⁶ These include high-dose IVIG, quinolone antibiotics, cidofovir, and leflunomide.^{349,351–353}

VARICELLA

Immunization against varicella-zoster virus is recommended before transplantation, as the use of live attenuated viral vaccine is controversial in immunocompromised patients.²¹² Loss of antibody response after transplantation has been

reported in 25% of children who were vaccinated before transplantation.²¹¹ In addition, these antibodies may have lower avidity and the cellular immune response may also be decreased in solid organ transplant recipients compared with healthy controls.³⁵⁴ Varicella infection in an unimmunized renal transplantation recipient may result in severe disease, with visceral involvement in some cases. Postexposure prophylaxis of unimmunized patients with varicella-zoster immunoglobulin should be administered as early as possible, and if clinical disease develops, treatment with intravenous acyclovir is initiated. Temporary discontinuation of the antimetabolite, or dose reduction, is usually recommended.³⁵⁵

OTHER INFECTIONS

Respiratory Viruses

Respiratory viruses, in particular influenza virus infection, may be associated with severe morbidity in the immunosuppressed host. Disease severity may be related to the degree of immunosuppression and clinical presentation may be atypical. Diagnosis of presumed respiratory viral infection can be made by nasopharyngeal swab, wash, or aspirate testing for pathogens. Neuraminidase inhibitors such as oral oseltamivir can be started empirically for suspected influenza infection before virological confirmation. Pretransplant vaccination is recommended as well as yearly seasonal inactivated influenza vaccine after transplantation. Young children (<9 years) receiving the vaccine for the first time should receive two doses, 4 weeks apart. Vaccine immunogenicity is variable, but recent data suggest that immunization reduces mortality and clinical severity as well as infection-related acute rejection.³⁵⁶

Cutaneous Warts

Cutaneous warts, usually due to infection with human papillomavirus, are common in children after renal transplantation, as are molluscum contagiosum lesions. Although malignant potential is low, these lesions can be numerous and cause distress due to cosmetic issues. Warts may be refractory to treatment or recurrent and may respond to adjustment of the immunosuppressive regimen if possible.³⁵⁷ Topical treatment is often ineffective and surgical removal may be required.

Pneumocystis jirovecii Pneumonia

P. jirovecii pneumonia is a classic example of an opportunistic infection, which is very rare in the immunocompetent host. With the widespread use of prophylaxis, usually with TMP-SMX, the true incidence of posttransplantation infection is unknown. Intravenous pentamidine can be given if TMP-SMX is contraindicated.³⁵⁸ Antipneumocystis prophylaxis is usually given for 6–12 months after kidney transplantation, after which net immunosuppression is lower. *Pneumocystis* infection may coexist with CMV or respiratory viruses, and should be considered in the immunosuppressed child with unexplained respiratory symptoms.³⁵⁹

Infectious Diarrhea

Diarrhea is a common symptom experienced by children after kidney transplantation. In some cases it can be attributed to medication such as mycophenolate or valganciclovir,

is usually mild, and resolves with a decrease in dose or change of formulation.³³⁸ Rarely, mycophenolate may cause severe colitis, which improves with discontinuation of the drug.³⁶⁰ In up to 60% of children with diarrhea, an infectious cause can be diagnosed, whether viral, bacterial, or parasitic.

Although in healthy children, diarrhea due to infection with the parasite cryptosporidium is common, it is almost always a mild self-limiting disease. In immunosuppressed patients, diarrhea may be protracted and can cause hypovolemia, perturbations in calcineurin inhibitor serum concentrations, and AKI. Treatment with nitazoxanide is often necessary as well as supportive measures.^{361,362}

Organ transplant recipients can also have persistent norovirus infection. Chronic infection with viral shedding may continue for several months.³⁶³ In many cases hospitalization for fluid therapy is necessary, and 30% still have diarrhea after 2 weeks.³⁶⁴

RECURRENCE OF PRIMARY RENAL DISEASE

According to NAPRTCS data, 7.8% of grafts in children are lost to disease recurrence.¹⁸⁸ The most common recurring kidney disease is FSGS, but MPGN (type II in particular), aHUS, PH1, and others can recur in the graft. A study from the ANZDATA database demonstrated that younger patients have a higher risk of recurrence of glomerular disease after transplantation, in comparison to older adults.³⁶⁵ It is important to identify patients at risk for recurrence to enable early diagnosis and therapeutic intervention (see also in “[Recipient Evaluation](#)” section above).

FSGS is responsible for approximately 10% of ESRD in children in the United States, presenting as SRNS, and carries an overall risk of 30% for recurrent nephrotic syndrome.¹¹¹ Native kidney nephrectomy before or during transplantation surgery, in children with persistent proteinuria, may enable prompt diagnosis of recurrence. Clinical presentation may include severe proteinuria immediately after transplantation or later in the course, but DGF or anuria may also be a sign of disease recurrence. Humoral factors cause damage to the glomerular barrier, which is the rationale for treatment strategies aimed at removal of such a factor. Serum soluble urokinase-type plasminogen activator receptor has been identified as a possible permeability factor responsible for recurrent FSGS, though this has not been supported in pediatric studies.³⁶⁶ Preemptive and therapeutic plasmapheresis for disease recurrence, as well as treatment with anti-CD-20 antibody rituximab, have shown some efficacy in achieving remission of proteinuria.^{221,367} Recurrent FSGS in the graft, particularly if unresponsive to therapy, is associated with a high rate of graft failure.

MPGN is the underlying diagnosis in 1.7% of children receiving their initial kidney transplant, may recur in 20% to 45% after transplantation, and is a cause of graft failure in many of the patients.³⁶⁸ When studying MPGN type I and type II separately, the risk of 5-year graft loss was 23.5% for MPGN type I and 67.5% among patients with type II, but the number of children with MPGN type II was very small.²²⁵ The newer classification of MPGN is based on glomerular injury mechanism rather than the location of and appearance of deposits. Under this system, most patients with MPGN had biopsies consistent with immune-complex-mediated

glomerulonephritis, characterized by the presence of immune complexes and complement components, and only 12% had complement-mediated glomerulonephritis in the absence of immune complexes.³⁶⁹ A recent study in adults, including some teenaged patients, showed a recurrence rate of 45% overall in MPGN patients, 55% of whom recurred in the first year after transplantation. Risk factors for recurrence included low complement level and the presence of monoclonal gammopathy, but not the subtype of the disease.³⁶⁹ Treatment with plasmapheresis, rituximab (anti-CD20), and corticosteroids has been used to treat MPGN recurrence without clear evidence of benefit.³⁶⁸

aHUS may recur in up to 60% of patients undergoing kidney transplantation without specific intervention. Most will progress to graft loss unless promptly treated with plasmapheresis or complement inhibition.³⁷⁰ Eculizumab, a monoclonal antibody of complement factor 5, inhibits terminal complement activation and was found to be effective in the treatment of aHUS.²²⁷ It has become the treatment of choice both to prevent and to treat recurrence of aHUS after transplantation with excellent results, as shown in a case series from France. Three patients in whom treatment was discontinued experienced a relapse, suggesting that long-term therapy is needed.²²⁷ Patients treated with complement inhibitors are at risk for meningococcal infection and preventive measures must be taken.³⁷⁰

In PH1 treatment of choice is combined or sequential liver–kidney transplantation. However, children who have been treated with dialysis for a long period prior to transplantation may have significant oxalate deposition, mostly in the bones. Despite the discontinuation of oxalate production, achieved by liver transplantation, mobilization of the systemic oxalate burden can cause damage to the kidney graft. Meticulous attention to hydration, use of crystallization inhibitors such as oral citrate, or even HD immediately after transplantation can help protect the new kidney.³⁷¹

Other diseases that may recur and may cause graft dysfunction or failure include lupus nephritis, IgA nephropathy, Henoch–Schönlein nephritis, membranous nephropathy, and diabetic nephropathy.

CHRONIC ALLOGRAFT NEPHROPATHY

The most common cause of graft loss after the first year is CAN, accounting for about 35% of failed kidney grafts, according to several pediatric databases worldwide.^{111,189} The term CAN replaces previously used terms, including chronic rejection, transplant nephropathy, and chronic renal allograft dysfunction or injury. It is primarily a histopathological diagnosis, used to denote features of chronic interstitial fibrosis and tubular atrophy within the renal allograft without evidence of any specific etiology. Pathological involvement also typically includes the blood vessels with arteriolar hyalinosis and fibrointimal hyperplasia, and chronic transplant glomerulopathy.³⁷² The pathogenesis is not completely understood, but numerous etiological factors have been implicated, both immune and nonimmune. Clinically, CAN is suspected when there is a slow deterioration in allograft function beyond the first 3 months after transplantation, with no evidence of acute rejection, drug toxicity, infection, or other pathological process. There may also be increasing

proteinuria and worsening HTN. The prevention and management of chronic renal allograft rejection remains one of the major challenges facing clinicians. Adequate long-term immunosuppression is important in prevention of immunologically mediated CAN, although chronic calcineurin inhibitor toxicity may be difficult to distinguish from CAN in some cases. Nonadherence to medical therapy in pediatric renal transplant recipients is associated with a higher rate of interstitial fibrosis and tubular atrophy.³⁷³ Changes in immunosuppression may occasionally ameliorate CAN, for example, treatment with mycophenolate in children with CAN may improve graft function.²⁷⁹ Optimal control of HTN, with possible preference for angiotensin-converting enzyme (ACE) inhibitors, and treatment of hyperlipidemia are also important. One future direction is clinical utilization of urinary biomarkers to help differentiate CAN from other graft injury phenotypes.³⁷⁴

PATIENT SURVIVAL

Survival of children after renal transplantation has improved in 2005 to 2013 compared with 1987 to 1996, and is 96.4% for recipients of deceased donor allografts and 97.4% for recipients of living donor allografts, at 5 years. Survival is lower in infants younger than 24 months at transplantation; however, this group of patients is small, and has also shown the most significant improvement in the past 2 decades.¹¹¹ The most common cause of death is infection (28.5%), followed by cardiopulmonary causes (14.6%) and malignancy (11.5%). Overall, death was more likely in patients with failed grafts, though 48.2% of patients who died had a functioning allograft. The USRDS 2017 annual report found a 30.8% decrease in adjusted 1-year all-cause mortality rate when comparing the 2005 to 2009 and 2010 to 2014 periods. Survival after renal transplantation is higher than in children remaining on chronic dialysis. Adjusted all-cause mortality is approximately fourfold higher in children on dialysis than in children after renal transplantation.²²

GRAFT SURVIVAL

Graft survival has improved greatly over the past 3 decades. Five-year deceased donor graft survival in the 2005 to 2013 cohort is 83%, compared with 78.1% in the cohort of 1996 to 2004 and only 62.4% in patients transplanted in 1986 to 1995. In living donor kidney allografts, an increase in 5-year graft survival was seen up until 2004, after which the 5-year graft survival stabilized at 83% to 85%, closing the gap with deceased donor kidney transplantation.¹¹¹ Reports from several developing countries that have a pediatric kidney transplantation program (some with living donor option only) have shown 5-year graft survival of 65% to 92%.³⁷⁵ Graft failure was defined by return to dialysis, retransplantation, or death. When assessing graft loss from January 2000 onward, the most common cause of graft failure was CAN or chronic rejection (40.5%), with various types of acute rejection responsible for 10.2% of graft loss. Disease recurrence was the cause of graft loss in 7.8% of pediatric patients, with FSGS accounting for almost half of these. In comparison to adults, death with functioning graft was less common (8.4%),

whereas vascular thrombosis or technical difficulties were more frequent (6.6%).¹¹¹ This is due to better general health in pediatric kidney recipients on the one hand, and smaller blood vessels which make vascular anastomoses technically challenging on the other.

Prognostic factors associated with shorter graft survival include race, prior transplantations, HLA mismatching, and blood transfusions. Patients receiving their graft at age 14 to 16 years were at risk for graft failure, in particular if they were African-American or did not have private insurance.³⁷⁶ It is difficult to evaluate the association of antibody induction therapy with graft outcome, due to patient selection bias. DGF (requiring dialysis) and slow-to-recover graft function are associated with a 20% reduction in 5-year graft survival in living and deceased donor kidney recipients.^{111,231}

Glomerular filtration rate, estimated by serum creatinine-based formula, has improved over the last decade: 74.7% of pediatric patients who received a kidney transplant in 2015 had an eGFR greater than 60 mL/min/1.73 m² at 1 year.¹¹¹

Irreversible graft failure may be associated with an inflammatory reaction, which may be severe enough to necessitate graft nephrectomy. This is more common in early graft loss or high-grade rejection by biopsy. The impact of graft nephrectomy on alloimmunization is difficult to determine due to multiple potential confounding factors.³⁷⁷

LONG-TERM FOLLOW-UP OF CHILDREN WITH KIDNEY TRANSPLANT

HYPERTENSION

BP monitoring is an integral part of regular follow-up of pediatric kidney transplant recipients. Appropriate size cuff should be used, and at least two or three measurements should be performed when the child is calm. BP measurements are compared with normal value for gender, age, and height percentile, as published in the fourth report on the diagnosis, evaluation, and treatment of high BP in children and adolescents, which provides 90th and 95th percentiles.³⁷⁸ The vast majority of children reaching ESRD have HTN, defined as values above the 95th percentile, and it remains a problem for many patients after kidney transplantation. In a European study reporting data from 10 countries, about two thirds of children receiving RRT had HTN, but uncontrolled HTN was more common in children on dialysis than in transplanted patients.⁵³

In the early period after transplantation HTN is common, due in part to volume overload, early graft dysfunction, higher doses of corticosteroids and calcineurin inhibitors, or kinking of the transplant renal artery. HTN may appear months or even years after transplantation and is more common in older recipients. In a study from the Netherlands, the prevalence of uncontrolled HTN, defined according to Kidney Disease Improving Global Outcomes guidelines, was 75.8% in adolescents and young adults compared with 38.3% in younger children, and was not found to be affected by transition to adult care facilities.³⁷⁹ Ambulatory 24-hour BP monitoring is a useful tool in diagnosing masked or nocturnal HTN, which are phenomena frequently seen in children with kidney transplantation.³⁸⁰

Any injury of the allograft, such as acute rejection or recurrence of glomerular disease, can cause or worsen HTN. CAN, like CKD of any cause, is also strongly associated with HTN. Increased prevalence of HTN is seen with use of calcineurin inhibitors, particularly cyclosporine, but also tacrolimus.³⁸¹ The most common etiology of secondary HTN after transplantation is transplant renal artery stenosis. This entity may present as new-onset resistant HTN, occasionally with hypokalemia or worsening allograft function. A bruit may be heard over the kidney allograft and duplex ultrasonography often detects the stenosis, although computed tomography, MRI, or angiography may be needed for diagnosis. Treatment of transplant renal artery stenosis depends on the severity and response to medication; however, angioplasty with or without stenting is the definitive treatment.³⁸¹

Although posttransplantation HTN may be partly due to native kidneys, it often improves with a well-functioning graft and only rarely warrants native nephrectomy. Obesity, which predates transplantation or develops subsequently, is also associated with HTN and metabolic syndrome.³⁸²

Treatment of posttransplantation HTN involves lifestyle modification including exercise, reducing salt intake, weight control, and smoking avoidance. Antihypertensive medication is prescribed as necessary, with target BP measurements under the 90th percentile for age and height, as in other high-risk pediatric groups.³⁸⁰ Choice of therapeutic agent depends on the underlying cause; ACE inhibitors and angiotensin receptor blockers may be appropriate in patients with HTN, particularly associated with proteinuria, but possibly hazardous in uncorrected allograft artery stenosis. Diuretics are helpful in treatment of early volume overload-associated HTN, but are not ideal agents later, as they induce a state of relative volume depletion. Calcium channel blockers may minimize calcineurin inhibitor-related renal vasoconstriction.³⁸⁰ Uncontrolled HTN may adversely affect graft function, as shown in adults in the Collaborative Transplant Study, which demonstrated that lowering systolic BP was associated with improved graft function.³⁸³

CARDIOVASCULAR DISEASE

LVH is the most frequent cardiac abnormality in children after renal transplantation, with a prevalence of 10% to 50% in various studies.^{384–386} LVH is correlated with HTN, and also with the presence of metabolic syndrome.³⁸⁷ In most children with LVH it appears prior to transplantation, and some studies have shown improvement in left ventricular mass index after transplantation.²¹⁸

Cardiovascular mortality is much higher in children with CKD compared with the general pediatric population. Although mortality is lower in children after kidney transplantation than in children on dialysis, it remains unacceptably high. According to the NAPRTCS database, 14.6% of mortality is due to cardiopulmonary causes, second only to infection, and ahead of malignancy.¹¹¹ Traditional risk factors such as HTN, dyslipidemia, diabetes mellitus, and obesity are more frequent due to impaired renal function as well as the adverse effects of immunosuppressive medications. Nontraditional risk factors such as abnormalities in calcium-phosphorus metabolism, anemia, hyperuricemia, and hyperhomocysteinemia are also common in children with CKD and are correlated with arterial stiffness and carotid

intimal-medial thickness.⁵⁴ Some of these risk factors may persist after transplantation, particularly if graft function is impaired. While vascular lesions progress in children with CKD, kidney transplantation can bring about improvement in some measures of endothelial function. However, cIMT may remain abnormal even years after kidney function is restored.^{388,389}

Interventions to reduce premature cardiovascular disease for young patients after renal transplantation must include lifestyle modification. Dietary interventions should be offered to all children, in particular to those with obesity and dyslipidemia. Encouraging physical activity is important, as is education to prevent smoking. Adherence to a healthy diet and physical activity is not straightforward, especially in adolescents. The Active Care after Transplantation (ACT) trial is investigating the effectiveness of an exercise intervention, or an exercise + diet intervention on physical functioning, weight gain, and cardiometabolic health in adults after transplantation.³⁹⁰ Similar studies could be helpful in determining the role of lifestyle modification in children and adolescents.

Optimization of graft function and BP remain major factors in cardiovascular risk after kidney transplantation. One of the main goals of steroid avoidance or withdrawal protocols is mitigating risk factors for cardiovascular disease. Several studies in children have shown decrease in the frequency of posttransplant diabetes mellitus, hyperlipidemia, and HTN in children on steroid-free protocols compared with triple immunosuppression.^{289,391} Children treated with tacrolimus were less hypertensive and had better lipid profiles than those on cyclosporine-based protocols.^{271,392} Treatment with β -hydroxy β -methylglutaryl-coenzyme A (HMG-CoA) reductase inhibitors has been shown to be effective and safe in reducing total and LDL-cholesterol in children.³⁹³ Although the ALERT study showed a reduction in cardiac deaths and nonfatal myocardial infarctions in adults, this has not been shown in other studies.³⁹⁴ In children and adolescents, movement between study groups of CKD, dialysis, and transplantation is rapid and makes long-term study of statins challenging. Currently, there are no pediatric data on posttransplantation statin therapy with endpoints of cardiovascular events; therefore, except in special circumstances, statin treatment is not recommended.³⁹⁵

METABOLIC SYNDROME AND DIABETES MELLITUS

Metabolic syndrome is less well defined in children and adolescents than in adults, but is based on the finding of at least three abnormalities such as obesity, elevated BP, low-HDL cholesterol, hypertriglyceridemia, and impaired glucose tolerance. It is increasingly common and its frequency correlates with age. In the United States, prevalence is estimated at 1.2% in 12-year-olds and up to 7.1% in 16- to 17-year-olds.³⁹⁶ Screening and prevention of metabolic syndrome in childhood is important, as it frequently continues into adulthood and carries an increased risk for cardiovascular disease, and even in childhood, there is evidence of significant atherosclerotic changes. The frequency of metabolic syndrome in children in the United States after kidney transplantation is high, up to 37% in the first year, most of which is of new onset.³⁸⁷ Dietary intervention and physical activity

are the mainstay of prevention and treatment. In patients treated with corticosteroids, late steroid-withdrawal has been shown to improve metabolic parameters and decrease the frequency of metabolic syndrome from 39% to 6% 2 years later.²⁸⁹

New-onset diabetes mellitus after transplant (NODAT) is less common in children than in adult patients, with an incidence of 7.1% at 3 years. However, its frequency may be increasing. It is associated with older age (adolescents vs. younger children), higher BMI, and treatment with tacrolimus.³⁹⁷ A study comparing early corticosteroid withdrawal with low-dose chronic prednisone treatment in adults did not show a significantly lower incidence of NODAT.³⁹⁸ Hypomagnesemia has emerged recently as a significant risk factor for NODAT in adults and in children. Although tacrolimus is known to cause renal magnesium wasting and is also a risk factor for NODAT, it appears that hypomagnesemia is independently associated with NODAT.³⁹⁹ Further studies are necessary to elucidate the effect of magnesium supplementation. Fasting blood glucose should be monitored regularly in the first months after transplantation; hemoglobin A1C is also helpful in diagnosing insidious-onset diabetes. When diabetes is diagnosed, optimizing tacrolimus levels and minimizing corticosteroid dose should be considered. Referral to a pediatric diabetes specialist clinic for patient and parent education and a combination of dietary modification and pharmacological therapy are needed. NODAT is not associated with decreased graft or patient survival in pediatric patients, and some cases may resolve spontaneously.³²³

CANCER

Immunosuppressive therapy for prevention of rejection after solid organ transplantation carries a risk for malignancy. In children, 2.6% of renal transplants develop a malignancy, of which 85% were classified as lymphoproliferative cancers.¹¹¹ The 3-year incidence of malignancy increased from 1% in 1987 to 1991 to 2.96% in 1997 to 2001; however, in the latest cohort an improvement has been noted: 1.78% in 2007 to 2013. This trend mirrors the introduction of newer, more potent immunosuppressive agents like tacrolimus in the early 1990s, followed by the use of lower doses and immunosuppression minimization protocols in subsequent years. Another study from the United States transplant registry showed a diagnosis of malignancy in 2.18% of pediatric solid organ transplant recipients. Compared with the general pediatric population, the incidence was significantly increased for hematological malignancies, particularly non-Hodgkin lymphoma, as well as other types of cancer including of the genitourinary tract.⁴⁰⁰ Long-term follow-up showed a sixfold increased rate of cancer compared with the general population. A larger single-center series in the United States found 13% of patients contracting some form of malignancy by 20 years after transplantation, with similar findings in a large study from the ANZDATA database – a cumulative incidence of nonskin cancer in 14% of pediatric patients at 25 years.^{401,402} A higher hazard of death or graft loss was seen in patients with posttransplantation cancer, compared with those with no malignancy.

PTLD is an uncontrolled proliferation of lymphocytes in the constellation of posttransplant immunosuppression, most often associated with EBV infection. The cumulative incidence

of PTLD in children in the United States is 1.7% at 5 years posttransplant, but is close to 3% in children who are EBV seronegative at transplantation.¹⁸⁸ The risk of PTLD is 30-fold higher than in healthy controls, and is higher in EBV-negative recipients of EBV-positive donor kidneys.^{401,402} By contrast, PTLD in children infected with EBV, who were previously seronegative, is associated with better survival.⁴⁰³ A high EBV viral load is correlated with the probability of developing PTLD.⁴⁰⁴ Diagnosis is made by histological examination of involved tissue, and is classified by the World Health Organization (WHO) into four categories: early lesions, polymorphic PTLD, monomorphic PTLD, and Hodgkin lymphoma. Early lesions are usually managed by immunosuppression reduction in order to restore the host's virus-specific immunity, and by monitoring of viral loads to assess response. In CD20-positive B-cell polymorphic disease, an anti-CD20 antibody, such as rituximab, is often effective together with immunosuppression reduction. Monomorphic disease is most frequently large B-cell lymphoma and may also respond to rituximab therapy, but many patients require the addition of chemotherapy. T-cell lymphoma is often more aggressive in nature, and Hodgkin lymphoma PTLD is very rare. The source of the lymphoma is almost always the recipient rather than the donor kidney.⁴⁰⁵ In a retrospective study of 45 consecutive PTLD patients younger than 25 years, 80% of tumors expressed EBV and 70% were CD20 positive. Both of these presented earlier after transplantation, and CD20-positive PTLD was associated with a higher 5-year survival.⁴⁰⁶ Patients with PTLD are at risk for graft dysfunction and loss, due to acute rejection or CAN.⁴⁰⁷

Other types of cancer in children after renal transplantation include the urinary tract, primarily renal cell or urothelial cell carcinoma, which may arise from the native kidneys or the allograft.⁴⁰⁷ Nonmelanoma skin cancer is common in transplant recipients, though more frequently diagnosed in young adults than in children. Regular use of high protection factor sunscreen can reduce the incidence of skin cancer, and referral for an annual dermatological examination may help in detection of malignant and premalignant lesions.⁴⁰⁸

ANEMIA

Anemia is usually defined in children as hemoglobin concentration more than 2 standard deviation below mean for age, or dependency on erythropoietin therapy. Blood transfusion is necessary in up to half of pediatric kidney transplantations, predicted by pretransplant hemoglobin.⁴⁰⁹ Anemia is common in children after renal transplantation, ranging from 25% to 50% of patients.^{410–412} Incidence is highest in the early months after transplantation, after which many patients have a gradual increase in hemoglobin.⁴¹⁰ Decreased graft function is strongly associated with posttransplant anemia, as anemia is a marker of allograft dysfunction due to impaired erythropoietin production. However, immunosuppressive and antiviral drugs are also implicated in its pathogenesis.^{413,414} Other causes of anemia include nutritional deficiencies and blood loss, particularly in the perioperative period, viral infections, chronic inflammation, severe hyperparathyroidism, and treatment with ACE inhibitors. CMV infection is associated with anemia as well as leukopenia and thrombocytopenia in some patients, and the

anemia resolves with antiviral treatment. Parvovirus B19 infection has been described in pediatric kidney transplant recipients, presenting as severe anemia with reticulocytopenia and bone marrow findings compatible with pure red cell aplasia.⁴¹⁵ Parvovirus infection can be confirmed by bone marrow and sometimes by peripheral blood PCR testing for the virus, and treatment with IVIG is successful in viral eradication and correction of anemia.⁴¹⁶

Posttransplant anemia should be treated, in order to optimize oxygen delivery to organs and improve patient well-being. It is important to avoid unnecessary blood transfusions, which may cause alloantigen sensitization. An iron-rich diet is insufficient and oral iron supplementation is usually required (often ESAs). In some patients, parenteral iron sucrose or sodium ferric gluconate may be beneficial, as it results in faster repletion of iron stores and is not dependent on patient compliance.⁴¹⁷ Complete blood count and iron stores parameters should be frequently monitored in pediatric kidney transplant recipients, and other causes of anemia, including B₁₂ and folate deficiencies, occult infection, inflammation, or blood loss, must be ruled out.

Leukopenia or neutropenia may also be seen after kidney transplantation, and are usually noted in the first few months after transplantation.²⁸⁵ They are associated with immunosuppressive medications, most frequently mycophenolate, though both viral infections and antiviral prophylactic therapy may also induce neutropenia.³³⁸ Drug withdrawal may increase the risk of rejection or viral infection, respectively, and granulocyte-colony stimulating factor therapy may accelerate recovery in severe cases.²⁸⁵

BONE HEALTH AND GROWTH

Children are unique in that they grow, and the effect of MBD on growing bones is enhanced. Preexisting renal osteodystrophy at the time of transplantation, impaired graft function and the effects of medications, particularly corticosteroids, are the main factors associated with bone disease and growth retardation. Successful kidney transplantation corrects many of the metabolic abnormalities which characterize CKD; however, some may persist, particularly when allograft function is impaired. New mineral disturbances may appear such as hypophosphatemia, due to residual hyperparathyroidism and elevated levels of FGF-23, a phosphaturic hormone produced by osteocytes, particularly in the early posttransplant period.^{418,419} Vitamin D deficiency is common among children with CKD and does not significantly improve after transplantation, whereas PTH levels decrease in most patients and are inversely correlated with 25-OH vitamin D concentrations.^{420,421} Bone mineral density may be measured by dual-energy X-ray absorptiometry (DXA); however, this modality is limited in differentiating between trabecular and cortical bone in CKD-MBD and thus does not predict fracture risk well in this population.⁴¹⁸ Peripheral quantitative computed tomography is another modality to assess bone mineralization, which can distinguish between cortical and trabecular bone. Adjustment of DXA spine and whole-body measures for height Z-scores can provide clinically relevant data on bone health that can be monitored over time.⁴²² Spine bone mineral density decrease is associated with higher corticosteroid doses. Therapeutic interventions to promote bone health include steroid minimization or

avoidance, promoting good nutrition, correction of metabolic abnormalities, and vitamin D deficiency. Weight-bearing physical activity can increase bone mass, especially in children and adolescents, who have not yet reached their peak bone mass, and should be encouraged in pediatric kidney transplant recipients.⁴²³

At the time of transplantation, most children have linear growth retardation, with a mean height deficit of -1.73 standard deviation according to NAPRTCS data.¹¹¹ After transplantation, reversal of the uremic milieu should allow normal GH/IGF-1 secretion and function; however, catch-up growth is not seen in all patients. Improved height deficit is observed in children younger than 6 years at transplantation, whereas other age groups tend to maintain a relatively normal growth rate without improvement in Z-scores. Final adult height of children undergoing kidney transplantation has improved over the past decades, with the most recent cohort (2007–2013) reaching a mean Z-score of -0.89 , compared with -1.93 20 years earlier.¹¹¹ This increase of height Z-score in children undergoing kidney transplantation over the past 20 years is a result of many factors, including optimizing rhGH use prior to and after transplantation, corticosteroid minimization and avoidance, and a lower incidence of acute rejection (necessitating additional steroid therapy).

Catch-up growth is limited in patients with poor graft function. Children with ESKD often have disproportionate growth stunting, with shorter leg length, whereas after transplantation a restoration of normal body proportions is noted.⁴²⁴

Corticosteroid treatment and impaired allograft function are associated with decreased linear growth. Steroid avoidance or early steroid-withdrawal protocols have shown improved linear growth particularly in younger children.²⁸⁰ Several studies have demonstrated the efficacy of rhGH treatment in pediatric kidney transplant recipients. Treatment with rhGH improves final adult height without adversely affecting graft function.⁴²⁵ Prescription of rhGH after kidney transplantation varies considerably between countries, as a result of differing reimbursement policies, but also diverse clinical practices. In many countries, rhGH is significantly less often prescribed than would be expected, even when patients are eligible.⁴²⁶

UROLOGICAL ISSUES

Ureteral stent placement at the time of transplantation has become standard of practice and may reduce early obstruction or leak.⁴²⁷ The stent is usually removed within a month to 6 weeks, but earlier removal may prevent stent-related adverse events, including UTI, pain, and hematuria.⁴²⁸ Late-onset ureteral obstruction is less frequent, and has been reported to be associated with BKV infection.⁴²⁹ Children with ESRD due to posterior urethral valves are at a higher risk for ureteral stenosis and VUR, most likely due to abnormal bladder wall.^{430,431} VUR to the allograft is not uncommon, as shown in a study in adults, which found early VUR to the graft in 40.7% of patients, despite using extravesical mucosal tunneling technique to prevent reflux. There was no increased incidence of UTI or impact on graft function.⁴³² In children with recurrent UTI after transplantation, voiding cystourethrography is recommended. In these patients, antibiotic prophylaxis

together with behavioral treatment of voiding dysfunction, if present, may be helpful. Endoscopic correction of reflux in symptomatic children is a feasible option; however post-procedure obstruction and graft dysfunction have been reported.^{433,434} Open surgical correction of VUR is an alternative approach that has been found safe.^{435,436}

Boys who undergo renal transplantation due to posterior urethral valves often have ongoing bladder dysfunction. Small defunctionalized bladders can increase in capacity and compliance after transplantation, but others might develop more severe bladder dysfunction over time and require later surgical intervention.⁴³⁷ Bladder dysfunction may manifest as increased postvoid volume; daytime frequency, urgency, or incontinence; and nighttime incontinence. Inadequate bladder drainage may adversely affect allograft function. Some children require intermittent catheterization, particularly if they have undergone bladder augmentation.⁴³⁸

PUBERTY AND REPRODUCTION

Delayed onset of puberty, menstrual irregularity, and infertility are common in patients with CKD, particularly ESKD. After transplantation many of the hormonal disturbances resolve and fertility is usually restored. Most adolescents who underwent kidney transplantation in childhood have normal onset of puberty, although a minority have delayed sexual maturation.⁴³⁹ No difference in growth and pubertal development was noted between patients treated with mTOR inhibitors compared with conventional calcineurin inhibitor-based immunosuppression.⁴⁴⁰ A recent study of young men who had undergone kidney transplantation in childhood suggested that although reproductive hormone levels were normal, testicular volume and sperm counts were lower than in healthy controls, raising concerns about fertility.^{439,440a}

Restoration of fertility in adolescent girls after kidney transplantation exposes them to the risk of unplanned pregnancy. Counseling regarding reproductive health is important in order to avoid unplanned pregnancy as well as sexually transmitted diseases. The form of contraception used must take into account safety and efficacy of the method and patient adherence.⁴⁴¹ Pregnancy after renal transplantation is high risk both to the mother and to the fetus, but with meticulous prenatal care in selected patients with stable graft function and well-controlled BP, excellent outcomes can be achieved.²⁷⁸ Timing is important, as pregnancy in the first 2 years after transplantation may adversely affect graft function.⁴⁴² Changes in medications including immunosuppressive regimen may be necessary before considering pregnancy, to avoid teratogenic effects of medications, such as mycophenolate, sirolimus, ACE inhibitors, and others.⁴⁴³ No increase in congenital malformations has been observed in infants fathered by kidney allograft recipients.⁴⁴⁴

NONADHERENCE

Alloimmunity necessitates lifelong immunosuppression in order to preserve graft function. Nonadherence to the medical regimen remains a major barrier to long-term allograft survival. In the case of young children, parents or caregivers are

responsible for medical care. Risk factors for nonadherence include health system factors such as insurance coverage for care and medications and treatment-related factors, like regimen complexity, side effects, and even taste and size of pills. Psychosocial factors such as poor social support, family distress, and lower health literacy also affect adherence.⁴⁴⁵ In adolescent patients, additional factors include the desire to be normal, poor planning, and insufficient parental monitoring.²⁰⁴ In some studies, deceased donor kidney transplantation was associated with a higher rate of nonadherence.²⁰⁶

Adherence can be measured by missed appointments, prescription-filling patterns, self-report or patient questionnaires, variance in immunosuppressive drug levels, or various electronic monitoring systems.⁴⁴⁶ The prevalence of drug nonadherence in pediatric renal transplant recipients varies in studies between 6% and 45% depending on the group and the method used to assess adherence.²⁰⁶ Adolescents are at high risk for nonadherence and have shorter graft survival; in fact, between the ages of 17 and 24 years the adjusted HR of graft loss was 1.61 compared with ages of 3 to 17 years and 1.28 compared with young adults aged older than 24 years.²⁰⁵ Persistent nonadherence is associated with acute rejection, graft loss, and death as well as increased medical costs.^{311,447} It is important to develop strategies to improve patient and family adherence to the medical regimen, using a multidisciplinary approach aimed at improving communication between the medical team and the family, as well as between the patient and caregivers. Attention to adverse effects and trying to minimize them, as well as simplifying the medication regimen, may help improve patient adherence. Empowering adolescents to gradually assume responsibility for their medical care, patient and family educational interventions at clinic visits, and availability of clinic staff to answer questions may all assist in improving adherence in pediatric kidney transplant recipients.

TRANSITION

Transition of adolescents or young adults from the pediatric to adult nephrology or transplant unit is a challenging process, which may be associated with increased risk of nonadherence and graft loss.⁴⁴⁸ Although some clinics may use chronological age alone to guide timing, other measures such as medical adherence, patient competency, and responsibility for care should be taken into account. Use of specialized transition clinics may assist in a smoother process, less medication changes, and higher patient satisfaction.⁴⁴⁹ Practices vary between centers as to age at transition (usually 16–21 years), the utilization of specialized transition clinics, structured assessment of patient readiness, and the use of educational tools.⁴⁵⁰ Strategies for avoiding the pitfalls of transition might include initiating the transition process early and involving the patient in an age-appropriate manner. Coordination with all members of the team and good communication, as well as transfer during a stable period are crucial for a successful process.⁴⁵¹

REHABILITATION AND QUALITY OF LIFE

Optimizing kidney transplantation in children cannot be defined only by excellent patient and graft survival rates. Measures of general health and well-being, psychosocial

adjustment, school attendance, level of education, neurodevelopmental outcome, psychological morbidity, and other parameters are important in evaluating overall quality of life. Several studies have utilized various questionnaires and scales to quantify psychosocial outcomes in these patients. Although health-related quality of life is higher in pediatric kidney transplant recipients compared with those remaining on dialysis, it is still lower than in healthy controls.⁴⁵² Many issues remain pertinent, for example, distress arising from physical symptoms and body image, especially height, adverse effects of medications, nonadherence, and psychological problems such as anxiety and depression. Neurocognitive development may be impaired in children with CKD, some of whom have lower scores on IQ testing than the general population, in particular in executive function and memory domains. Although children after kidney transplantation compared favorably with their counterparts treated with dialysis, their scores were below those of matched healthy children. Mild deficits in academic skills were also found.⁴⁵³ This difference persisted when children with neurological comorbidities were excluded.⁴⁵⁴ School performance is also affected by prolonged absences for medical reasons, which impact on academic achievements as well as peer relationships.

Kidney transplantation in childhood has a long-term influence on socioeconomic rehabilitation in adulthood. Two studies report on adult outcomes of children who underwent transplantation in an earlier era (1970s and early 1980s) and reached adulthood. A study from France found a lower level of education than in the general population and a lower rate of independent living, marriage, and parenthood, although most of the patients were employed. A significant correlation was found between education level, employment, marriage, independent living, and final height.⁴⁵⁵ A similar study from the United States found a high rate of employment, as well as education levels and overall satisfaction equivalent to the general population, despite significant physical morbidity.⁴⁵⁶ Of note, in both these studies a large percentage of the patients were severely short, and therefore different from recent cohorts who are treated with lower doses of corticosteroids or steroid-avoidance protocols, as well as rhGH. One study compared young adults with ESKD which presented in childhood with a group with later-onset disease. Patients with later clinical presentation were more likely to have a higher educational attainment, to be employed, and to be living independently than young adults with early onset disease.⁴⁵⁷ A recent study which looked at long-term rehabilitation in kidney-transplanted children in France found lower rates of living with a partner and of parenthood than the general population, and this was correlated with disease severity. Approximately 10% had intellectual disability and the remainder were within the normal range of intelligence. When corrected for the parental educational level, most educational levels were comparable to the general population as were professional occupational categories. Adult height was much improved in the pediatric kidney transplant recipients in comparison to the earlier cohort, and was no longer associated with most social outcomes.⁴⁵⁸

Although most adults with CKD who underwent kidney transplantation in childhood have good long-term outcomes, some challenges persist in integrating into work, independent living, and building a family. A multidisciplinary team approach helps provide individualized care to children and

adolescents with CKD on dialysis and after kidney transplantation. The goal in these young patients is to achieve the best outcomes, not just of kidney function, but in all aspects of health maintenance and rehabilitation.



Complete reference list available at [ExpertConsult.com](http://www.expertconsult.com).

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BOARD REVIEW QUESTIONS

1. Which of the following statements regarding the incidence of end-stage kidney disease (ESKD) in children is FALSE?
- The incidence in children is approximately 10% of the incidence in adults across various registries.
 - There is a higher incidence in boys, due to the increased prevalence of congenital anomalies of the kidney and urinary tract (CAKUT).
 - In societies with high rates of consanguineous marriages, autosomal recessive inherited diseases constitute a high proportion of ESKD.
 - The etiology of ESKD is not affected by patient ethnicity.
 - Varying definitions of diseases such as focal segmental glomerulosclerosis (FSGS) account in part for discrepancies between registries

Answer: d

2. Which of the following statements regarding choice of renal replacement modality is TRUE?
- A 6-year-old child with mild developmental delay and ESKD due to renal dysplasia should initially be referred for peritoneal dialysis rather than preemptive kidney transplantation.
 - Peritoneal dialysis enables better school attendance than in-center hemodialysis.
 - The most frequent renal replacement therapy in infants in most registries is hemodialysis.
 - In adolescents, central venous catheters are preferred over arteriovenous fistulas for hemodialysis access, because they have fewer complications.
 - A severely nephrotic child with ESKD should receive preemptive kidney transplantation.

Answer: b

3. Special considerations should be taken when performing kidney transplantation in these cases EXCEPT:
- When ESKD is due to primary hyperoxaluria type 1, combined liver-kidney transplantation is indicated.
 - A full urinary tract evaluation should be performed before transplantation when the underlying disease is posterior urethral valves.
 - In young children with ESKD due to CAKUT, most childhood vaccinations should be postponed until after transplantation.
 - Recent Epstein-Barr seroconversion should prompt postponement of kidney transplantation.
 - Pretransplantation bilateral nephrectomy should be considered in patients with heavy proteinuria and ESKD due to FSGS.

Answer: c

4. The following statements are FALSE regarding dialysis in children?

- Dialysis should be initiated once the eGFR falls below 15 mL/min/1.73 m², or earlier, if other metabolic derangements, volume overload, or medical signs and symptoms are refractory to conventional treatment.
- There is obligatory exposure to multiple blood products, which may increase the risk of infection or HLA sensitization.
- Disequilibrium syndrome is a common potentially dangerous complication of HD.
- Hypothermia may occur if the dialysate is not warmed, particularly in small children,
- Hypoglycemia may occur during HD as described in adults. It is associated with the use of glucose-free dialysate and high dialysate flow, leading to excessive glucose clearance

Answer: c

5. Which of the statements about long-term follow-up of children after kidney transplantation is FALSE?

- Frequency of clinic visits can be decreased during adolescence, because compliance improves.
- Anemia is fairly frequent and often correlates with poor graft function.
- Risk factors for cardiovascular disease are common in pediatric kidney transplant recipients and should be monitored and treated.
- Lymphoproliferative disease is the most frequent malignancy in pediatric kidney transplant recipients.
- Treatment with recombinant growth hormone improves growth velocity, without adversely affecting graft function.

Answer: a

6. Which statements are FALSE regarding renal replacement therapy in children?

- HD is the most common modality of dialysis treatment in children worldwide.
- Some contraindications to performing PD are specific to the pediatric patient. These include various abdominal congenital anomalies such as gastroschisis, omphalocele, diaphragmatic hernia or bladder exstrophy, and usually polycystic kidney disease in addition to previous abdominal surgery, ventriculoperitoneal shunt, and abdominal adhesions.
- Survival is longer following kidney transplantation in all age groups and in both genders.
- The main forms of vascular access for hemodialysis are central venous catheter (CVC), AV fistula, or rarely an AV graft.

Answer: a